From Block Orthogonality to Decidability in Complex-Weighted Counting CSP

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Abstract

In a landmark JACM paper recognized with the 2021 Gödel Prize, Cai and Chen established a complexity dichotomy for counting CSPs over arbitrary finite domains with algebraic complex weights. Their tractability criterion consists of three conditions on the infinite family of functions obtained from CSP instances by summing out some variables: the Block Orthogonality and Type Partition conditions, and a Mal'tsev polymorphism condition on support and row-equivalence relations. They left open whether this criterion is decidable from the finite set of constraint functions. We resolve this question by giving a total exact algorithm for every q-element domain ($q \geq 1$), with algebraic weights encoded exactly. We also prove that the Block Orthogonality condition implies the other two conditions, reducing the tractability criterion to a single structural property. We obtain similar decidability and structural results for Lin's tractability criterion for $\#\mathrm{CSP}^d$, where each variable occurs a multiple of d times.

Our method first shows that every failure of Block Orthogonality can be witnessed by a generated function of arity at most $q^2 + 1$. At each such arity, we reduce checking infinitely many generated functions to finitely many exact algebraic tests, using tensor products and a pinned extension of Young's all-instance isomorphism theorem. For the structural implication, we use powers of generated functions to control complex cancellation and construct functions whose supports encode row equivalence. These constructions yield the Type Partition and Mal'tsev polymorphism conditions.

1 Introduction

Counting constraint satisfaction problems provide a common language for partition functions, weighted graph homomorphisms, and many other counting problems. Cai and Chen established a complete complexity dichotomy for counting CSPs over arbitrary finite domains with algebraic complex weights: for every fixed finite set $\mathcal{F}$ of constraint functions, $\#\mathsf{CSP}(\mathcal{F})$ is either polynomial-time computable or #P-hard [CC17]. Such a classification rules out intermediate complexity throughout this class. Ladner's theorem shows that intermediate complexity can occur between P and NP-completeness if P $\neq$ NP [Lad75].

The dichotomy leaves a separate algorithmic question: given $\mathcal{F}$, can one decide whether its tractability criterion holds? Cai and Chen's criterion concerns an infinite family $W_{\mathcal{F}}$ of functions obtained from instances by retaining some variables and summing over the others. Both the arity of a generated function and the size of an instance representing it are unbounded. The resulting family can contain infinitely many distinct functions even at a fixed arity. The criterion imposes the Block Orthogonality, Type Partition, and Mal'tsev polymorphism conditions on this entire family and its derived relations. Cai and Chen explicitly left decidability of their criterion open [CC17, Section 9].

We resolve this question and show that the three conditions reduce to one. Fix a domain $D = \{0, \ldots, q-1\}$, where $q \geq 1$, and write $\mathsf{BO}(\mathcal{F})$, $\mathsf{TP}(\mathcal{F})$, and $\mathsf{Mal}(\mathcal{F})$ for the three conditions, defined in Section 2. Their conjunction is denoted by $\mathsf{CC}(\mathcal{F})$. The input consists of the arities and algebraic complex values of finitely many functions on D.

Theorem 1.1 (Uniform decidability)**.** *There is a deterministic algorithm that, given $q \geq 1$ and a finite set $\mathcal{F}$ of algebraic complex-valued functions on D, halts and decides whether $\mathsf{CC}(\mathcal{F})$ holds. Moreover,*

$$\mathsf{CC}(\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{BO}(\mathcal{F}).$$

Thus one can decide from the finite input which side of Cai and Chen's dichotomy applies.

Our result extends to Lin's dichotomy for $\#\mathsf{CSP}^d(\mathcal{F})$, where every variable occurs a multiple of d times, counting all positions in all constraints [Lin21, Theorem 3]. Let $W_{\mathcal{F}}^d$ be the family generated by instances satisfying this restriction. Write $\mathsf{BO}^d(\mathcal{F})$, $\mathsf{TP}^d(\mathcal{F})$, and $\mathsf{Mal}^d(\mathcal{F})$ for the corresponding conditions on $W_{\mathcal{F}}^d$, and let $\mathsf{CC}^d(\mathcal{F})$ be their conjunction. The extension requires an identity test for the restricted instance class and constructions that preserve the degree restriction.

Corollary 1.2 (Uniform decidability for $\#\mathrm{CSP}^d$)**.** *For every $d \geq 1$, there is a deterministic algorithm that, given $q \geq 1$ and a finite set $\mathcal{F}$ of algebraic complex-valued functions on D, halts and decides whether $\mathsf{CC}^d(\mathcal{F})$ holds. Moreover,*

$$\mathsf{CC}^d(\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{BO}^d(\mathcal{F}).$$

From infinitely many functions to a finite test. An algorithm accepting $\mathcal{F}$ must certify Block Orthogonality throughout the generated family, despite unbounded arity and instance size. A failure is witnessed by two rows. Identifying variable positions with the same pair of values in their row indices preserves both rows and leaves at most q^2 row variables. Purification invariance then bounds the witness arity by $q^2 + 1$. This bounds the number of entries in each function, but leaves infinitely many possible representing instances at each fixed arity.

At each remaining arity, Block Orthogonality combines support and magnitude restrictions with a choice between dependence and blockwise orthogonality for each row pair. All generated values lie in the number field determined by the input. Its finite group of roots of unity lets us express the purified condition by finite systems of polynomial equations in the original values. Combining these

alternatives gives one finite set of identities. The task becomes deciding whether each identity holds on every generated function (Section 3).

To handle this universal quantifier, we use how instances produce function values. Gluing two instances at their retained variables multiplies the corresponding values. Consequently, each monomial M defines a multiplicative function χ_M on instances, with its input tuples fixed by M. These functions are multiplicative characters. Distinct characters are linearly independent, so an identity holds on all instances exactly when its coefficients sum to zero within each group of monomials defining the same character.

This leaves one question: whether two characters agree on all instances. Tensor products express them as pinned partition functions on finite domains. A pinned extension of Young's isomorphism theorem makes their agreement equivalent to the existence of a domain bijection satisfying explicit function-value and pin conditions. Enumerating these finitely many bijections therefore determines the character groups. Together with the coefficient sums, this gives a finite test for the original universal identities (Section 4).

From row conditions to a common operation. The Mal'tsev polymorphism condition requires one ternary operation φ satisfying $\varphi(a, b, b) = \varphi(b, b, a) = a$ and preserving both the supports and the row-equivalence relations of all generated functions. Block Orthogonality constrains rows without specifying such an operation. We connect the two by making the operation's values the variables of a constraint system.

For finitely many generated supports, the constraints require that applying the candidate operation to three tuples in any chosen support produce another tuple in it. Products of representing functions realize the solution relation as a generated support. We also need projections of this relation, but summing over omitted variables may cancel the contributions from satisfying extensions. Taking a suitable common positive power before summing prevents this cancellation, simultaneously for every retained assignment, by a consequence of the Skolem–Mahler–Lech theorem.

These projections are therefore generated supports, so Block Orthogonality forces their overlapping nonempty row supports to be equal. Starting with the third projection operation, which preserves every relation and already satisfies one Mal'tsev identity, we use this property to impose the other identity one input triple at a time. The previously prescribed values remain extendible to a solution of all preservation constraints. This gives a common operation for each finite collection of supports. Since there are finitely many candidate operations, one preserves all generated supports: otherwise finitely many supports would exclude every candidate.

The same operation must also preserve row equivalence, which relates indices of nonzero linearly dependent rows. We construct a generated function on pairs of row indices whose support is exactly this relation. Suitable positive powers and summation use the phase restrictions from Block Orthogonality to give zero on unrelated pairs, while a power-sum argument ensures nonzero values on every related pair. The operation already constructed for all generated supports thus preserves row equivalence too. Its preservation of row equivalence also gives Type Partition (Section 5).

Relation to earlier effective classifications. Effective dichotomies were known for unweighted counting CSP [Bul13, DR13], for nonnegative weighted counting CSP [CCL16], and for complex-weighted counting CSP on the Boolean domain [CLX14]. For graph homomorphisms with algebraic complex weights, Cai, Chen, and Lu gave a polynomial-time algorithm for deciding the tractable case [CCL13, Theorem 1.2]. Our result makes the tractability criteria of Cai and Chen [CC17] and Lin [Lin21] decidable over arbitrary finite domains with algebraic complex weights, including Lin's degree restriction. Our decision procedure uses finite exhaustive searches. The claimed algorithmic

guarantee is termination on every exactly represented input.

Organization. After the definitions in Section 2, Sections 3 and 4 give the two reductions and assemble the decision procedure. Section 5 proves the structural implication, and Section 6 treats the degree restriction. The full paper follows this exposition. References with prefix F point to its detailed statements and proofs.

2 Problems and tractability conditions

Write $[n] = \{1, \ldots, n\}$ and $[0] = \varnothing$. As in the full paper, $\mathbb{C}$ denotes the algebraic complex numbers, with input values represented exactly. Arithmetic, conjugation, and equality are effective [Coh93, Section 4.2]. Each $f \in \mathcal{F}$ has an arity $r_f \geq 1$. An instance I has a variable set $V(I) = \{x_1, \ldots, x_n\}$, with $n = n(I) \geq 1$, and a finite multiset $\mathcal{C}_I$ of constraints $(f; i_1, \ldots, i_{r_f})$. Indices may repeat within a constraint. Its instance function and partition value are

$$F_I(a_1, \ldots, a_n) = \prod_{(f;i_1,\ldots,i_{r_f}) \in \mathcal{C}_I} f(a_{i_1}, \ldots, a_{i_{r_f}}), \qquad Z(I) = \sum_{\mathbf{a} \in D^n} F_I(\mathbf{a}).$$

The problem $\#\mathsf{CSP}(\mathcal{F})$ asks for $Z(I)$ on input I. For $1 \leq t \leq n$, retaining the first t variables gives

$$F_I^{[t]}(\mathbf{x}) = \sum_{\mathbf{y} \in D^{n-t}} F_I(\mathbf{x}, \mathbf{y}), \qquad \mathbf{x} \in D^t.$$

Define

$$W_{\mathcal{F}} = \{F_I^{[t]} : I \text{ is a finite } \#\mathsf{CSP}(\mathcal{F}) \text{ instance}, \quad 1 \leq t \leq n(I)\}.$$

Here I ranges over all variable orderings, so any ordered subset of variables may be retained. For $t = n(I)$, the sum has one term. There is no bound on n or on the number of constraints. In particular, fixing t fixes the number of function values but leaves infinitely many possible representing instances. Products with fresh internal variables, identification of retained variables, and summation over some variables are again represented by instances. These constructions allow us to apply the global conditions to auxiliary functions.

For $F : D^r \to \mathbb{C}$, write $\mathrm{arity}(F) = r$. When $r \geq 2$, regard F as a matrix with rows $F(\mathbf{x}, *) = (F(\mathbf{x}, z))_{z \in D}$ indexed by $\mathbf{x} \in D^{r-1}$. The function is *block-rank one* if any two nonzero rows have disjoint supports or proportional magnitude vectors. For two nonzero rows $\mathbf{u}, \mathbf{v}$ with proportional magnitudes, partition their common support into the level sets B of $|u_z|$. They are *block-orthogonal* if $\sum_{z \in B} u_z \overline{v_z} = 0$ on every such set. The function F is block-orthogonal if it is block-rank one and every pair of nonzero rows with proportional magnitudes is either linearly dependent or block-orthogonal.

The global condition uses Cai and Chen's *purification* [CC17, Lemma 7]. This transformation replaces multiplicative generators of the nonzero values by distinct positive primes, retains their root-of-unity factors, and leaves zeros fixed. After scaling, the values are nonnegative integers times roots of unity. It puts the multiplicative structure into a form on which the row conditions above are imposed. We use this theory through its consequences for those conditions. The precise construction is given in the full paper at Definition F.2.2 and the subsequent definition of joint purification.

Definition 2.1. The *Block Orthogonality condition* $\mathsf{BO}(\mathcal{F})$ holds if, for every nonempty finite tuple of distinct functions in $W_{\mathcal{F}}$, each component of arity at least two is block-orthogonal after joint purification.

Thus $\mathsf{BO}(\mathcal{F})$ requires the row conditions throughout the generated family, including functions represented by arbitrarily large instances.

For a function $F : D^r \to \mathbb{C}$, its support is $\Phi_F = \{\mathbf{a} \in D^r : F(\mathbf{a}) \neq 0\}$. For $r \geq 2$, partition its nonzero row indices into classes $S_1, \ldots, S_s$ of linearly dependent rows. Its row-equivalence relation is $\Omega_F = \bigcup_{j \in [s]} S_j \times S_j$. For a prefix $\boldsymbol{\alpha} \in D^\ell$, where $0 \leq \ell \leq r-1$, define

$$\mathrm{type}_F(\boldsymbol{\alpha}) = \{j \in [s] : \text{some row index in } S_j \text{ extends } \boldsymbol{\alpha}\}.$$

For the zero function, there are no row classes, and these relations and types are empty.

An operation $\varphi : D^3 \to D$ preserves a relation if applying φ to the corresponding entries of any three tuples in the relation produces another tuple in it. Such an operation is a *polymorphism* of the relation. It is a *Mal'tsev operation* if $\varphi(a, b, b) = \varphi(b, b, a) = a$ for all $a, b \in D$. Define the relation family

$$\Lambda_{\mathcal{F}} = \{\Phi_F : F \in W_{\mathcal{F}}\} \cup \{\Omega_F : F \in W_{\mathcal{F}},\ \mathrm{arity}(F) \geq 2\}.$$

Definition 2.2. The *Type Partition condition* $\mathsf{TP}(\mathcal{F})$ holds if, for every $F \in W_{\mathcal{F}}$ of arity at least two, any two types of prefixes of the same length are equal or disjoint. The *Mal'tsev polymorphism condition* $\mathsf{Mal}(\mathcal{F})$ holds if a single Mal'tsev operation preserves every relation in $\Lambda_{\mathcal{F}}$.

For supports, applying the operation to three tuples where F is nonzero must give another such tuple. For row equivalence, applying it to the indices of three pairs of nonzero dependent rows must again give indices of such a pair. One operation must satisfy both requirements for all $F \in W_{\mathcal{F}}$. We establish them by different constructions in Section 5.

Cai and Chen's dichotomy makes $\#\mathsf{CSP}(\mathcal{F})$ polynomial-time computable when $\mathsf{CC}(\mathcal{F}) = \mathsf{BO}(\mathcal{F}) \wedge \mathsf{TP}(\mathcal{F}) \wedge \mathsf{Mal}(\mathcal{F})$ holds and #P-hard otherwise [CC17, Theorem 1 and Sections 3.1–3.2]. Lin's conditions use the same definitions with $W_{\mathcal{F}}$ replaced by $W^d_{\mathcal{F}}$ [Lin21].

3 From Block Orthogonality to polynomial identities

We first eliminate the need to purify arbitrary tuples jointly, then bound the arity of a counterexample. To obtain a finite test at each remaining arity, we describe the purified condition by equations in the original function values.

3.1 Bounding the arity

We first reduce joint purification to a property of individual functions. The Block Orthogonality row conditions are invariant under the legal choice of purification, including purification as part of a larger tuple. Consequently, $\mathsf{BO}(\mathcal{F})$ holds exactly when each generated function of arity at least two has a block-orthogonal singleton purification (Lemma F.3.1 and Corollary F.3.2). The same invariance lets us transfer a failure between a function and the lower-arity function constructed next.

Lemma 3.1. *If $\mathsf{BO}(\mathcal{F})$ fails, some $F \in W_{\mathcal{F}}$ of arity between two and q^2+1 has a singleton purification that is not block-orthogonal.*

Proof. Choose two rows witnessing failure in a purified r-ary function, with row indices $\mathbf{x}, \mathbf{y} \in D^{r-1}$. Identify positions i, j whenever $(x_i, y_i) = (x_j, y_j)$, keeping the column variable separate. There are at most q^2 resulting row variables. Identifying the retained variables in an instance representing F gives a generated function H of arity at most q^2+1. Assigning the first or second entry of each ordered pair to the new variables recovers exactly the two original rows.

To transfer the failure through purification, purify F and H jointly before normalization. The same map is applied to their shared values, so the two row equalities persist. Purification invariance transfers the failure from the original singleton purification to this joint purification and then to a singleton purification of H. □

3.2 Finite algebraic conditions

All generated values lie in the computable number field $K = \mathbb{Q}(f(\mathbf{a}), \overline{f(\mathbf{a})} : f \in \mathcal{F},\ \mathbf{a} \in D^{r_f})$, generated over $\mathbb{Q}$ by the input values and their conjugates. Its finite group of roots of unity $\mu(K)$ is effectively computable [Coh93, Algorithm 4.9.10]. Purification relates magnitude comparisons to equations with factors in $\mu(K)$, preserves row dependence, and preserves ratios that are roots of unity (Lemma F.3.5). Thus the purified row conditions can be expressed using the original function values and finitely many root-of-unity choices. We now give that construction.

Fix $k \geq 2$, with row indices $\mathbf{x} \in D^{k-1}$ and columns $c \in D$, and fix a total order on the row indices. Introduce one variable $X_{\mathbf{x},c}$ for each function value. A certificate specifies how the alternatives in Block Orthogonality are to hold. Each choice contributes polynomial equations.

Definition 3.2 (Block Orthogonality certificate)**.** A certificate consists of the following finite choices.

1. Choose a possibly empty family of rectangles $R_i \times C_i$, where the nonempty row sets $R_i \subseteq D^{k-1}$ are pairwise disjoint, as are the nonempty column sets $C_i \subseteq D$. Impose $X_{\mathbf{x},c} = 0$ outside their union. These rectangles describe the possible groups of rows sharing a support.

2. For distinct rows $\mathbf{x}, \mathbf{y} \in R_i$ and distinct columns $c, e \in C_i$, choose $\tau_{\mathbf{x},\mathbf{y};c,e} \in \mu(K)$ and impose

$$X_{\mathbf{x},c}X_{\mathbf{y},e} - \tau_{\mathbf{x},\mathbf{y};c,e}X_{\mathbf{x},e}X_{\mathbf{y},c} = 0.$$

   These equations enforce proportional magnitudes after purification.

3. For each pair of rows $\mathbf{x} < \mathbf{y}$ in R_i, choose dependence or orthogonality. For dependence, impose all ordinary 2×2 minors on C_i, that is, the preceding equations with $\tau_{\mathbf{x},\mathbf{y};c,e} = 1$. For orthogonality, choose a partition $\mathcal{T}_{\mathbf{x},\mathbf{y}}$ of C_i. For each group $T \in \mathcal{T}_{\mathbf{x},\mathbf{y}}$, choose an anchor $c_T \in T$ and roots $\rho_{T,c}, \sigma_{T,c} \in \mu(K)$ for $c \in T$, with $\rho_{T,c_T} = \sigma_{T,c_T} = 1$, such that $\sum_{c \in T} \rho_{T,c}\overline{\sigma_{T,c}} = 0$. Impose

$$X_{\mathbf{x},c} = \rho_{T,c}X_{\mathbf{x},c_T}, \qquad X_{\mathbf{y},c} = \sigma_{T,c}X_{\mathbf{y},c_T} \qquad (c \in T \setminus \{c_T\}).$$

   Within each group these equations fix the root-of-unity ratios, so its purified inner product is a common factor times the chosen zero root sum.

We evaluate these equations on K-valued functions. The row and column sets, their partitions, and $\mu(K)$ are finite. We enumerate all choices and retain those whose root sums are zero. The resulting equations have degree one or two. In particular, the construction uses only equations: it does not require entries inside the chosen rectangles to be nonzero, or distinct groups to have different magnitudes. This is essential for the polynomial identity test that follows.

Every block-orthogonal singleton purification supplies a certificate by its actual support rectangles, magnitude ratios, and magnitude blocks. Conversely, the equations with root-of-unity factors force any overlapping nonzero rows to have the same support and proportional purified magnitudes. In orthogonal mode, a group whose anchor is zero disappears from that row. Every surviving group lies within one purified magnitude block and has zero inner product there. Several groups may

merge into one magnitude block, and their zero inner products add to zero. Hence allowing zeros and merged groups introduces no false solutions (Theorem F.4.2).

Different functions may satisfy different certificates. We replace this disjunction by a common system of equations for the identity test. Let $E_1, \dots, E_s$ be the retained certificates' sets of polynomial left-hand sides, with syntactically zero polynomials removed. If some E_j is empty, put $Q_k = \varnothing$. Otherwise put

$$Q_k = \left\{ \prod_{j \in [s]} e_j : e_j \in E_j \text{ for all } j \in [s] \right\}.$$

Write $P(F)$ for evaluation at $X_{\mathbf{x},c} = F(\mathbf{x}, c)$. A function satisfies at least one certificate exactly when every $P \in Q_k$ satisfies $P(F) = 0$: a satisfied certificate supplies a zero factor to every product, whereas if none is satisfied, choosing a nonzero equation value from each gives a nonzero product. Writing $W_{\mathcal{F}}^{(k)}$ for the k-ary members of $W_{\mathcal{F}}$, we obtain

$$\mathsf{BO}(\mathcal{F}) \iff \forall\, 2 \le k \le q^2 + 1\ \forall P \in Q_k\ \forall F \in W_{\mathcal{F}}^{(k)}, \quad P(F) = 0. \tag{1}$$

The remaining universal quantifier ranges over functions represented by instances of arbitrary size. We now give a finite test for it.

4 Testing identities on all instances

The remaining task in (1) is to decide one polynomial identity over all generated functions of a fixed arity. We establish a general test, also allowing conjugates of function values.

Theorem 4.1 (Universal identity test, Theorem F.5.8)**.** *Given a finite set $\mathcal{F}$ of algebraic complex-valued functions on D, an integer $k \ge 1$, and a polynomial P with algebraic coefficients in variables $X_{\mathbf{a}}, Y_{\mathbf{a}}$ for $\mathbf{a} \in D^k$, one can decide whether*

$$P\big((F(\mathbf{a}))_{\mathbf{a} \in D^k}, (\overline{F(\mathbf{a})})_{\mathbf{a} \in D^k}\big) = 0 \quad \textit{for every } F \in W_{\mathcal{F}}^{(k)}.$$

The test uses the behavior of monomials under gluing. It compares the finitely many monomials of P as functions on instances and then tests sums of their coefficients. No Block Orthogonality assumption is needed.

4.1 Gluing and multiplicative characters

Fix $k \ge 1$ and label the k retained variables of an instance by $1, \dots, k$. For a labelled instance I, let $\mathbf{z}_I(\mathbf{a})$ be its partition value with these variables fixed to $\mathbf{a} \in D^k$. Thus $\mathbf{z}_I$ is exactly its generated k-ary function. To multiply two such functions, take disjoint copies of their representing instances and identify equally labelled variables. All unlabelled variables remain separate. If IJ is the glued instance, independence of the two hidden assignments gives

$$\mathbf{z}_{IJ}(\mathbf{a}) = \mathbf{z}_I(\mathbf{a}) \mathbf{z}_J(\mathbf{a}).$$

Up to label-preserving isomorphism, this operation forms a commutative monoid $\mathfrak{M}_k$. Its identity has just the k labelled variables and no constraints.

Introduce formal variables $X_{\mathbf{a}}, Y_{\mathbf{a}}$, evaluated as $\mathbf{z}_I(\mathbf{a}), \overline{\mathbf{z}_I(\mathbf{a})}$. For a monomial, let $p_+, p_- \ge 0$ count its X- and Y-factors, respectively, with multiplicity. Write it as

$$M = \prod_{i=1}^{p_+} X_{\mathbf{a}^{(i)}} \prod_{j=1}^{p_-} Y_{\mathbf{b}^{(j)}}, \qquad \mathbf{a}^{(i)}, \mathbf{b}^{(j)} \in D^k.$$

Evaluating M at the generated values gives

$$\chi_M(I) = \prod_{i=1}^{p_+} \mathbf{z}_I(\mathbf{a}^{(i)}) \prod_{j=1}^{p_-} \overline{\mathbf{z}_I(\mathbf{b}^{(j)})}.$$

The input tuples and their multiplicities are fixed by M. As I varies, χ_M is a function on instances satisfying $\chi_M(IJ) = \chi_M(I)\chi_M(J)$ and $\chi_M(1) = 1$. Such a function is a *multiplicative character*.

A polynomial identity is therefore a linear relation among finitely many characters. We first decide which monomials define the same character. We then show that distinct characters cannot cancel in a nontrivial linear combination.

4.2 Tensor realization and finite comparison

The product defining $\chi_M(I)$ involves $p_+ + p_-$ independent assignments to the same instance. Collect their values at each variable into one element of $D_M = D^{p_+ + p_-}$. Define the indexed family $\mathcal{H}_M = (h_f : f \in \mathcal{F})$ on D_M, where $h_f = T_{p_+,p_-}(f)$ is given by

$$h_f(\mathbf{u}_1, \ldots, \mathbf{u}_{r_f}) = \prod_{\ell=1}^{p_+} f(u_{1,\ell}, \ldots, u_{r_f,\ell}) \prod_{\ell=p_++1}^{p_++p_-} \overline{f(u_{1,\ell}, \ldots, u_{r_f,\ell})}.$$

Pin label t to $\beta_M(t) = (a_t^{(1)}, \ldots, a_t^{(p_+)}, b_t^{(1)}, \ldots, b_t^{(p_-)})$. Replacing every occurrence of f by h_f gives an instance I^M. Its partition value with pins β_M, denoted by $Z_{\beta_M}^{\mathcal{H}_M}(I^M)$, equals $\chi_M(I)$ by expanding the sum over tuple-valued assignments. For a constant monomial, $D_M = D^0$ is a singleton and the transformed functions are constant one. If a second monomial N has r_+ unconjugated factors and r_- conjugated factors, the same construction gives $g_f = T_{r_+,r_-}(f)$ on $D_N = D^{r_+ + r_-}$, the indexed family $\mathcal{H}_N = (g_f : f \in \mathcal{F})$, and pins β_N (Definition F.5.3 and Lemma F.5.4).

We compare these pinned partition functions using the following extension of Young's all-instance isomorphism theorem [You25, Theorem 8].[1] Two domain elements are *twins* if replacing one by the other in any position of any constraint function leaves its value unchanged. Two finite sets of functions are *similar* if an arity-preserving bijection between them is specified. The same instance is evaluated in the two families by replacing each constraint function with its corresponding function.

Theorem 4.2 (Pinned isomorphism, Theorem F.5.2)**.** *Let $\mathcal{H}, \mathcal{G}$ be similar finite sets of constraint functions on finite domains $D_\mathcal{H}$ and $D_\mathcal{G}$, respectively, with values in a common field of characteristic zero. Fix $k \geq 1$ and pins $\beta : [k] \to D_\mathcal{H}$, $\gamma : [k] \to D_\mathcal{G}$. Their pinned partition values agree on every finite k-labelled instance if and only if there is a bijection $\sigma : D_\mathcal{H} \to D_\mathcal{G}$ preserving all corresponding function values such that $\sigma(\beta(t))$ and $\gamma(t)$ are twins in $\mathcal{G}$ for every $t \in [k]$.*

The proof is given in Theorem F.5.2, following Young's interpolation approach. We use the theorem to compare the tensor realizations. Its pin condition reflects what instances can distinguish: replacing a pinned domain element by a twin leaves every constraint value unchanged.

Tensorization may identify distinct functions, whereas Theorem 4.2 compares sets. At positive tensor degree, a transformed function is zero exactly when its original is. Degree zero gives constant-one functions. Thus each family has at most one zero function per arity. Common positive rational scalars λ_f can make $\lambda_f h_f$ and $\lambda_f g_f$ distinct within each arity in their respective families: each

[1] The first arXiv version of this paper used Young's simple-instance strengthening [You25, Corollary 35]. During the Lean formalization, we found that its premise does not imply the asserted isomorphism. The all-instance pinned theorem below supplies the conclusion required by our identity test. Appendix F.B states Young's results, gives an explicit counterexample to the simple-instance corollary, and identifies the gap in its proof.

collision excludes at most one choice for the next scalar. Both partition values are multiplied by the same nonzero factor $\prod_f \lambda_f^{N_f(I)}$, where $N_f(I)$ counts occurrences of f. Their equality is therefore preserved (Lemma F.5.5).

The tagged functions correspond bijectively to $\mathcal{F}$, so every labelled instance over either tagged family comes from an original instance. We may therefore apply Theorem 4.2: enumerate the bijections $D_M \to D_N$ and test corresponding function values and pin-twin conditions exactly. These finite tests decide equality on all instances. If the domain sizes differ, no bijection exists (Lemma F.5.6).

4.3 The identity test and the decision procedure

For any polynomial P in generated values and their conjugates, expand it into monomials, compare their characters, and group equal ones. Distinct multiplicative characters on a monoid are linearly independent over a field. To prove this, suppose a nontrivial relation $\sum_{i=1}^{s} c_i\chi_i = 0$ has the fewest possible nonzero terms. Evaluation at the identity excludes $s = 1$. Choose J with $\chi_1(J) \neq \chi_s(J)$. Subtracting $\chi_s(J)$ times the relation at I from the relation at IJ gives

$$\sum_{i=1}^{s-1} c_i(\chi_i(J) - \chi_s(J))\chi_i(I) = 0 \qquad \text{for every } I.$$

This is a nontrivial relation with fewer terms, a contradiction (Lemma F.5.7). Consequently, writing $P = \sum_M c_M M$, the finite tests

$$\sum_{M \in E} c_M = 0 \quad \text{for every class } E \text{ of monomials defining the same character} \tag{2}$$

decide whether $P(\mathbf{z}_I, \overline{\mathbf{z}_I}) = 0$ for all $I \in \mathfrak{M}_k$. All comparisons and coefficient sums take place in effectively constructed number fields. This proves Theorem 4.1.

Equation (2) involves only the finitely many coefficients of P. The proof of linear independence supplies the link to all instances: if one class sum is nonzero, the resulting linear combination is a nonzero function on $\mathfrak{M}_k$, so some labelled instance violates the identity.

Combining this test with (1) gives the algorithm for $\mathsf{BO}(\mathcal{F})$:

1. Construct K and $\mu(K)$ from the input functions.
2. For each $2 \leq k \leq q^2 + 1$, construct the finite set Q_k.
3. Test every $P \in Q_k$ on all k-labelled instances by character comparison and coefficient sums. Accept exactly when every test succeeds.

Each step is finite. A rejected identity has a generated counterexample at a tested arity. Conversely, any failure of $\mathsf{BO}(\mathcal{F})$ compresses to a tested arity and violates one of its identities. This proves exact decidability of $\mathsf{BO}(\mathcal{F})$ (Theorem F.6.2). When $\mathcal{F} = \varnothing$, all generated functions are positive constants, and the algorithm accepts immediately.

5 From Block Orthogonality to the other conditions

Assume $\mathsf{BO}(\mathcal{F})$. The Mal'tsev polymorphism condition requires an operation $\varphi : D^3 \to D$ that simultaneously preserves all generated supports and all row-equivalence relations. Block Orthogonality

constrains rows, but does not specify the values of such an operation. Even though there are finitely many ternary operations on D, each candidate faces infinitely many preservation requirements.

We first construct a common operation for supports. We express preservation as constraints on its values and use Block Orthogonality to impose the Mal'tsev identities while keeping those constraints satisfiable. To make Block Orthogonality applicable, we realize the solution relation and its projections as generated supports. We then realize row equivalence as another generated support, which makes it preserved by the same operation. Type Partition follows from this second preservation property.

5.1 Realizing relations as generated supports

We realize conjunctions and projections of generated supports as generated supports, making Block Orthogonality applicable. Fix $h \geq 1$, functions $F_1, \ldots, F_h \in W_{\mathcal{F}}$, and free and existential variable tuples $\mathbf{x}, \mathbf{y}$ of lengths $\ell \geq 1, s \geq 0$, respectively. For a finite set $\mathcal{E}$ of atom indices, choose $i(e) \in [h]$ and an ordered scope $\mathbf{z}_e$ of length $\mathrm{arity}(F_{i(e)})$ from $(\mathbf{x}, \mathbf{y})$ for each $e \in \mathcal{E}$. Define $R \subseteq D^\ell$ by $R(\mathbf{x}) \iff \exists \mathbf{y} \bigwedge_{e \in \mathcal{E}} \Phi_{F_{i(e)}}(\mathbf{z}_e)$. Variables may repeat within and across the scopes $\mathbf{z}_e$. Such a definition is an *equality-free primitive-positive definition*: it uses conjunction and existential quantification without adding equality atoms. Multiplication realizes conjunction. Summation realizes projection only if the nonzero contributions do not cancel. We address this by taking a common positive power before summing:

$$H_m(\mathbf{x}) = \sum_{\mathbf{y} \in D^s} \prod_{e \in \mathcal{E}} F_{i(e)}(\mathbf{z}_e)^m, \qquad m \geq 1. \tag{3}$$

Taking m copies of each representing instance, sharing the specified variables and keeping its internal variables fresh, realizes H_m inside $W_{\mathcal{F}}$.

For each fixed assignment $\mathbf{a} \in R$, let $\mathcal{A}_{\mathbf{a}}$ be the nonempty multiset of nonzero products contributed by its satisfying extensions $\mathbf{b} \in D^s$. Then $H_m(\mathbf{a}) = \sum_{w \in \mathcal{A}_{\mathbf{a}}} w^m$. For any finite collection of nonempty multisets of nonzero algebraic numbers, there is a common positive m making all these power sums nonzero (Lemma F.7.1). Each power sum satisfies a linear recurrence with constant coefficients. By the Skolem–Mahler–Lech theorem [Lec53], its zeros form finitely many arithmetic progressions and a finite exceptional set. A progression of period p cannot have residue zero: along sufficiently large multiples of p, grouping equal pth powers would give a Vandermonde system forcing positive integer multiplicities to be zero. A sufficiently large common multiple of all periods therefore avoids both the progressions and the finite exceptional sets. Thus enumerating m and testing the finitely many sums terminates.

With this choice, (3) is nonzero exactly on R, since every term is zero outside R. Thus these relations, including projections retaining at least one free variable, are generated supports (Lemma F.7.2). Purification preserves support, so $\mathsf{BO}(\mathcal{F})$ implies the following rectangularity property: when one free variable is put last, any two nonempty row supports of such a relation are equal or disjoint (Corollary F.7.3). This is the property used to impose the Mal'tsev identities.

5.2 Constructing a common Mal'tsev operation

Fix a finite family Δ of generated supports. Discard empty relations. If none remains, any Mal'tsev operation suffices. We treat the values of a candidate operation as an assignment to a constraint system. For every input triple $\tau = (a, b, c)$ occurring in the preservation tests below, introduce a variable x_τ whose value is the operation's output at τ. Let $T \subseteq D^3$ be the set of these triples. The

constraints are

$$\bigwedge_{R\in\Delta}\ \bigwedge_{\mathbf{u},\mathbf{v},\mathbf{w}\in R} R((x_{(u_i,v_i,w_i)})_{i\in[r_R]}), \qquad R\subseteq D^{r_R}. \tag{4}$$

They require one common operation to preserve all relations in Δ. Write $U_\Delta \subseteq D^T$ for their solution relation. Replacing each relation atom in (4) by its representing function and multiplying gives a function with support U_Δ. The construction in (3) also realizes every projection retaining at least one variable as a generated support. Hence Block Orthogonality supplies rectangularity for precisely the relations in which we will construct the operation's values.

Every projection operation $\pi_i(a_1, a_2, a_3) = a_i$ satisfies (4). These supply initial solutions and values with which to modify them. The third projection satisfies $\pi_3(a, a, b) = b$, whereas the first satisfies $\pi_1(a, b, b) = a$. We combine these two properties without losing extendibility to a solution of the full constraint system.

The required values are b on $A = \{(a, a, b) \in T\}$ and a on $C = \{(a, b, b) \in T : a \neq b\}$. The set A is nonempty: repeating a tuple of a nonempty relation three times supplies a constant triple in T. Enumerate $C = \{c_1, \dots, c_s\}$ and start with $p_0 = \pi_3|_{A\cup C}$. At step j, the assignment p_j on $A \cup C$ extends to a solution of (4), keeps all prescribed values on A, and has the required values at $c_1, \dots, c_j$. The following step imposes the next value while maintaining this invariant.

Suppose $0 \leq j < s$ and write $c_{j+1} = (a, b, b)$ for the next triple. Let $K_j = A \cup \{c_i : i \in [j]\}$. Lift p_j to a solution $\widehat{p}_j \in U_\Delta$, then extend it arbitrarily outside T to a common polymorphism $t_j : D^3 \to D$. The operation $\widehat{t}_j(x, y, z) = t_j(x, y, y)$ also preserves every relation in Δ. It agrees with π_1 on K_j and with p_j at c_{j+1}. For $(x, x, z) \in A$, repeating a relation tuple containing x gives $(x, x, x) \in A$, where the prescribed value forces $t_j(x, x, x) = x$. On the previously treated triples of C, it follows from the values already imposed. Write $X = \pi_1|_{K_j}$, $Y = p_j|_{K_j}$, and $v = p_j(c_{j+1})$. In the projection of U_Δ to $K_j \cup \{c_{j+1}\}$, the restrictions of $\widehat{t}_j$, t_j, and π_1 give respectively

$$(X, v), \qquad (Y, v), \qquad (X, a).$$

These are three corners of a rectangle: rows X, Y share column v, and row X contains a, so rectangularity gives (Y, a). Lift this tuple to U_Δ and restrict to $A \cup C$ to obtain p_{j+1}, retaining the values on K_j and setting $p_{j+1}(c_{j+1}) = a$. Each lift exists by membership in the projection and is found by finite search. After all triples are treated, lift to U_Δ and extend outside T with the prescribed Mal'tsev values, obtaining a common Mal'tsev polymorphism of Δ (Lemma F.7.4).

There are only finitely many Mal'tsev operations on D. If none preserved every generated support, one could choose a support excluding each candidate. That finite family would have no common Mal'tsev operation, contradicting the construction above. Hence one operation φ preserves Φ_F for every $F \in W_\mathcal{F}$ (Theorem F.7.5).

5.3 Realizing row equivalence

For $F \in W_\mathcal{F}$ of arity $r \geq 2$, we realize Ω_F as a generated support, which the same operation φ will preserve. The function must be zero on unrelated row pairs and simultaneously nonzero on all pairs of nonzero linearly dependent rows.

Although Block Orthogonality involves conjugates, we use positive powers, products, and summation, realized by instances over the original $\mathcal{F}$. Let e_K be the exponent of $\mu(K)$ and let $L_q = \operatorname{lcm}(1, \dots, q)$. Thus $e_K \geq 2$ and $\xi^{e_K} = 1$ for every $\xi \in \mu(K)$. Consider

$$C_n(\mathbf{x}, \mathbf{y}) = \sum_{z\in D} F(\mathbf{x}, z)^{(e_K-1)n} F(\mathbf{y}, z)^n, \qquad n = 1 + tL_q, \quad t \geq 0. \tag{5}$$

Copies of an instance representing F realize both positive powers, sharing the displayed variables and keeping all internal variables fresh. Summing over z gives $C_n \in W_{\mathcal{F}}$.

The exponent e_K removes root-of-unity ratios within purified magnitude blocks. The progression $n = 1 + tL_q$ preserves the relative phases between rows, so that their required zero sums remain zero. Finally, choosing t avoids cancellation on dependent pairs. We verify these properties in turn.

The relative phases of overlapping rows have bounded order. For such rows $\mathbf{u}, \mathbf{v}$, choose z_0 in their common support and write $v_z = \lambda \zeta_z u_z$, where $\lambda = v_{z_0}/u_{z_0}$ and $\zeta_{z_0} = 1$. We show $\zeta_z \in \mu(K)$ and $\zeta_z^{L_q} = 1$. Every positive entrywise power of F is generated. Its singleton purification is block-orthogonal by $\mathsf{BO}(\mathcal{F})$, and the Purification Lemma [CC17, Lemma 7 and Corollary 1] gives Block Orthogonality of the original power. Fix a nonempty magnitude block B of the original row $\mathbf{u}$. If the powered rows were independent for every exponent $1, \ldots, q$, the first $|B|$ power sums of $u_z\overline{v_z}$ on B would be zero, since positive powering preserves these blocks. Newton's identities would force the product of these nonzero numbers to be zero. Thus the rows become dependent at some exponent $1 \le h \le q$, giving $\zeta_z^h = 1$. As $\zeta_z \in K$ and $h \mid L_q$, the phase bound follows (Lemma F.8.2).

We now apply this phase bound to (5). Zero rows or disjoint supports contribute zero term by term. For overlapping independent rows, now let B be a magnitude block of the row corresponding to $\mathbf{u}$ in a singleton purification of F. Relative to an anchor $z_B \in B$, the original ratios u_z/u_{z_B} and v_z/v_{z_B} belong to $\mu(K)$. Write these ratios as $\eta_z = u_z/u_{z_B}$ and $\theta_z = v_z/v_{z_B}$. With the relative phases anchored at z_B, we have $v_z = \lambda\zeta_z u_z$ with $\lambda = v_{z_B}/u_{z_B}$, and hence $\theta_z = \zeta_z\eta_z$. Purified block orthogonality and preservation of root-of-unity ratios give $\sum_{z\in B} \eta_z\overline{\theta_z} = 0$. Conjugating yields $\sum_{z\in B} \zeta_z = 0$. Since $n \equiv 1 \pmod{L_q}$, the contribution of this block is

$$\sum_{z\in B} u_z^{(e_K-1)n} v_z^n = \lambda^n u_{z_B}^{e_K n} \sum_{z\in B} \zeta_z = 0.$$

Thus every independent pair gives zero for every allowed n.

For a nonzero dependent pair $\mathbf{v} = \lambda\mathbf{u}$, the value is $\lambda^n \sum_z u_z^{e_K n}$, with the sum over the common support. Along $n = 1 + tL_q$, group equal bases in $\sum_z u_z^{e_K}(u_z^{e_K L_q})^t$. Equal bases at z, w imply $u_z/u_w \in \mu(K)$, hence equal coefficients $u_z^{e_K} = u_w^{e_K}$. Grouped coefficients are thus positive integer multiples of nonzero values. A root-of-unity quotient between distinct bases would likewise force $u_z/u_w \in \mu(K)$ and equality of those bases, a contradiction. The Skolem–Mahler–Lech theorem and a Vandermonde argument therefore give only finitely many zeros. Choose t beyond the exceptional values for all the finitely many dependent row pairs. Then

$$\Phi_{C_n} = \Omega_F.$$

Existence of such a common exponent ensures that enumeration of t and exact tests on the finitely many row pairs terminate (Theorem F.8.4). Since C_n is generated, the operation φ already constructed for all supports also preserves Ω_F. This establishes both preservation requirements of $\mathsf{Mal}(\mathcal{F})$ with the same operation.

5.4 Type Partition and the main theorem

For equal-length tuples $\mathbf{u}, \mathbf{v}, \mathbf{w}$, write $\varphi(\mathbf{u}, \mathbf{v}, \mathbf{w})$ for componentwise application of φ. Fix F and two equal-length prefixes $\boldsymbol{\alpha}, \boldsymbol{\beta}$ whose types intersect. Choose row indices $\mathbf{a}, \mathbf{b}$ in one common class extending these prefixes. For any row index $\mathbf{c}$ extending $\boldsymbol{\alpha}$, the pairs $(\mathbf{c}, \mathbf{c})$, $(\mathbf{a}, \mathbf{a})$, and $(\mathbf{b}, \mathbf{a})$ lie in Ω_F whenever $\mathbf{c}$ indexes a nonzero row. Preservation by φ gives

$$(\varphi(\mathbf{c}, \mathbf{a}, \mathbf{b}), \varphi(\mathbf{c}, \mathbf{a}, \mathbf{a})) = (\varphi(\mathbf{c}, \mathbf{a}, \mathbf{b}), \mathbf{c}) \in \Omega_F.$$

The first row index extends $\boldsymbol{\beta}$, because $\varphi(\alpha_i, \alpha_i, \beta_i) = \beta_i$ at each prefix position, and it belongs to the same class as $\mathbf{c}$. Every class reached from $\boldsymbol{\alpha}$ is therefore reached from $\boldsymbol{\beta}$. Interchanging the prefixes proves that their types are equal (Lemma F.9.1).

We have proved $\mathsf{BO}(\mathcal{F}) \Rightarrow \mathsf{TP}(\mathcal{F}) \wedge \mathsf{Mal}(\mathcal{F})$. The converse implication $\mathsf{CC}(\mathcal{F}) \Rightarrow \mathsf{BO}(\mathcal{F})$ holds by definition. For the empty input family, generated functions are positive constants, so the same conclusion holds directly. The algorithm of Section 4 consequently decides $\mathsf{CC}(\mathcal{F})$, proving Theorem 1.1.

6 The degree restriction

For Corollary 1.2, restrict to instances whose variable degrees are divisible by d. Gluing adds degrees of identified variables, so these form a submonoid $\mathfrak{M}_{k,d} \subseteq \mathfrak{M}_k$. We reduce character equality here to finite comparisons on $\mathfrak{M}_k$.

Let $C_d = \mathbb{Z}/d\mathbb{Z}$ and choose a primitive dth root of unity ω_d, taking $\omega_1 = 1$. The identity

$$\sum_{s \in C_d} \omega_d^{sm} = \begin{cases} d, & d \mid m, \\ 0, & d \nmid m \end{cases}$$

tests one variable's degree. Augment the tensor functions for M to the domain $D_M \times C_d$ by defining, for $a_i \in D_M$ and $s_i \in C_d$,

$$h_f^{(d)}((a_1, s_1), \ldots, (a_{r_f}, s_{r_f})) = h_f(a_1, \ldots, a_{r_f}) \omega_d^{s_1 + \cdots + s_{r_f}}.$$

Each variable occurrence contributes its auxiliary root once. For $\mathbf{s} \in C_d^k$, pin label t to $(\beta_M(t), s_t)$. The resulting partition function $\rho_{\mathbf{s}}^M$ on original instances is a multiplicative character. It sums auxiliary values at hidden variables, and summing over $\mathbf{s}$ covers the labels, giving

$$\chi_M(I) \prod_{v \in V(I)} \left(\sum_{s \in C_d} \omega_d^{s \deg_I(v)} \right),$$

where $\deg_I(v)$ counts all occurrences of v, including repeated positions within a constraint. Consequently,

$$R_M(I) := \sum_{\mathbf{s} \in C_d^k} \rho_{\mathbf{s}}^M(I) = \begin{cases} d^{|V(I)|} \chi_M(I), & I \in \mathfrak{M}_{k,d}, \\ 0, & I \notin \mathfrak{M}_{k,d}. \end{cases} \tag{6}$$

Construct R_N similarly. The common nonzero factor in (6) gives $\chi_M = \chi_N$ on $\mathfrak{M}_{k,d}$ exactly when $R_M = R_N$ on $\mathfrak{M}_k$. The added factors are nonzero, so each augmented family still has at most one zero function per arity. Lemma F.5.6 uses common scalar tags to compare the $2d^k$ terms. Character independence makes $R_M = R_N$ equivalent to equal multiplicities in every character class. Grouping monomials by this restricted equality and testing their coefficient sums, using character independence on $\mathfrak{M}_{k,d}$, gives the identity test on $W_{\mathcal{F}}^d$ (Lemma F.10.1 and Theorem F.10.2).

The purification-invariance, singleton, and certificate arguments apply unchanged to $W_{\mathcal{F}}^d$. All instance constructions preserve degree divisibility: identifying retained variables adds degrees, fresh internal copies retain theirs, and summation preserves occurrences. Thus the same arity bound, polynomials Q_k, and restricted identity test decide $\mathsf{BO}^d(\mathcal{F})$. The support and detector constructions (3) and (5) stay in $W_{\mathcal{F}}^d$, so the structural proof gives $\mathsf{BO}^d(\mathcal{F}) \Rightarrow \mathsf{TP}^d(\mathcal{F}) \wedge \mathsf{Mal}^d(\mathcal{F})$. This proves Corollary 1.2, with the full argument in Section F.10.2.

AI methodology statement

We used OpenAI's GPT-5.6 Sol to assist with mathematical exploration and proof development throughout this paper. Its most substantial role was in developing the approach presented in Sections 3.2 and 4, with full details given in Sections F.4–F.5. This approach expresses Block Orthogonality through polynomial identities and decides their validity over the infinite generated family using multiplicative characters and finite isomorphism tests.

We used OpenAI's GPT-6 Astra to improve the writing and presentation and to assist with the Lean formalization. During formalization, we identified a gap in an early version of the paper: the argument relied on a simple-instance corollary of Young's isomorphism theorem whose conclusion does not follow from its stated hypotheses. We repaired the argument by proving the all-instance pinned version needed here and completed the Lean formalization of the proof. Appendix F.B explains the gap and gives an explicit counterexample to Young's corollary. The authors take full responsibility for the correctness and originality of the paper.

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Full paper

F.1 Introduction

Counting constraint satisfaction problems provide a common language for partition functions, weighted graph homomorphisms, and many other counting problems. In a landmark JACM paper recognized with the 2021 Gödel Prize, Cai and Chen established a complete complexity dichotomy for counting CSPs over arbitrary finite domains with algebraic complex weights: for every fixed finite set $\mathcal{F}$ of constraint functions, $\#\mathsf{CSP}(\mathcal{F})$ is either solvable in polynomial time or #P-hard [CC17]. This rules out intermediate complexity throughout this broad class of counting problems. Such a dichotomy is far from automatic: Ladner's theorem [Lad75] shows that, if P $\neq$ NP, there are problems in NP that are neither polynomial-time solvable nor NP-complete.

Cai and Chen's tractability criterion consists of three conditions on an infinite family of generated functions. They explicitly left open whether one can decide, from the finite set of constraint functions alone, whether all three conditions hold [CC17, Section 9]. We resolve this question by giving a total exact decision procedure, making their classification effective.

To state our result, write $[n] = \{1, \dots, n\}$ for $n \geq 0$, with $[0] = \varnothing$. Fix the domain $D = \{0, \dots, q-1\}$, where $q \geq 1$, and let $\mathcal{F}$ be a finite set of algebraic complex-valued constraint functions on D. For a finite $\#\mathsf{CSP}(\mathcal{F})$ instance I, choose an ordering of its $n = |V(I)| \geq 1$ variables. Let $F_I : D^n \to \mathbb{C}$ be the function obtained by multiplying its constraint functions applied to their specified variables. A variable may occur more than once within a constraint. For $1 \leq t \leq n$, define

$$F_I^{[t]}(\mathbf{x}) = \sum_{\mathbf{y} \in D^{n-t}} F_I(\mathbf{x}, \mathbf{y}), \qquad \mathbf{x} \in D^t.$$

Here $\mathbf{x}$ assigns values to the first t variables, and the sum ranges over all assignments to the remaining $n - t$ variables. In particular, $F_I^{[n]} = F_I$. Following Cai and Chen [CC17, Section 3], define

$$W_{\mathcal{F}} = \left\{ F_I^{[t]} : \begin{array}{l} I \text{ is a finite } \#\mathsf{CSP}(\mathcal{F}) \text{ instance} \\ \text{with an ordering of its variables,} \quad 1 \leq t \leq |V(I)| \end{array} \right\}. \tag{F.1}$$

Both the size of the representing instances and the arity of the resulting functions are unbounded.

Cai and Chen's tractability criterion imposes the Block Orthogonality, Type Partition, and Mal'tsev polymorphism conditions on the entire family $W_{\mathcal{F}}$ and the relations derived from it. We denote them by $\mathsf{BO}(\mathcal{F})$, $\mathsf{TP}(\mathcal{F})$, and $\mathsf{Mal}(\mathcal{F})$, respectively. Their precise definitions are given in Definitions F.2.3 and F.2.4. Their conjunction

$$\mathsf{CC}(\mathcal{F}) = \mathsf{BO}(\mathcal{F}) \wedge \mathsf{TP}(\mathcal{F}) \wedge \mathsf{Mal}(\mathcal{F})$$

is the tractability criterion in their dichotomy: when $\mathsf{CC}(\mathcal{F})$ holds, $\#\mathsf{CSP}(\mathcal{F})$ is polynomial-time computable. Otherwise, it is #P-hard [CC17, Theorem 1 and Sections 3.1–3.2]. In addition to deciding this criterion, we prove that the Block Orthogonality condition alone implies the other two conditions.

Theorem F.1.1 (Uniform decidability)**.** *There is a deterministic algorithm that, given $q \geq 1$ and a finite set $\mathcal{F}$ of algebraic complex-valued functions on D, halts and decides whether $\mathsf{CC}(\mathcal{F})$ holds. Moreover,*

$$\mathsf{CC}(\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{BO}(\mathcal{F}). \tag{F.2}$$

For $d \geq 1$, let $\#\mathsf{CSP}^d(\mathcal{F})$ denote the restriction of $\#\mathsf{CSP}(\mathcal{F})$ to instances in which the total number of occurrences of each variable in the ordered tuples of variables in the constraints is divisible by d. Define $W_{\mathcal{F}}^d$ by (F.1), restricting I to $\#\mathsf{CSP}^d(\mathcal{F})$ instances. Lin proved a dichotomy theorem for this class, with a tractability criterion given by the Block Orthogonality, Type Partition, and Mal'tsev polymorphism conditions on $W_{\mathcal{F}}^d$ and its derived relations [Lin21, Theorem 3]. Denote these conditions by $\mathsf{BO}^d(\mathcal{F})$, $\mathsf{TP}^d(\mathcal{F})$, and $\mathsf{Mal}^d(\mathcal{F})$, respectively, and write

$$\mathsf{CC}^d(\mathcal{F}) = \mathsf{BO}^d(\mathcal{F}) \wedge \mathsf{TP}^d(\mathcal{F}) \wedge \mathsf{Mal}^d(\mathcal{F}).$$

We obtain the following result by developing an identity test for $W_{\mathcal{F}}^d$ based on a root-of-unity filter (Section F.10.1) and proving that the support and row-equivalence constructions preserve the degree restriction (Section F.10.2).

Corollary F.1.2 (Uniform decidability for $\#\mathrm{CSP}^d$)**.** *For every $d \geq 1$, there is a deterministic algorithm that, given $q \geq 1$ and a finite set $\mathcal{F}$ of algebraic complex-valued functions on D, halts and decides whether $\mathsf{CC}^d(\mathcal{F})$ holds. Moreover,*

$$\mathsf{CC}^d(\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{BO}^d(\mathcal{F}).$$

F.1.1 Proof overview

The proof has two parts: deciding $\mathsf{BO}(\mathcal{F})$ and showing that it implies $\mathsf{TP}(\mathcal{F})$ and $\mathsf{Mal}(\mathcal{F})$. The first part must handle generated functions of unbounded arity, represented by instances of unbounded size. The second must overcome cancellation when summing over variables, so that we can construct functions whose supports express the required relations.

From arbitrary arity to finitely many polynomial conditions. We first show that the Block Orthogonality condition $\mathsf{BO}(\mathcal{F})$ can be checked by purifying each generated function separately. For a k-ary function F, write $F(\mathbf{x}, z)$, where $\mathbf{x} \in D^{k-1}$ and $z \in D$. We view F as a matrix whose rows are indexed by $\mathbf{x}$ and whose columns are indexed by z. A failure of Block Orthogonality is witnessed by two rows, indexed by tuples $\mathbf{x}$ and $\mathbf{y}$. Grouping their positions according to $(x_i, y_i) \in D^2$ and identifying the corresponding variables leaves at most q^2 row variables. Keeping the column variable separate preserves both rows, so every failure has a witness of arity at most $q^2 + 1$ (Section F.3).

At each fixed arity, all generated function values lie in the number field determined by the input functions. Using the finitely many roots of unity in this field, we express Block Orthogonality after purification as finitely many alternatives, each given by polynomial equations in the original function values. Products of these equations then give a finite set of polynomial identities. Block Orthogonality holds at this arity exactly when every one of these identities holds on every generated function of this arity (Section F.4).

Testing identities through multiplicative characters. For a fixed arity, label the variables that remain unsummed. The values of a generated function are then the partition values of its representing instance with these labelled variables fixed. Gluing two instances at equally labelled variables multiplies the corresponding values.

Each monomial M in the polynomials constructed above is a product of powers of function values and their complex conjugates, with the input tuples and exponents fixed by M. Evaluating this expression for each labelled instance I defines a function χ_M on labelled instances. The gluing property gives

$$\chi_M(IJ) = \chi_M(I)\chi_M(J),$$

where IJ denotes the glued instance. These functions are called multiplicative characters. Distinct characters are linearly independent, so a polynomial identity holds on every generated function exactly when its coefficients sum to zero within each group of monomials defining the same character.

It remains to decide which characters are equal. Tensor products represent the monomials as pinned partition functions over finite domains. After distinguishing the resulting functions by scalar factors, we apply a pinned extension of Young's all-instance isomorphism theorem [You25, Theorem 8], proved here as Theorem F.5.2.[1] The theorem replaces agreement on infinitely many labelled instances by the existence of a single domain bijection preserving the corresponding constraint functions and satisfying the pin conditions. Since the tensor constructions have finite domains and finitely many constraint functions, we can enumerate all candidate bijections and check each one using finitely many exact comparisons. We obtain an exact identity test and, together with the arity bound, a terminating decision procedure for $\mathsf{BO}(\mathcal{F})$ (Sections F.5 and F.6).

From Block Orthogonality to the other tractability conditions. The Mal'tsev polymorphism condition requires a single ternary operation $\varphi : D^3 \to D$ satisfying

$$\varphi(a, b, b) = \varphi(b, b, a) = a \qquad \text{for all } a, b \in D,$$

and preserving every support and row-equivalence relation of the generated functions. The support of a generated function is defined as the set of input tuples at which it takes a nonzero value. An operation preserves a relation if applying it to the corresponding entries of any three tuples in the relation produces another tuple in that relation.

Assume $\mathsf{BO}(\mathcal{F})$. We first construct a common Mal'tsev operation for the generated supports. For a finite collection of these supports, we introduce variables $x_{a,b,c}$ representing the outputs $\varphi(a, b, c)$ of a candidate ternary operation. For each relation $R \subseteq D^r$ in the collection and every three tuples $\mathbf{u}, \mathbf{v}, \mathbf{w} \in R$, we impose the constraint

$$\left(x_{u_i, v_i, w_i}\right)_{i \in [r]} \in R.$$

These constraints require one common operation to preserve all the chosen relations. To realize the system, choose a generated function F_R with support R and replace each constraint by an instance representing F_R, sharing the specified variables and keeping internal variables separate. The resulting function is a product of the corresponding F_R-values, so it is nonzero exactly when all constraints are satisfied. We also need projections of this solution relation. Summing over the omitted variables may cancel contributions from satisfying extensions, so we first use repeated copies of the instances to raise each contribution to a common positive power. A consequence of the Skolem–Mahler–Lech theorem [Lec53] guarantees an exponent for which the sum is nonzero whenever a satisfying extension exists, simultaneously for all retained assignments. Thus both the solution relation and its required projections are generated supports.

Since purification preserves support, Block Orthogonality forces any two overlapping nonempty rows of these projected relations to have the same support. We use this property to construct the desired operation, starting with the third projection $(a, b, c) \mapsto c$, which preserves every relation and already satisfies $\varphi(b, b, a) = a$. The row-support property allows us to impose the remaining requirements $\varphi(a, b, b) = a$ one input triple at a time, retaining the previously prescribed outputs

[1] The first arXiv version of this paper relied on Young's simple-instance strengthening [You25, Corollary 35]. During the Lean formalization, we found that its simple-instance premise does not imply the asserted isomorphism. Theorem F.5.2 establishes the all-instance pinned statement needed here: our character comparisons require equality on all finite labelled instances. Appendix F.B gives an explicit counterexample to the simple-instance statement and identifies the gap in its proof.

and their extendibility to an operation preserving all the chosen relations. This gives a common Mal'tsev operation for each finite collection of generated supports. Since there are only finitely many Mal'tsev operations on D, this yields one preserving every generated support. Otherwise, finitely many supports would rule out all candidates (Section F.7).

To complete the Mal'tsev polymorphism condition, we must show that the same operation also preserves every row-equivalence relation, which relates indices of nonzero, linearly dependent rows. We do this by realizing each such relation as a generated support. For a generated function F, we multiply suitable positive powers of $F(\mathbf{x}, z)$ and $F(\mathbf{y}, z)$, then sum over the shared variable z. Copies of an instance representing F, with separate internal variables, realize this construction within $W_{\mathcal{F}}$. The phase restrictions imposed by Block Orthogonality allow us to choose exponents for which the resulting function is zero outside the row-equivalence relation. A power-sum argument then selects a choice making it nonzero on every pair of nonzero, linearly dependent rows. Its support is therefore exactly the required relation, which is preserved by the common Mal'tsev operation from Section F.7 (Section F.8).

Preservation of row equivalence also yields the Type Partition condition. For each prefix of a fixed length, its type records the row-equivalence classes reached by completing that prefix. The Mal'tsev identities and preservation of row equivalence show that, whenever two types overlap, every class reached from one prefix can also be reached from the other. Thus the types are either equal or disjoint, establishing the Type Partition condition and completing the implication from the Block Orthogonality condition to the other two tractability conditions (Section F.9).

The degree restriction. For $\#\mathrm{CSP}^d$, we must compare characters on instances in which every variable's occurrence degree is divisible by d. We reduce this restricted comparison to the unrestricted identity test using a root-of-unity filter. Assign an auxiliary dth root of unity to each variable and include it once for each occurrence of that variable. Summing over these auxiliary choices gives zero whenever any variable violates the degree condition, and a nonzero factor otherwise. Applying the filter to two characters therefore makes both zero outside the permitted class and multiplies them by the same nonzero factor within it. The filtered expressions are finite sums of characters on unrestricted instances, so their equality can be decided by the character comparison and linear-independence arguments above (Section F.10.1).

We must also ensure that the arity compression and the constructions used to establish the Type Partition and Mal'tsev polymorphism conditions respect the degree restriction. Copying instances with separate internal variables preserves their occurrence degrees, identifying retained variables adds degrees divisible by d, and summing out a variable does not change its occurrence degree. Thus the compressed functions and the functions realizing supports and row equivalence remain in $W_{\mathcal{F}}^d$. Together with the filtered identity test, this gives uniform decidability of $\mathsf{BO}^d(\mathcal{F})$ and shows that it implies the other two tractability conditions. Hence Lin's criterion $\mathsf{CC}^d(\mathcal{F})$ is uniformly decidable, as stated in Corollary F.1.2 (Section F.10.2).

F.1.2 Related work

Counting graph homomorphisms (#GH). For a symmetric $q \times q$ matrix A indexed by D, on an input graph $G = (V, E)$, the problem $\#\mathrm{GH}(A)$ evaluates

$$Z_A(G) = \sum_{\sigma : V(G) \to D} \prod_{\{u,v\} \in E(G)} A_{\sigma(u),\sigma(v)}.$$

The problem $\#\mathrm{GH}(A)$ is the special case of #CSP with one symmetric binary constraint and is therefore covered by the complex-weighted #CSP dichotomy theorem of Cai and Chen [CC17].

Dyer and Greenhill classified $\{0,1\}$-valued matrices, Bulatov and Grohe extended the dichotomy to nonnegative matrices, Goldberg, Grohe, Jerrum, and Thurley handled arbitrary real matrices, and Cai, Chen, and Lu completed the classification for algebraic complex matrices [DG00, DG04, BG05, GGJT10, CCL13]. For algebraic-complex matrices, the dichotomy theorem of Cai, Chen, and Lu also provides a polynomial-time decision algorithm for its tractable case [CCL13, Theorem 1.2].

Planar input restrictions admit further tractable cases through holographic algorithms and matchgates. Relevant classifications include real symmetric matrices on three elements [CM23] and nonnegative full-rank matrices on four elements [CM24]. Cai, Maran, and Young connect more general planar classifications to vertex separation and quantum automorphism groups [CMY26].

Counting CSP. For general constraint languages, Dyer, Goldberg, and Jerrum classified nonnegative weighted Boolean #CSP [DGJ09]. Bulatov proved the finite-domain unweighted dichotomy, and Dyer and Richerby subsequently gave an elementary effective criterion [Bul13, DR13]. For algebraic weights, Cai, Chen, and Lu obtained an effective dichotomy in the nonnegative setting. Cai, Lu, and Xia classified complex-weighted Boolean languages, and Cai and Chen established the dichotomy for every finite set of functions over arbitrary finite domains with algebraic complex weights [CCL16, CLX14, CC17]. Lin proved a dichotomy theorem for the degree-divisibility restriction $\#\mathrm{CSP}^d$ whose tractable side is characterized by the corresponding three infinitary conditions on $W_{\mathcal{F}}^d$ [Lin21, Theorem 3]. We establish uniform decidability of the tractability criteria in both dichotomies.

Holant. Valiant's holographic algorithms introduced basis transformations for tensor-network partition functions [Val08], and Cai, Lu, and Xia formulated Holant as a general counting framework containing #CSP and graph homomorphisms [CLX09]. On the Boolean domain, a sequence of dichotomies progressed from symmetric and auxiliary-signature settings to arbitrary nonnegative signatures, complex-valued Holant^c, and finally arbitrary real-valued signatures [CGW16, LW18, Bac21, Bac25, SC20]. For complex-valued Boolean Holant problems, recent work establishes a dichotomy in the presence of a nontrivial odd-arity signature [MWXZ25, GSS26].

Higher-domain results include single-signature and edge-coloring classifications [CLX13, CGW14, LFC25], and a dichotomy with all unary signatures available for arbitrary sets of symmetric real-valued signatures on three elements [CI25].

Decision CSP. For the decision problem, Schaefer established the Boolean-domain dichotomy [Sch78]. Feder and Vardi placed finite-domain CSP in a broad logical framework and formulated the dichotomy conjecture that every fixed finite constraint language yields either a polynomial-time problem or an NP-complete one [FV98]. The universal-algebraic approach relates CSP complexity to polymorphisms [BJK05]. Mal'tsev polymorphisms also underlie effective counting-CSP algorithms [DR13]. Bulatov and Zhuk independently proved the full finite-domain dichotomy in 2017 [Bul17, Zhu17]. See also Zhuk's full journal account [Zhu20].

F.1.3 Organization

Section F.2 defines the problems and the three tractability conditions. Section F.3 reduces the Block Orthogonality condition to singleton purification at bounded arity and establishes the purification tests used in Section F.4 to construct polynomial conditions. Section F.5 gives a universal identity test, and Section F.6 combines these results into the Block Orthogonality decision procedure. Sections F.7 to F.9 construct a common Mal'tsev polymorphism, establish the Type Partition condition, and complete the proof of Theorem F.1.1. Section F.10 extends the argument to $\#\mathrm{CSP}^d$.

Section F.11 summarizes the results. Appendix F.A illustrates the tractability conditions, variable identification, multiplicative characters, and tensor realization. Appendix F.B explains the failure of Young's simple-instance strengthening.

F.2 Preliminaries and the Three Tractability Conditions

This section formalizes the three tractability conditions in the complex-weighted #CSP dichotomy theorem of Cai and Chen [CC17, Sections 2.8–2.9 and 3.1]. Examples are collected in Appendix F.A.

F.2.1 Algebraic and group notation

For an integer $n \geq 0$, write $[n] = \{1, \ldots, n\}$, with $[0] = \varnothing$. We use $[n]$ to index variables and finite lists. The CSP domain is $D = \{0, \ldots, q-1\}$, where $q \geq 1$, and $D^0 = \{\epsilon\}$, with ϵ the empty tuple. Any nonempty finite domain can be put in this form by a bijection. Transporting constraint functions along the bijection preserves partition values and all three tractability conditions. The integer $d \geq 1$ denotes Lin's occurrence-divisibility parameter. Throughout, $\mathbb{C}$ denotes the field of algebraic complex numbers. Input values have standard exact representations specifying the chosen complex root. Arithmetic, conjugation, and equality tests are effective [Coh93, Section 4.2].

The following group notation describes the exponent maps in Lemma F.3.1 and the root-of-unity tests in Lemma F.3.5. If L is a field, then $L^\times = L \setminus \{0\}$ is its multiplicative group. For a possibly infinite subset $S \subseteq L^\times$, the subgroup generated by S is

$$\langle S \rangle = \Big\{ \prod_{a \in T} a^{n_a} : T \subseteq S \text{ is finite, } n_a \in \mathbb{Z} \text{ for every } a \in T \Big\}.$$

We write $H \leq \Gamma$ when H is a subgroup of the abelian group Γ. A homomorphism from a multiplicative group to an additive group satisfies $e(xy) = e(x) + e(y)$ and has kernel $\ker e = \{x : e(x) = 0\}$. For $\mathbf{v} \in \mathbb{Z}^m$ and $N \geq 1$, $N\mathbf{v} = 0$ implies $\mathbf{v} = 0$.

Let $\mu_\infty \leq \mathbb{C}^\times$ be the group of all roots of unity, namely the elements ξ satisfying $\xi^N = 1$ for some $N \geq 1$. For a number field K, the group $\mu(K) = K^\times \cap \mu_\infty$ is finite and cyclic. For $\Gamma \leq \mathbb{C}^\times$, a homomorphism $\kappa : \Gamma \to \mathbb{C}^\times$ fixes every root of unity in Γ if

$$\kappa(\xi) = \xi \qquad \text{for every } \xi \in \Gamma \cap \mu_\infty. \tag{F.3}$$

Every homomorphism $e : \Gamma \to \mathbb{Z}^m$ maps each root of unity in Γ to 0. The *exponent* of a finite group is the least positive integer e such that $x^e = 1$ for every group element x, equivalently the least common multiple of the element orders. We write lcm for least common multiple.

If a homomorphism $e : \Gamma \to \mathbb{Z}^m$ has kernel $\Gamma \cap \mu_\infty$, it induces the injective homomorphism $\Gamma/(\Gamma \cap \mu_\infty) \hookrightarrow \mathbb{Z}^m$ given by $x(\Gamma \cap \mu_\infty) \mapsto e(x)$.

F.2.2 Constraint functions and instances

Let $\mathcal{F}$ be a finite set of functions $f : D^{r_f} \to \mathbb{C}$ with $r_f \geq 1$, each given together with its arity and exact values. Such a set is also called a *weighted constraint language* [CC17]. If the input is a list, we identify functions with the same arity and the same values. This leaves $W_{\mathcal{F}}$ unchanged. An instance I of $\#\mathsf{CSP}(\mathcal{F})$ consists of a variable set $V(I) = \{x_1, \ldots, x_n\}$, where $n(I) = |V(I)| = n \geq 1$, and a finite multiset $\mathcal{C}_I$ of constraints $(f; i_1, \ldots, i_{r_f})$, where $f \in \mathcal{F}$ and $i_1, \ldots, i_{r_f} \in [n]$. A scope is ordered and its indices need not be distinct. The instance function $F_I : D^n \to \mathbb{C}$ is

$$F_I(a_1, \ldots, a_n) = \prod_{(f; i_1, \ldots, i_{r_f}) \in \mathcal{C}_I} f(a_{i_1}, \ldots, a_{i_{r_f}}), \tag{F.4}$$

where multiplicities are retained and the empty product equals 1. For an assignment $\sigma : V(I) \to D$, we also write $F_I(\sigma) = F_I(\sigma(x_1), \ldots, \sigma(x_n))$. For $1 \le t \le n$, define

$$F_I^{[t]}(a_1, \ldots, a_t) = \sum_{(a_{t+1}, \ldots, a_n) \in D^{n-t}} F_I(a_1, \ldots, a_n). \tag{F.5}$$

Following Cai and Chen [CC17], define the set of functions

$$W_{\mathcal{F}} = \left\{ F_I^{[t]} : \begin{array}{l} I \text{ is a finite } \#\mathsf{CSP}(\mathcal{F}) \text{ instance,} \\ 1 \le t \le n(I) \end{array} \right\}.$$

The variable ordering in an instance is arbitrary, so (F.5) retains any prescribed ordered subset after relabeling. Repeated scope indices permit diagonal minors and arbitrary identifications. Instances may contain arbitrarily many variables, constraints, hidden variables, and retained variables.

For $d \ge 1$, the *occurrence degree* of a variable is the number of positions in all constraint scopes occupied by that variable, counting constraint multiplicity and repeated positions. Following Lin [Lin21], $\#\mathsf{CSP}^d(\mathcal{F})$ denotes the restriction of $\#\mathsf{CSP}(\mathcal{F})$ to instances in which every variable has occurrence degree divisible by d. The corresponding set of functions is

$$W_{\mathcal{F}}^{d} = \left\{ F_I^{[t]} : \begin{array}{l} I \text{ is a finite } \#\mathsf{CSP}^d(\mathcal{F}) \text{ instance,} \\ 1 \le t \le n(I) \end{array} \right\}.$$

For $d = 1$ this is $W_{\mathcal{F}}$. Isolated variables have degree zero and are permitted.

F.2.3 Rows and Block Orthogonality

For every function $F : D^r \to \mathbb{C}$, write $\mathrm{arity}(F) = r$. Following Cai and Chen [CC17, Sections 2.3 and 2.5], suppose $r \ge 2$ and regard F as the $q^{r-1} \times q$ matrix whose row indexed by $\mathbf{x} \in D^{r-1}$ is $F(\mathbf{x}, *) = \big(F(\mathbf{x}, z)\big)_{z \in D}$. The nonzero row indices carry the equivalence relation

$$\mathbf{x} \sim_F \mathbf{y} \quad \Longleftrightarrow \quad F(\mathbf{x}, *) \text{ and } F(\mathbf{y}, *) \text{ are linearly dependent over } \mathbb{C}. \tag{F.6}$$

Let $S_1, \ldots, S_s$ be its equivalence classes. For each $j \in [s]$, let $\mathbf{v}_j$ be the unique vector proportional to the rows in S_j whose first nonzero entry, with respect to a fixed ordering of D, is 1. The family

$$\mathrm{Row}(F) = \{(S_1, \mathbf{v}_1), \ldots, (S_s, \mathbf{v}_s)\}$$

is the *row representation* of F. If F is the zero function, then $s = 0$ and $\mathrm{Row}(F) = \varnothing$.

For a function or vector H, write $|H|$ for its entrywise absolute value. For a vector $\mathbf{u} \in \mathbb{C}^D$, write $\mathrm{supp}(\mathbf{u}) = \{z \in D : u_z \ne 0\}$. The function F is *block-rank one* if the nonnegative matrix $|F|$ is block-rank one: for every two nonzero rows, their magnitude vectors are positively proportional or their supports are disjoint. Equivalently, the nonzero support is a disjoint union of complete row-by-column rectangles, and within each rectangle the magnitude rows are proportional. See Appendix F.A.1 for an example.

Let $\mathbf{u}, \mathbf{v} \in \mathbb{C}^D$ be nonzero with $|\mathbf{u}|$ and $|\mathbf{v}|$ proportional, and set $T = \mathrm{supp}(\mathbf{u}) = \mathrm{supp}(\mathbf{v})$. Let $m_1 > \cdots > m_h > 0$ be the distinct values of $|u_z|$ on T, and put $T_i = \{z \in T : |u_z| = m_i\}, \quad i \in [h]$. The same partition is obtained from $\mathbf{v}$. The two vectors are *block-orthogonal*, written $\mathbf{u} \perp_{\mathrm{B}} \mathbf{v}$, if

$$\sum_{z \in T_i} u_z \overline{v_z} = 0 \qquad \text{for every } i \in [h]. \tag{F.7}$$

Definition F.2.1 (Block-orthogonal function)**.** A function $F : D^r \to \mathbb{C}$, $r \ge 2$, is *block-orthogonal* if it is block-rank one and, whenever two nonzero complex rows have proportional magnitude vectors, those rows are either linearly dependent or block-orthogonal in the sense of (F.7).

See Appendix F.A.2 for an example.

F.2.4 Joint purification

A function is *pure* if each of its values is a nonnegative integer times a root of unity. We use the notions of pure functions and generating sets, together with the Purification Lemma, from Cai and Chen [CC17, Definitions 2–3 and Lemma 7].

Definition F.2.2 (Generating set)**.** Given a finite tuple $\mathbf{F} = (F_1, \ldots, F_h)$ with $h \geq 1$, let $A(\mathbf{F})$ be the set of its nonzero values. Write $\mathbb{Q}(A(\mathbf{F}))$ for the smallest subfield of $\mathbb{C}$ containing $\mathbb{Q}$ and $A(\mathbf{F})$. A *generating set* for $A(\mathbf{F})$ is a tuple $g_1, \ldots, g_m$, where $m \geq 0$, satisfying the following three conditions:

1. $g_i \in \mathbb{Q}(A(\mathbf{F}))^\times$ for every $i \in [m]$.
2. For every nonzero $\mathbf{k} = (k_1, \ldots, k_m) \in \mathbb{Z}^m$,
$$\prod_{i \in [m]} g_i^{k_i} \notin \mu_\infty.$$
3. Every $a \in A(\mathbf{F})$ has a unique exponent vector $e(a) = (e_1(a), \ldots, e_m(a)) \in \mathbb{Z}^m$ such that
$$a = \vartheta(a) \prod_{i \in [m]} g_i^{e_i(a)}, \qquad \vartheta(a) \in \mu_\infty. \tag{F.8}$$

When $A(\mathbf{F})$ is nonempty, such a generating set exists [CCL13, Lemma 7.3]. See also [CC17, Lemma 8]. Let $p_1, \ldots, p_m$ be the first m positive primes. For every nonzero value a, put

$$\varrho(a) = \prod_{i \in [m]} p_i^{e_i(a)} \in \mathbb{Q}_{>0}.$$

The transformation displayed in the proof of the Purification Lemma [CC17, Lemma 7] replaces each nonzero a by $\vartheta(a)\varrho(a)$. Zeros remain zero. Write $\widetilde{F}_i$ for the resulting function corresponding to F_i. Negative exponents in $\varrho(a)$ may introduce denominators. To clear them, let N_i be the least positive integer such that $N_i \varrho(a) \in \mathbb{Z}_{>0}$ for every nonzero value a of F_i, taking $N_i = 1$ when F_i is zero, and put $F_i' = N_i \widetilde{F}_i$. Every value of F_i' is a nonnegative integer times a root of unity, so F_i' is pure. We call $\widetilde{F}_i$ and F_i' the *unnormalized* and *normalized purifications*, respectively. Multiplication by a positive scalar changes neither support nor any clause of Block Orthogonality, so the unnormalized and normalized forms define the same global condition. Appendix F.A.2 illustrates this purification.

Fix a legal generating-set choice whenever $A(\mathbf{F})$ is nonempty and define $\mathrm{Pure}(\mathbf{F}) = (F_1', \ldots, F_h')$, using the pure outputs, normalized separately for each function. If $A(\mathbf{F}) = \varnothing$, define $\mathrm{Pure}(\mathbf{F})$ to be the all-zero tuple. We call any unnormalized or normalized output obtained from a legal generating-set choice a *legal purification*. Lemma F.3.1 proves that the relevant structure is independent of these choices.

Definition F.2.3 (Global Block Orthogonality condition)**.** The set $\mathcal{F}$ satisfies the *Block Orthogonality condition* of Cai and Chen [CC17, Section 3.1], denoted $\mathsf{BO}(\mathcal{F})$, if for every nonempty finite tuple $(F_1, \ldots, F_h)$ of pairwise distinct members of $W_{\mathcal{F}}$, writing $\mathrm{Pure}(F_1, \ldots, F_h) = (F_1', \ldots, F_h')$, the function F_i' is block-orthogonal for every i with $\mathrm{arity}(F_i) \geq 2$.

All functions in a finite tuple are purified using a common generating set, with a separate normalization factor for each function. Corollary F.3.2 proves that checking one function at a time gives the same condition.

F.2.5 Types and derived relations

Following Cai and Chen [CC17, Sections 2.8–2.9 and 3.1], let $F : D^r \to \mathbb{C}$, $r \geq 2$, have row representation $\{(S_j, \mathbf{v}_j) : j \in [s]\}$. For $0 \leq \ell \leq r-1$ and $\boldsymbol{\alpha} \in D^\ell$, define

$$\mathrm{type}_F(\boldsymbol{\alpha}) = \{j \in [s] : \text{some } \mathbf{x} \in S_j \text{ has prefix } \boldsymbol{\alpha}\}.$$

The map is a *type-partition map* if

$$\forall\, 0 \leq \ell \leq r-1\ \forall \boldsymbol{\alpha}, \boldsymbol{\beta} \in D^\ell,$$
$$\mathrm{type}_F(\boldsymbol{\alpha}) = \mathrm{type}_F(\boldsymbol{\beta}) \quad \text{or} \quad \mathrm{type}_F(\boldsymbol{\alpha}) \cap \mathrm{type}_F(\boldsymbol{\beta}) = \varnothing.$$

Set $\mathrm{type}_F(\epsilon) = [s]$. If F is the zero function, then $s = 0$, so every type is empty. See Appendix F.A.3 for an example.

For every $F : D^r \to \mathbb{C}$, define its support relation $\Phi_F = \{\mathbf{a} \in D^r : F(\mathbf{a}) \neq 0\}$. For $r \geq 2$, define the $2(r-1)$-ary row-equivalence relation

$$\Omega_F = \bigcup_{j \in [s]} (S_j \times S_j) = \{(\mathbf{x}, \mathbf{y}) \in D^{r-1} \times D^{r-1} : \mathbf{x} \sim_F \mathbf{y}\}.$$

If F is the zero function, then $\Phi_F = \varnothing$ and, when $r \geq 2$, $\Omega_F = \varnothing$.

An operation $\varphi : D^3 \to D$ *preserves* a relation $R \subseteq D^r$ if coordinatewise application of φ to any three tuples of R returns a tuple of R. In this case φ is a *polymorphism* of R. It is a *Mal'tsev operation* if

$$\varphi(a, b, b) = \varphi(b, b, a) = a \qquad \text{for all } a, b \in D.$$

Set

$$\Lambda_{\mathcal{F}} = \{\Phi_F : F \in W_{\mathcal{F}}\} \cup \{\Omega_F : F \in W_{\mathcal{F}},\ \mathrm{arity}(F) \geq 2\}.$$

$\Lambda_{\mathcal{F}}$ is a family of relations of varying arities, and preservation is required separately for every Φ_F and every applicable Ω_F.

Definition F.2.4 (Type Partition and Mal'tsev polymorphism conditions)**.** The set $\mathcal{F}$ satisfies the *Type Partition condition*, denoted $\mathsf{TP}(\mathcal{F})$, if every function in $W_{\mathcal{F}}$ of arity at least two has a type-partition map. It satisfies the *Mal'tsev polymorphism condition*, denoted $\mathsf{Mal}(\mathcal{F})$, if

$$\exists \varphi : D^3 \to D : \quad \varphi \text{ is a Mal'tsev operation} \quad \text{and} \quad \forall R \in \Lambda_{\mathcal{F}},\ \varphi \text{ preserves } R.$$

Equivalently, $\Lambda_{\mathcal{F}}$ has a common Mal'tsev polymorphism.

Appendix F.A.4 illustrates how one Mal'tsev operation preserves both the support and row-equivalence relations of a function.

For $d \geq 1$, the conditions $\mathsf{BO}^d(\mathcal{F})$, $\mathsf{TP}^d(\mathcal{F})$, and $\mathsf{Mal}^d(\mathcal{F})$ are given by Definitions F.2.3 and F.2.4 with $W_{\mathcal{F}}$ replaced throughout by $W_{\mathcal{F}}^d$.

We now consider $\mathcal{F} = \varnothing$. In this case $F_I \equiv 1$, and every function $F = F_I^{[t]}$ in $W_{\mathcal{F}}$ is the positive constant function $F(a_1, \ldots, a_t) = q^{n(I)-t}$. Consequently $\Phi_F = D^t$. When $t \geq 2$, all rows are nonzero and proportional, there is exactly one complex row class, and $\Omega_F = D^{2(t-1)}$. Every legal purification is again a nonzero constant function, so Block Orthogonality holds. Since F itself is constant, Type Partition holds as well. The operation

$$\varphi_0(a, b, c) = \begin{cases} c, & a = b, \\ a, & a \neq b \end{cases}$$

is Mal'tsev and preserves every full relation. Thus $\mathsf{CC}(\mathcal{F})$ holds. The same argument applies to every $\mathsf{CC}^d(\mathcal{F})$ because constraint-free instances have degree zero. The decision procedures accept this case immediately, and henceforth we may assume $\mathcal{F} \neq \varnothing$.

F.3 Purification Tests and Arity Compression

To bound the arity of a counterexample to the Block Orthogonality condition, we will identify variables while preserving two rows that witness its failure. This requires the relevant tests to be independent of the other functions purified jointly with the given function. We first establish this invariance and reduce to singleton purification, then compress the arity and express the purified tests in the original values.

F.3.1 Singleton purification

Joint purification uses one generating set for the nonzero values of the whole tuple. We show that the tests defining Block Orthogonality for a given function are independent of both the generator choice and the other functions in the tuple. This allows us to test each generated function separately and later to transfer a failed test through variable identification. We also establish the properties of the purification maps used in Section F.3.3.

Lemma F.3.1. *Let $F : D^r \to \mathbb{C}$, where $r \geq 2$, occur in each of two finite tuples of functions. Fix an occurrence of F in each tuple, and let F' and F'' be the corresponding outputs of two legal joint purifications. Then:*

1. $\Phi_{F'} = \Phi_{F''}$.
2. *For all $\mathbf{x}, \mathbf{y} \in D^{r-1}$, $F'(\mathbf{x}, *)$ and $F'(\mathbf{y}, *)$ have disjoint supports if and only if $F''(\mathbf{x}, *)$ and $F''(\mathbf{y}, *)$ do. The same equivalence holds for positive proportionality of their magnitude vectors.*
3. $\Omega_{F'} = \Omega_{F''}$.
4. *If the rows indexed by $\mathbf{x}$ and $\mathbf{y}$ have proportional magnitude vectors in F', then the partition of their common support into equal-magnitude blocks is the same for F' and F''. On each block, the inner product of these two rows restricted to that block is zero for F' if and only if it is zero for F''.*

In particular, F' is block-orthogonal if and only if F'' is block-orthogonal.

For either chosen purification, let $e(a)$ and $\vartheta(a)$ be the exponent vector and root-of-unity factor of a nonzero value a of F in (F.8)*. With m generators, the purification replaces a by*

$$N\kappa(a), \qquad \kappa(a) = \vartheta(a) \prod_{i \in [m]} p_i^{e_i(a)},$$

where $N > 0$ is common to all values of F, with $N = 1$ for an unnormalized output. Let Γ be the multiplicative group generated by the nonzero values of F. These assignments extend to well-defined homomorphisms $e : \Gamma \to \mathbb{Z}^m$, $\vartheta : \Gamma \to \mu_\infty$, and $\kappa : \Gamma \to \mathbb{C}^\times$. The extensions satisfy the same formula for κ and have the following properties:

- $\ker e = \Gamma \cap \mu_\infty$.
- *ϑ fixes every root of unity in Γ.*
- *κ is injective, and $|\kappa(x)| = 1$ if and only if $x \in \mu_\infty$.*

For a singleton purification, the root-of-unity map also satisfies $\vartheta(\Gamma) \subseteq \mu(K)$ whenever K is a number field containing the values of F.

Proof. If F is the zero function, both purified functions are zero and all assertions hold. Assume that F has a nonzero value, and write $\Gamma = \langle a : a \text{ is a nonzero value of } F \rangle$.

We first prove that the assignments in the purification formula extend to homomorphisms with the stated properties. Fix one of the two joint purifications, and let $c_1, \ldots, c_t$ be the nonzero values in its tuple, with legal generators $g_1, \ldots, g_m$ and exponent vectors $e(c_j) = (e_1(c_j), \ldots, e_m(c_j))$. Put $H = \langle c_1, \ldots, c_t \rangle$. Every nonzero value of F occurs among the c_j, so $\Gamma \le H$. To extend the prescribed exponent vectors to H, set

$$e\left(\prod_{j\in[t]} c_j^{n_j}\right) = \sum_{j\in[t]} n_j e(c_j) \qquad (n_1, \ldots, n_t \in \mathbb{Z}). \tag{F.9}$$

To check that this is well-defined, suppose $\prod_{j\in[t]} c_j^{n_j} = 1$. Then (F.8) gives

$$\prod_{i\in[m]} g_i^{\sum_{j\in[t]} n_j e_i(c_j)} = \prod_{j\in[t]} \vartheta(c_j)^{-n_j} \in \mu_\infty.$$

Condition 2 of Definition F.2.2 forces $\sum_{j\in[t]} n_j e_i(c_j) = 0$ for every $i \in [m]$. If two products represent the same element of H, their quotient is 1, so this calculation shows that their exponent sums agree. Thus (F.9) is well-defined. Multiplication of products adds exponent vectors, so e is the unique homomorphism on H taking each c_j to its prescribed vector. Restricting it to Γ gives $e : \Gamma \to \mathbb{Z}^m$.

We next determine its kernel. For $x = \prod_j c_j^{n_j} \in \Gamma$, if $e(x) = 0$, the decompositions of the c_j give $x = \prod_{j\in[t]} \vartheta(c_j)^{n_j} \in \mu_\infty$. Conversely, if $x^N = 1$ for some $N \ge 1$, then $Ne(x) = e(x^N) = 0$, which implies $e(x) = 0$ in $\mathbb{Z}^m$. Hence $\ker e = \Gamma \cap \mu_\infty$, and e induces an injective homomorphism $\nu : \Gamma/(\Gamma \cap \mu_\infty) \hookrightarrow \mathbb{Z}^m$, given by $\nu(x(\Gamma \cap \mu_\infty)) = e(x)$. For $x \in \Gamma$, define

$$\vartheta(x) = x \prod_{i\in[m]} g_i^{-e_i(x)}.$$

The decomposition of x shows that $\vartheta(x) \in \mu_\infty$, and additivity of e makes ϑ a homomorphism. For $\xi \in \Gamma \cap \mu_\infty$, we have $e(\xi) = 0$ and hence $\vartheta(\xi) = \xi$, as in (F.3). It follows that

$$\kappa(x) = \vartheta(x) \prod_{i\in[m]} p_i^{e_i(x)}$$

is a homomorphism $\Gamma \to \mathbb{C}^\times$ that fixes every root of unity in Γ. Unique factorization in $\mathbb{Q}_{>0}$ gives the multiplicative independence of the distinct primes: for integers $k_1, \ldots, k_m$, $\prod_{i\in[m]} p_i^{k_i} = 1$ implies $k_1 = \cdots = k_m = 0$. Consequently, κ is injective. Indeed, if $\kappa(x) = 1$, taking absolute values gives $e(x) = 0$. Then $x \in \mu_\infty$ and $\kappa(x) = \vartheta(x) = x$, so $x = 1$.

The magnitude of $\kappa(a)$ is $\prod_i p_i^{e_i(a)}$, which may have denominators when exponents are negative. Since F has finitely many values, there is a positive integer N_F such that $N_F \prod_{i\in[m]} p_i^{e_i(a)} \in \mathbb{Z}_{>0}$ for every nonzero value a of F. Thus replacing a by $N_F\kappa(a)$ gives a pure function. Each product of two entries and each block sum below is multiplied by N_F^2. This preserves equality of the products and whether a block sum equals zero.

The magnitude comparisons below follow from

$$|\kappa(x)| = 1 \quad \Longleftrightarrow \quad e(x) = 0 \quad \Longleftrightarrow \quad x \in \mu_\infty. \tag{F.10}$$

Indeed, $\vartheta(x)$ has modulus one, and unique factorization in $\mathbb{Q}_{>0}$ gives $\prod_i p_i^{e_i(x)} = 1$ exactly when every exponent equals zero.

For a singleton purification of a K-valued function, condition 1 of Definition F.2.2 places every g_i in the field generated by its values, and hence in K. Since $\Gamma \subseteq K^\times$, the formula for ϑ gives $\vartheta(\Gamma) \subseteq \mu(K)$.

We now compare the two purifications. Applying this construction to each gives homomorphisms κ_1 and κ_2 and positive factors N_1 and N_2. For a value a of F, write $\widehat{a}_1$ and $\widehat{a}_2$ for its images in F' and F'', respectively. Both are zero when $a = 0$. For $a \neq 0$, they satisfy $\widehat{a}_1 = N_1\kappa_1(a)$ and $\widehat{a}_2 = N_2\kappa_2(a)$. Since both homomorphisms fix roots of unity, for $\xi \in \Gamma \cap \mu_\infty$ and $x \in \Gamma$ we have

$$\kappa_1(\xi x) = \xi\kappa_1(x), \qquad \kappa_2(\xi x) = \xi\kappa_2(x). \tag{F.11}$$

Consequently, if both a and ξa occur as values of F, then $\widehat{\xi a}_1 = \xi\widehat{a}_1$ and $\widehat{\xi a}_2 = \xi\widehat{a}_2$. Thus root-of-unity ratios are preserved. A root-of-unity value ξ itself is replaced by $N_1\xi$ or $N_2\xi$.

We now prove (1) and (2), and establish the common blocks required in (4). For nonzero values a, b, c, u of F, applying (F.10) to a/b gives

$$|\widehat{a}_1| = |\widehat{b}_1| \quad\Longleftrightarrow\quad a/b \in \mu_\infty \quad\Longleftrightarrow\quad |\widehat{a}_2| = |\widehat{b}_2|. \tag{F.12}$$

Applying it to $au/(bc)$ gives

$$|\widehat{a}_1\widehat{u}_1| = |\widehat{b}_1\widehat{c}_1| \quad\Longleftrightarrow\quad au/(bc) \in \mu_\infty \quad\Longleftrightarrow\quad |\widehat{a}_2\widehat{u}_2| = |\widehat{b}_2\widehat{c}_2|.$$

Zeros remain zero and nonzero values remain nonzero, proving (1) and preservation of disjoint row supports. A product with a zero factor remains zero, while a product of nonzero entries remains nonzero. Together with the second equivalence, this preserves whether each 2×2 minor of the magnitude rows equals zero. These equalities and the unchanged supports preserve positive proportionality of magnitude rows, completing (2). The first equivalence gives the common equal-magnitude blocks in (4).

To prove (3), we use the fact that complex linear dependence of two rows is determined by their 2×2 minors. Write the original rows as $\mathbf{a} = (a_z)_{z\in D}$ and $\mathbf{b} = (b_z)_{z\in D}$. For columns $z, w \in D$, the minor is $a_z b_w - a_w b_z$. When these four entries are nonzero, injectivity and multiplicativity give

$$\widehat{a_z}_1\widehat{b_w}_1 = \widehat{a_w}_1\widehat{b_z}_1 \quad\Longleftrightarrow\quad a_z b_w = a_w b_z \quad\Longleftrightarrow\quad \widehat{a_z}_2\widehat{b_w}_2 = \widehat{a_w}_2\widehat{b_z}_2.$$

If an entry is zero, preservation of zero and nonzero products gives the same equivalence. Thus every minor equals zero under F' exactly when it equals zero under F''. Together with the common support from (1), this gives $\Omega_{F'} = \Omega_{F''}$, proving (3).

We complete (4) by comparing the inner products on the common blocks established above. Suppose the two rows have proportional magnitudes after purification. Let T be one of their common equal-magnitude blocks and fix $z_0 \in T$. By (F.12), there are roots of unity $\xi_z, \eta_z \in \Gamma \cap \mu_\infty$ such that $a_z = \xi_z a_{z_0}$ and $b_z = \eta_z b_{z_0}$ for $z \in T$. Equation (F.11) and the common scaling factor within each purified function preserve these ratios. Consequently,

$$\sum_{z\in T} \widehat{a_z}_1\overline{\widehat{b_z}_1} = \widehat{a_{z_0}}_1\overline{\widehat{b_{z_0}}_1}\sum_{z\in T}\xi_z\overline{\eta_z},$$
$$\sum_{z\in T} \widehat{a_z}_2\overline{\widehat{b_z}_2} = \widehat{a_{z_0}}_2\overline{\widehat{b_{z_0}}_2}\sum_{z\in T}\xi_z\overline{\eta_z}.$$

The two prefactors are nonzero, so the two block sums equal zero simultaneously, completing (4). □

The lemma makes each function's Block Orthogonality test independent of the other functions purified with it. The global condition therefore reduces to the following singleton test.

Corollary F.3.2 (Singleton purification suffices)**.** *The joint condition* $\mathsf{BO}(\mathcal{F})$ *holds if and only if every* $F \in W_{\mathcal{F}}$ *of arity at least two has a block-orthogonal singleton purification.*

Proof. If the joint condition holds, apply it to the one-element tuple (F) for each $F \in W_{\mathcal{F}}$ of arity at least two.

Conversely, assume every $F \in W_{\mathcal{F}}$ of arity at least two has a block-orthogonal singleton purification. Fix a nonempty finite tuple $\mathbf{F} = (F_1, \ldots, F_h)$ from $W_{\mathcal{F}}$ and a legal joint purification $\mathrm{Pure}(\mathbf{F}) = (F'_1, \ldots, F'_h)$. For each i with $\mathrm{arity}(F_i) \geq 2$, choose a block-orthogonal singleton purification F''_i of F_i. Apply Lemma F.3.1 to F_i in the tuples (F_i) and $(F_1, \ldots, F_h)$. It gives that F'_i is block-orthogonal if and only if F''_i is block-orthogonal, so every jointly purified component of arity at least two is block-orthogonal. Since $\mathbf{F}$ was arbitrary, the joint condition $\mathsf{BO}(\mathcal{F})$ holds. □

F.3.2 Arity compression

We now show that a failure of $\mathsf{BO}(\mathcal{F})$ has a witness of arity at most $q^2 + 1$. By Corollary F.3.2, we can start with one generated function whose singleton purification fails Block Orthogonality. The construction identifies variables while retaining the two rows that witness this failure. We first show that identifying retained variables keeps the resulting function in $W_{\mathcal{F}}$.

Lemma F.3.3. *Let* $F \in W_{\mathcal{F}}$ *have arity* r*, let* $1 \leq s \leq r$*, and let* $\pi : [r] \twoheadrightarrow [s]$ *be a surjection. Then*

$$H(y_1, \ldots, y_s) = F(y_{\pi(1)}, \ldots, y_{\pi(r)})$$

belongs to $W_{\mathcal{F}}$*.*

Proof. Choose a $\#\mathsf{CSP}(\mathcal{F})$ instance I with $F = F_I^{[r]}$ and retained variables $x_1, \ldots, x_r$. Introduce new retained variables $y_1, \ldots, y_s$ and replace every occurrence of x_i in a constraint by $y_{\pi(i)}$. Keep all variables being summed out unchanged and distinct from the y_j. Repeated variables are allowed in constraint scopes, so this gives a $\#\mathsf{CSP}(\mathcal{F})$ instance I_π. Summing over the same hidden variables gives $F_{I_\pi}^{[s]} = H$, hence $H \in W_{\mathcal{F}}$. □

Appendix F.A.5 illustrates this construction. To preserve two selected rows, we group their variable positions by the ordered pairs of domain values assigned there. There are at most q^2 such pairs. Each group needs one retained variable, and the column variable remains separate, giving the following bound.

Lemma F.3.4. *If the singleton Block Orthogonality condition fails on* $W_{\mathcal{F}}$*, then it fails for some function in* $W_{\mathcal{F}}$ *of arity at most* $q^2 + 1$*.*

Proof. Let $F \in W_{\mathcal{F}}$ have arity $r \geq 2$, and choose a legal singleton purification F' that fails Block Orthogonality. There are two row indices $\mathbf{x}, \mathbf{y} \in D^{r-1}$ witnessing this failure in F'. Either their magnitude rows have overlapping supports but are not positively proportional, or their magnitude rows are proportional while the complex rows are neither linearly dependent nor block-orthogonal. Group the positions $i, j \in [r-1]$ by

$$i \equiv j \quad \Longleftrightarrow \quad (x_i, y_i) = (x_j, y_j).$$

There is at least one class because $r \geq 2$, and at most q^2 classes because each corresponds to a distinct pair in D^2. Identify the variables within each class and keep the column variable separate. By Lemma F.3.3, the resulting function H belongs to $W_{\mathcal{F}}$ and satisfies $2 \leq \mathrm{arity}(H) \leq q^2 + 1$.

For each new row variable, assign the first component of its class's ordered pair to obtain one row of H, and the second component to obtain the other. These assignments recover $\mathbf{x}$ and $\mathbf{y}$ when expanded to the original positions. Since the column variable is unchanged, the two rows of H equal $F(\mathbf{x}, *)$ and $F(\mathbf{y}, *)$.

To transfer the failed test, take a legal joint unnormalized purification $(\widetilde{F}, \widetilde{H})$ of (F, H). By Lemma F.3.1, the test that fails in F' also fails in the corresponding rows of $\widetilde{F}$. Joint unnormalized purification applies the same map to the values of both functions, so the two row equalities persist in $(\widetilde{F}, \widetilde{H})$. Thus the same test fails in $\widetilde{H}$. Applying Lemma F.3.1 to H transfers this failure to every legal singleton purification of H. Hence H is a counterexample with the required arity bound. □

Appendix F.A.6 illustrates the grouping of positions and the two recovered rows.

F.3.3 Intrinsic purification tests

Lemma F.3.4 reduces the decision problem to finitely many arities. At each arity, we need equations in the original function values that test Block Orthogonality after purification. We first express comparisons of purified magnitudes and products in this form. A common number field restricts the root-of-unity factors in these equations to a finite set.

Fix the computable conjugation-stable number field

$$K = \mathbb{Q}\left(\{f(\mathbf{a}), \overline{f(\mathbf{a})} : f \in \mathcal{F},\ \mathbf{a} \in D^{r_f}\}\right), \tag{F.13}$$

and let $\mu(K)$ be its finite, effectively computable group of roots of unity [Coh93, Algorithm 4.9.10]. Every input value $f(\mathbf{a})$ belongs to K. By Equations (F.4) and (F.5), each value of every $F \in W_{\mathcal{F}}$ is a finite sum of finite products of input values, so it also belongs to K.

The next lemma gives the tests used in the certificates of Section F.4: magnitude comparisons become equations with a root-of-unity factor, ordinary minors retain their original equations, and root-of-unity ratios are preserved.

Lemma F.3.5. *Let F be a K-valued function on a finite domain, let F' be any legal singleton purification of F, and, for every nonzero value a occurring in F, write $\widehat{a}$ for its purified value. For nonzero values a, b, c, u occurring in F,*

$$|\widehat{a}| = |\widehat{b}| \iff a/b \in \mu(K), \tag{F.14}$$

$$|\widehat{a}||\widehat{u}| = |\widehat{b}||\widehat{c}| \iff au/(bc) \in \mu(K) \iff au = \zeta bc \text{ for some } \zeta \in \mu(K), \tag{F.15}$$

$$\widehat{a}\,\widehat{u} = \widehat{b}\,\widehat{c} \iff au = bc. \tag{F.16}$$

Moreover, if $a = \rho b$ for $\rho \in \mu(K)$, then

$$\widehat{a} = \rho\,\widehat{b}. \tag{F.17}$$

Proof. If F is the zero function, the assertions are vacuous. Assume that F has a nonzero entry. Let $\Gamma \leq K^\times$ be generated by the nonzero entries of F. The construction in Lemma F.3.1 gives an injective homomorphism $\kappa : \Gamma \to K^\times$ fixing every root of unity in Γ, with $|\kappa(x)| = 1$ exactly when $x \in \mu_\infty$, and a positive factor N such that $\widehat{a} = N\kappa(a)$ for every nonzero entry a.

We first prove Equations (F.14) and (F.15). Multiplicative independence of the replacement primes gives

$$|\kappa(a/b)| = 1 \iff a/b \in \Gamma \cap \mu(K).$$

The common factor N cancels when comparing $|\widehat{a}|$ and $|\widehat{b}|$, giving (F.14). Applying the same argument to $au/(bc)$, with the common factor N^2, gives (F.15).

For (F.16), cancel the common factor N^2 and use injectivity of κ. Finally, if $a = \rho b$, then $\rho = a/b \in \Gamma$. Since κ fixes every root of unity in Γ as in (F.3), multiplying by the common factor N gives (F.17). □

Together with support preservation, (F.16) shows that all ordinary 2×2 minors of two original rows equal zero if and only if the corresponding purified minors do. Thus row dependence can be tested directly in the original values. We now combine these tests with the magnitude and root-of-unity tests to describe Block Orthogonality by polynomial equations.

F.4 Algebraic Certificates for Block Orthogonality

At each fixed arity, we construct finitely many algebraic certificates from the tests in Lemma F.3.5. A function has a block-orthogonal singleton purification exactly when it satisfies the equations of at least one certificate. We then combine these alternatives into one finite set of polynomial identities, which will be tested on all generated functions in Section F.5. Throughout, K is the field in (F.13), and $\mu(K)$ is its group of roots of unity.

F.4.1 Finite certificates

Fix an arity $k \geq 2$ and put $\mathcal{R}_k = D^{k-1}$. We use A and A' for the matrix forms of a K-valued k-ary function F and its singleton purification F', respectively: their entries are $a_{\mathbf{x},c} = F(\mathbf{x}, c)$ and $a'_{\mathbf{x},c} = F'(\mathbf{x}, c)$ for $\mathbf{x} \in \mathcal{R}_k$ and $c \in D$. This identifies the original function with a point of K^{D^k}. Let $\mathbf{X} = \{X_{\mathbf{x},c} : \mathbf{x} \in \mathcal{R}_k,\ c \in D\}$ be the formal variables corresponding to its entries. Fix arbitrary total orders on $\mathcal{R}_k$ and D.

Definition F.4.1 (Block Orthogonality certificate)**.** A *Block Orthogonality certificate* of arity k consists of the following choices and their associated polynomial equations.

Support rectangles. Choose a rectangle system $(R_1, C_1), \ldots, (R_t, C_t)$, where every $R_i \subseteq \mathcal{R}_k$ and $C_i \subseteq D$ is nonempty, the row sets R_i are pairwise disjoint, and the column sets C_i are pairwise disjoint. The empty system $t = 0$ is allowed. Impose the support equations

$$X_{\mathbf{x},c} = 0 \qquad \text{whenever } (\mathbf{x}, c) \notin \bigcup_{i \in [t]} (R_i \times C_i). \tag{F.18}$$

Magnitude comparisons. For every i, every two distinct rows $\mathbf{x}, \mathbf{y} \in R_i$, and every two distinct columns $c, e \in C_i$, choose $\tau_{\mathbf{x},\mathbf{y};c,e} \in \mu(K)$ and impose the twisted minor

$$X_{\mathbf{x},c} X_{\mathbf{y},e} - \tau_{\mathbf{x},\mathbf{y};c,e} X_{\mathbf{x},e} X_{\mathbf{y},c} = 0. \tag{F.19}$$

Row comparisons. For every unordered pair $\mathbf{x} < \mathbf{y}$ in one R_i, choose one of the following two modes.

Dependent mode. Impose every ordinary minor

$$X_{\mathbf{x},c} X_{\mathbf{y},e} - X_{\mathbf{x},e} X_{\mathbf{y},c} = 0 \qquad (c, e \in C_i). \tag{F.20}$$

Orthogonal mode. Choose a set partition $\mathcal{T}_{\mathbf{x},\mathbf{y}}$ of C_i. For every $T \in \mathcal{T}_{\mathbf{x},\mathbf{y}}$, choose an anchor $c_T \in T$ and roots $\rho_{T,c}, \sigma_{T,c} \in \mu(K) \quad (c \in T)$, with $\rho_{T,c_T} = \sigma_{T,c_T} = 1$. Retain this discrete choice only if

$$\sum_{c \in T} \rho_{T,c} \overline{\sigma_{T,c}} = 0 \qquad \text{for every } T \in \mathcal{T}_{\mathbf{x},\mathbf{y}}. \tag{F.21}$$

For every $T \in \mathcal{T}_{\mathbf{x},\mathbf{y}}$ and every $c \in T \setminus \{c_T\}$, impose the anchor equations

$$X_{\mathbf{x},c} - \rho_{T,c} X_{\mathbf{x},c_T} = 0, \tag{F.22}$$

$$X_{\mathbf{y},c} - \sigma_{T,c} X_{\mathbf{y},c_T} = 0. \tag{F.23}$$

For a retained certificate $\mathfrak{c}$, let $E_{\mathfrak{c}} \subseteq K[\mathbf{X}]$ be the set of left-hand sides of its selected equations among (F.18)–(F.20), (F.22), and (F.23). We use these polynomials to describe the common solution set of the equations.

The root sums in (F.21) are checked when enumerating the choices in Definition F.4.1. Each retained $E_{\mathfrak{c}}$ consists of homogeneous polynomials in $\mathbf{X}$: support and anchor equations have degree one, and minor equations have degree two. Conjugation occurs only in the root sums over $\mu(K)$.

The row and column sets, their partitions, and $\mu(K)$ are finite, and the root sums can be checked by exact number-field equality tests. Hence the family $\mathfrak{C}_k$ of retained certificates is finite and effectively enumerable. Homogeneity makes the zero function a solution of every certificate's equations.

For $E \subseteq K[\mathbf{X}]$, write $V_K(E) = \{A \in K^{D^k} : e(A) = 0 \text{ for every } e \in E\}$. Let $\mathcal{B}_k$ be the set of K-valued k-ary functions whose singleton purifications are block-orthogonal, a property independent of the purification chosen by Lemma F.3.1.

The next theorem shows that these certificates describe $\mathcal{B}_k$. The forward direction uses Lemma F.3.5 to choose the roots and row comparisons from a block-orthogonal purification. For the reverse direction, the support and twisted-minor equations determine the surviving support and give block-rank one after purification. The chosen mode then gives linear dependence or block orthogonality for each pair of overlapping nonzero rows.

Theorem F.4.2 (Algebraic description of the BO locus)**.** *For every $k \geq 2$, the effectively enumerable certificate family $\mathfrak{C}_k$ satisfies*

$$\mathcal{B}_k = \bigcup_{\mathfrak{c} \in \mathfrak{C}_k} V_K(E_{\mathfrak{c}}). \tag{F.24}$$

Proof. For the forward inclusion, let $A \in \mathcal{B}_k$ represent F, and fix a singleton purification F' with matrix $A' = (a'_{\mathbf{x},c})$. The support of A' equals that of A. Because $|A'|$ is block-rank one, the nonempty connected components of its bipartite support graph are complete bipartite graphs $R_1 \times C_1, \ldots, R_t \times C_t$, with pairwise disjoint row sets and pairwise disjoint column sets. Choose these components as the certificate rectangles. This gives (F.18).

To choose the twisted minors, use proportionality of the purified magnitude rows within each rectangle. For $\mathbf{x}, \mathbf{y} \in R_i$ and $c, e \in C_i$,

$$|a'_{\mathbf{x},c}||a'_{\mathbf{y},e}| = |a'_{\mathbf{x},e}||a'_{\mathbf{y},c}|.$$

All four original entries are nonzero. By (F.15), there is a root $\tau_{\mathbf{x},\mathbf{y};c,e} \in \mu(K)$ for which the corresponding twisted minor equals zero.

It remains to choose a mode for each pair $\mathbf{x} < \mathbf{y}$ in R_i. If their purified rows are linearly dependent, then (F.16) shows that every ordinary minor of the original rows equals zero. We choose dependent mode.

For a linearly independent pair, choose orthogonal mode and partition C_i into the nonempty level sets T of $c \mapsto |a'_{\mathbf{x},c}|$. Since the magnitude rows are proportional, these are also the level sets for the other row. Choose any anchor $c_T \in T$. For every $c \in T$,

$$\rho_{T,c} = \frac{a_{\mathbf{x},c}}{a_{\mathbf{x},c_T}}, \qquad \sigma_{T,c} = \frac{a_{\mathbf{y},c}}{a_{\mathbf{y},c_T}}$$

belong to $\mu(K)$ by (F.14). These roots satisfy the anchor equations. By (F.17) and block orthogonality on T,

$$0 = \sum_{c \in T} a'_{\mathbf{x},c}\overline{a'_{\mathbf{y},c}} = a'_{\mathbf{x},c_T}\overline{a'_{\mathbf{y},c_T}} \sum_{c \in T} \rho_{T,c}\overline{\sigma_{T,c}}.$$

The prefactor is nonzero, so (F.21) holds. Thus the choices form a retained certificate $\mathfrak{c}$ with $A \in V_K(E_{\mathfrak{c}})$.

For the reverse inclusion, fix a certificate $\mathfrak{c}$, let $A \in V_K(E_{\mathfrak{c}})$, and let A' be the matrix of a singleton purification of the corresponding function F. We verify block-rank one first, then the two row-comparison modes of Definition F.4.1.

The support equations place every nonzero row in a unique certified row set R_i, and its support is contained in C_i. Rows belonging to different certified rectangles have disjoint supports because the corresponding column sets are disjoint.

Consider two nonzero rows $\mathbf{x}, \mathbf{y} \in R_i$. If their supports are disjoint, they already satisfy the block-rank-one alternative. Suppose instead that they overlap, and choose $e \in C_i$ with $a_{\mathbf{x},e}a_{\mathbf{y},e} \neq 0$. For every $c \in C_i \setminus \{e\}$, the twisted minor gives

$$a_{\mathbf{x},c}a_{\mathbf{y},e} = \tau_{\mathbf{x},\mathbf{y};c,e}a_{\mathbf{x},e}a_{\mathbf{y},c}.$$

Consequently, $a_{\mathbf{x},c} \neq 0$ if and only if $a_{\mathbf{y},c} \neq 0$. Thus overlapping rows have the same support. On that common support, (F.15) applied to the same twisted equations gives

$$|a'_{\mathbf{x},c}||a'_{\mathbf{y},e}| = |a'_{\mathbf{x},e}||a'_{\mathbf{y},c}|,$$

so their purified magnitude rows are positively proportional. Hence $|A'|$ is block-rank one, including when a chosen rectangle has zero support or its nonzero support splits into smaller rectangles.

For an overlapping nonzero pair in dependent mode, (F.20) and (F.16) show that all purified ordinary minors equal zero. Since both rows are nonzero, they are linearly dependent.

Now consider an overlapping nonzero pair in orthogonal mode. The anchor equations make each certified group T entirely nonzero or entirely zero in each row. We first show that the groups with nonzero entries are the same for both rows. If T survives only in row $\mathbf{x}$, if U survives in both, and if $c \in T$ and $e \in U$, then the twisted minor has a nonzero left product and a zero right product. The symmetric case is identical. Since the two rows overlap, at least one common surviving group exists. Hence every surviving group is common to both rows.

For every common surviving group T, Equations (F.17) and (F.21) give

$$\sum_{c \in T} a'_{\mathbf{x},c}\overline{a'_{\mathbf{y},c}} = a'_{\mathbf{x},c_T}\overline{a'_{\mathbf{y},c_T}} \sum_{c \in T} \rho_{T,c}\overline{\sigma_{T,c}} = 0. \tag{F.25}$$

It remains to pass from the certified groups to the actual magnitude blocks. The anchor equations give all purified entries of each row within T the same magnitude, so each surviving group lies wholly in one magnitude block. Distinct groups may have the same purified magnitude. Because the two magnitude rows are proportional, every magnitude block is a disjoint union of whole common surviving groups. Summing (F.25) over those groups shows that the Hermitian sum on the merged block is still zero.

Thus every overlapping nonzero pair is linearly dependent or block-orthogonal. Together with block-rank one, this proves that A' is block-orthogonal by Definition F.2.1, so $A \in \mathcal{B}_k$. □

Remark F.4.3 (The role of the working field). Equation (F.24) holds in K^{D^k}. Every generated function is K-valued, as established in Section F.3.3, so this equality applies to all generated functions of arity k. Their root-of-unity ratios belong to the finite group $\mu(K)$ used in the certificates.

F.4.2 Polynomial identities

Theorem F.4.2 expresses membership in $\mathcal{B}_k$ as satisfaction of at least one certificate's equations. We now express the same property by the simultaneous equations $P(A) = 0$ for P in a single finite set Q_k. Each polynomial in Q_k takes one equation from every certificate and multiplies their left-hand sides.

Enumerate the certificates as $\mathfrak{C}_k = \{\mathfrak{c}_1, \ldots, \mathfrak{c}_s\}$ and write $E_j = E_{\mathfrak{c}_j}$, after deleting syntactically zero polynomials. If some E_j is empty, then $V_K(E_j) = K^{D^k}$ and $\mathcal{B}_k = K^{D^k}$. In that case define $Q_k = \varnothing$. Otherwise define the finite set

$$Q_k = \left\{ \prod_{j \in [s]} e_j : (e_1, \ldots, e_s) \in E_1 \times \cdots \times E_s \right\} \subseteq K[\mathbf{X}].$$

Corollary F.4.4 (Polynomial identities for the BO locus)**.** *For every $A \in K^{D^k}$,*

$$A \in \mathcal{B}_k \quad \Longleftrightarrow \quad P(A) = 0 \text{ for every } P \in Q_k. \tag{F.26}$$

The finite set Q_k is effectively computable from D, k, and $\mu(K)$.

Proof. For the forward implication, suppose $A \in V_K(E_j)$ for some j. Every polynomial $P \in Q_k$ contains a factor from E_j, and that factor equals zero at A. Hence every $P(A)$ is zero.

For the converse, suppose that $A \notin V_K(E_j)$ for every j. For each j, choose $e_j \in E_j$ with $e_j(A) \neq 0$. Since K is a field, $\left(\prod_{j \in [s]} e_j\right)(A) \neq 0$. The resulting product belongs to Q_k, contradicting all these products being zero. Together with Theorem F.4.2, this proves (F.26).

For effective construction, enumerate the certificates in Definition F.4.1 and their sets E_j, then form the products defining Q_k. All choices range over finite explicitly represented sets. □

For fixed $k \geq 2$, write $W_{\mathcal{F}}^{(k)}$ for the k-ary members of $W_{\mathcal{F}}$. By Theorem F.4.2 and Corollary F.4.4, all these functions have block-orthogonal singleton purifications exactly when every polynomial in Q_k equals zero on $W_{\mathcal{F}}^{(k)}$. The next section gives an exact test for such universal identities.

F.5 Universal Identities on $W_{\mathcal{F}}$

The polynomial conditions from Section F.4 must hold on every generated function of a fixed arity, with no bound on the size of its representing instance. We give a finite test for these identities. Labelled gluing makes each monomial a multiplicative character on instances. We compare these characters by representing them as pinned partition values and applying an isomorphism theorem. Linear independence then reduces each identity to finitely many coefficient sums. The generated values lie in the field K from (F.13).

F.5.1 Labelled instances and characters

We represent generated functions by instances with labelled retained variables, then construct a product of instances that realizes pointwise multiplication of these functions.

Fix $k \geq 1$. A k-labelled $\mathcal{F}$-instance is a finite $\mathcal{F}$-instance with k distinct variables carrying the labels $1, \ldots, k$. Unlabelled isolated variables are permitted. For such an instance I, define its pinned partition vector $\mathbf{z}_I \in K^{D^k}$ by

$$\mathbf{z}_I(\mathbf{a}) = \sum_{\substack{\sigma : V(I) \to D \\ \sigma \text{ assigns } \mathbf{a} \text{ to the labels}}} \prod_{\substack{(f; v_1, \ldots, v_{r_f}) \\ \in \mathcal{C}_I}} f(\sigma(v_1), \ldots, \sigma(v_{r_f})). \tag{F.27}$$

When the labelled variables are ordered first, $\mathbf{z}_I(\mathbf{a}) = F_I^{[k]}(\mathbf{a})$. Thus $\mathbf{z}_I$ records the values of the generated function $F_I^{[k]}$, indexed by the assignments $\mathbf{a} \in D^k$ to the labels.

To multiply two generated functions at the same input, we glue their representing instances along the labelled variables. For k-labelled instances I and J, form IJ by taking their disjoint union and identifying variables with the same label. All unlabelled variables remain separate, and all constraint occurrences are retained. The common labels specify the same input tuple in both instances. Lemma F.5.1 verifies that this construction multiplies the corresponding function values.

Up to label-preserving isomorphism, this product is associative and commutative. Its identity is the instance consisting of the k labelled variables and no constraints. Thus the label-preserving isomorphism classes form a *commutative monoid*, denoted $\mathfrak{M}_k$, with multiplication IJ. All monoids below are commutative. Unless displayed explicitly in additive notation, they are written multiplicatively.

Lemma F.5.1. *Let $W_{\mathcal{F}}^{(k)}$ be the set of k-ary members of $W_{\mathcal{F}}$. Then*

$$\{\mathbf{z}_I : I \in \mathfrak{M}_k\} = W_{\mathcal{F}}^{(k)}.$$

Moreover, for $I, J \in \mathfrak{M}_k$, we have

$$\mathbf{z}_{IJ} = \mathbf{z}_I \odot \mathbf{z}_J,$$

where $\odot$ is entrywise multiplication.

Proof. For the realization statement, take $F = F_I^{[k]} \in W_{\mathcal{F}}^{(k)}$ and label the retained variables of its representing instance. The vector in (F.27) is then F. Conversely, order the labelled variables of a labelled instance first and sum all unlabelled variables to obtain a member of $W_{\mathcal{F}}^{(k)}$.

For the product statement, the unlabelled variables of the two factors remain independent after the common labels are pinned, which gives the entrywise product formula. □

For each fixed label assignment $\mathbf{a}$, the function $I \mapsto \mathbf{z}_I(\mathbf{a})$ is multiplicative by Lemma F.5.1 and equals 1 on the unit instance. A unital multiplicative *character* from a monoid M to a field L is a map

$$\chi : M \to L, \qquad \chi(1) = 1, \qquad \chi(xy) = \chi(x)\chi(y).$$

Thus each map $I \mapsto \mathbf{z}_I(\mathbf{a})$ is a character on $\mathfrak{M}_k$. Characters may take the value zero. Pointwise products of characters are characters, as are complex conjugates of complex-valued characters. Consequently, monomials in the values $\mathbf{z}_I(\mathbf{a})$ and their conjugates define characters on $\mathfrak{M}_k$, with the input tuples and exponents fixed by the monomial. Equality of characters means agreement on every element of the monoid, so different monomial expressions can define the same character.

F.5.2 Pinned constraint-function isomorphism

To compare monomial characters, we will represent them as pinned partition values of tensorized constraint functions. Theorem F.5.2 replaces agreement on all labelled instances by the existence of a domain isomorphism satisfying conditions on the pins. Over finite domains with exactly represented algebraic values, this becomes a finite search over bijections.

We use the following terminology from Young [You25, Definitions 3, 5–7 and Section 4.1]. A constraint-function family $\mathcal{H}$ is defined over a common finite domain, denoted $D_{\mathcal{H}}$. Two finite families $\mathcal{H}$ and $\mathcal{G}$ are *similar* if they are equipped with an arity-preserving bijection between their

functions. A domain bijection $\sigma : D_{\mathcal{H}} \to D_{\mathcal{G}}$ is an *isomorphism* of similar families if, for every corresponding pair $h \leftrightarrow g$ of arity r,

$$h(a_1, \ldots, a_r) = g(\sigma(a_1), \ldots, \sigma(a_r)) \qquad (a_1, \ldots, a_r \in D_{\mathcal{H}}).$$

Elements $u, v \in D_{\mathcal{H}}$ are *twins in* $\mathcal{H}$ if, for every $h \in \mathcal{H}$ of arity r, every $j \in [r]$, and every $(a_1, \ldots, a_r) \in D_{\mathcal{H}}^r$,

$$h(a_1, \ldots, a_{j-1}, u, a_{j+1}, \ldots, a_r) = h(a_1, \ldots, a_{j-1}, v, a_{j+1}, \ldots, a_r).$$

For a k-labelled $\mathcal{H}$-instance I and a pin $\beta : [k] \to D_{\mathcal{H}}$, write $Z_\beta^{\mathcal{H}}(I)$ for its pinned partition value. In particular, for the original family $\mathcal{F}$ and the pin $\beta(i) = a_i$, we have $Z_\beta^{\mathcal{F}}(I) = \mathbf{z}_I(\mathbf{a})$. The superscript records the function family used to evaluate the instance. If $\mathcal{H}$ and $\mathcal{G}$ are similar, let $I_{\mathcal{H}\to\mathcal{G}}$ be the $\mathcal{G}$-instance obtained by replacing every function of $\mathcal{H}$ by its corresponding function of $\mathcal{G}$, while retaining the variables, labels, and constraint scopes. The following theorem characterizes equality of the two pinned partition functions on all labelled instances.

Theorem F.5.2 (Pinned constraint-function isomorphism)**.** *Let $\mathcal{H}$ and $\mathcal{G}$ be similar finite sets of constraint functions over a field of characteristic zero. Fix $k \geq 1$ and pins $\beta : [k] \to D_{\mathcal{H}}$ and $\gamma : [k] \to D_{\mathcal{G}}$. If*

$$Z_\beta^{\mathcal{H}}(I) = Z_\gamma^{\mathcal{G}}(I_{\mathcal{H}\to\mathcal{G}})$$

for every k-labelled $\mathcal{H}$-instance I, then the two domains have the same size and there is an isomorphism $\sigma : D_{\mathcal{H}} \to D_{\mathcal{G}}$ of the two families such that $\sigma(\beta(i))$ and $\gamma(i)$ are twins in $\mathcal{G}$ for every $i \in [k]$.

Conversely, such a domain bijection and twin replacements preserve pinned partition values on every k-labelled instance.

Proof. Let L be the coefficient field. For any $\ell \geq k$, call two pin tuples $a : [\ell] \to D_{\mathcal{H}}$ and $b : [\ell] \to D_{\mathcal{G}}$ *indistinguishable* if their pinned partition values agree on every finite ℓ-labelled instance, with the functions retargeted from $\mathcal{H}$ to $\mathcal{G}$ on the second side. The hypothesis says that β and γ are indistinguishable. We prove that pins can be extended while preserving this property, then use extensions covering both domains to recover the function values and the sizes of twin classes.

We show that a prescribed new pin on either side can be matched on the other side while preserving indistinguishability. Suppose a and b are indistinguishable. For $x \in D_{\mathcal{H}}$ and $y \in D_{\mathcal{G}}$, define their extension profiles on finite $(\ell+1)$-labelled $\mathcal{H}$-instances J by

$$p_x(J) = Z_{(a,x)}^{\mathcal{H}}(J), \qquad \widetilde{p}_y(J) = Z_{(b,y)}^{\mathcal{G}}(J_{\mathcal{H}\to\mathcal{G}}).$$

Here (a, x) extends a by pinning the new label to x. Let $\mathcal{P} = \{p_x : x \in D_{\mathcal{H}}\} \cup \{\widetilde{p}_y : y \in D_{\mathcal{G}}\}$ be the finite set of *distinct* profiles. Each profile is a function on all such J. For a fixed $\rho \in \mathcal{P}$ and every $\tau \in \mathcal{P} \setminus \{\rho\}$, choose an instance $J_{\rho,\tau}$ for which $\rho(J_{\rho,\tau}) \neq \tau(J_{\rho,\tau})$. Such an instance exists because the profiles differ as functions. On the profiles in $\mathcal{P}$, the polynomial expression

$$Q_\rho(\eta) = \prod_{\tau \in \mathcal{P}\setminus\{\rho\}} \frac{\eta(J_{\rho,\tau}) - \tau(J_{\rho,\tau})}{\rho(J_{\rho,\tau}) - \tau(J_{\rho,\tau})}$$

is 1 when $\eta = \rho$ and 0 otherwise. All denominators are nonzero. If $\mathcal{P} = \{\rho\}$, the empty product is 1.

For a fixed pin tuple, labelled gluing multiplies partition values, and the constraint-free instance with just the labelled variables has value 1. Expanding Q_ρ therefore expresses it as a finite L-linear combination of partition values of $(\ell+1)$-labelled instances formed by gluing. The coefficients

multiply the resulting values, and each instance uses the original constraint family. If J^- is obtained from J by forgetting the last label while retaining its variable, indistinguishability gives

$$\sum_{x\in D_{\mathcal{H}}} p_x(J) = Z_a^{\mathcal{H}}(J^-) = Z_b^{\mathcal{G}}((J^-)_{\mathcal{H}\to\mathcal{G}}) = \sum_{y\in D_{\mathcal{G}}} \widetilde{p}_y(J).$$

Applying this equality to the linear combination defining Q_ρ yields

$$\left|\{x \in D_{\mathcal{H}} : p_x = \rho\}\right| 1_L = \sum_{x\in D_{\mathcal{H}}} Q_\rho(p_x) = \sum_{y\in D_{\mathcal{G}}} Q_\rho(\widetilde{p}_y) = \left|\{y \in D_{\mathcal{G}} : \widetilde{p}_y = \rho\}\right| 1_L.$$

Since L has characteristic zero, the two cardinalities are equal as integers. In particular, for every prescribed $x \in D_{\mathcal{H}}$ there is a $y \in D_{\mathcal{G}}$ with $p_x = \widetilde{p}_y$: the extended tuples (a, x) and (b, y) remain indistinguishable. The same argument works in the reverse direction.

We use the extension step to make every domain element accessible through a label, then recover function values with single-constraint instances. Starting with (β, γ), apply the extension step once for each element of $D_{\mathcal{H}}$. Then apply it in the reverse direction once for each element of $D_{\mathcal{G}}$. We obtain indistinguishable tuples $a : [\ell] \to D_{\mathcal{H}}$ and $b : [\ell] \to D_{\mathcal{G}}$ that extend the original pins and are both surjective. Each extension adds a distinct labelled variable, whose pinned value may repeat a value already used. For corresponding functions $h \leftrightarrow g$ of arity r, the instance consisting of the ℓ labelled variables and the single constraint $(h; i_1, \dots, i_r)$ gives

$$h(a(i_1), \dots, a(i_r)) = g(b(i_1), \dots, b(i_r)) \qquad (i_1, \dots, i_r \in [\ell]). \tag{F.28}$$

Thus corresponding constraints have equal values on the paired pins, including when scope positions repeat.

We first match the twin classes by their function values, then use interpolation to show that corresponding classes have equal size. For $x \in D_{\mathcal{H}}$, form the finite vector $\theta_{\mathcal{H}}(x)$ whose entries are

$$h\big(a(i_1), \dots, a(i_{j-1}), x, a(i_{j+1}), \dots, a(i_r)\big),$$

indexed by $h \in \mathcal{H}$, its positions $j \in [r]$, and all choices of the other $r - 1$ indices in $[\ell]$. Define $\theta_{\mathcal{G}}(y)$ with the same index set, replacing h and a by g and b. Surjectivity of a implies that $\theta_{\mathcal{H}}(x) = \theta_{\mathcal{H}}(x')$ exactly when x and x' are twins in $\mathcal{H}$: the entries test every function position with every possible assignment to the other positions. The corresponding statement holds for $\theta_{\mathcal{G}}$ because b is surjective. Equation (F.28) gives $\theta_{\mathcal{H}}(a(i)) = \theta_{\mathcal{G}}(b(i))$ for every i. Hence the two maps have the same finite range, and their common values specify a bijection between the twin classes of the two families.

To obtain a bijection between domain elements, we next show that corresponding twin classes have equal cardinality. For each vector u in this common range, choose, for every other vector v, an entry on which u and v differ. The same product of normalized linear factors used to extend pins gives a polynomial R_u that is 1 at u and 0 at every other vector in the range. Each entry of $\theta_{\mathcal{H}}(x)$, with its corresponding entry of $\theta_{\mathcal{G}}(y)$, is the partition value of a single-constraint instance whose new label occupies position j and whose other positions are old labels. Labelled gluing realizes products of these entries. After expanding R_u and forgetting the new label, indistinguishability of a, b therefore gives

$$\sum_{x\in D_{\mathcal{H}}} R_u(\theta_{\mathcal{H}}(x)) = \sum_{y\in D_{\mathcal{G}}} R_u(\theta_{\mathcal{G}}(y)).$$

Both sides count the elements of the corresponding twin class in L. Characteristic zero again makes their integer cardinalities equal. This also handles an empty constraint family: the vectors are

empty, there is one class on each side, and the constant polynomial 1 counts the two domains by means of one isolated unlabelled variable.

Choose a bijection between each pair of corresponding twin classes. Their union is a bijection $\sigma : D_{\mathcal{H}} \to D_{\mathcal{G}}$, and $\sigma(a(i))$ is a twin of $b(i)$ for every $i \in [\ell]$. To check preservation of function values, take $x_1, \dots, x_r \in D_{\mathcal{H}}$ and choose $i_1, \dots, i_r$ with $a(i_j) = x_j$. By (F.28) and successive replacement of input values by twins,

$$h(x_1, \dots, x_r) = g(b(i_1), \dots, b(i_r)) = g(\sigma(x_1), \dots, \sigma(x_r)).$$

Thus σ is an isomorphism, and its restriction to the original pins satisfies the asserted twin condition.

For the converse, an isomorphism σ bijectively transports assignments with pins β to assignments with pins $\sigma \circ \beta$, preserving every constraint value. Replacing these pins by the prescribed twins γ also preserves every constraint value, one input position at a time, including when a labelled variable occurs repeatedly in a scope. Hence

$$Z^{\mathcal{H}}_{\beta}(I) = Z^{\mathcal{G}}_{\sigma\circ\beta}(I_{\mathcal{H}\to\mathcal{G}}) = Z^{\mathcal{G}}_{\gamma}(I_{\mathcal{H}\to\mathcal{G}})$$

for every finite labelled instance, as required. □

For exactly represented algebraic values, Theorem F.5.2 gives a finite decision procedure for equality on all labelled instances. If the domains have different sizes, equality fails. Otherwise, enumerate their finitely many bijections and test preservation of all function values and the prescribed twin relations exactly. Equality holds precisely when one bijection passes these tests.

F.5.3 Tensor realization and character comparison

We first construct an instance whose pinned partition value is a given monomial in generated function values and their conjugates. We then compare the constructions for two monomials using Theorem F.5.2.

Fix $k \geq 1$. For each assignment to the labels $\mathbf{a} = (a_1, \dots, a_k) \in D^k$, introduce two independent formal variables $X_{\mathbf{a}}$ and $Y_{\mathbf{a}}$. When a polynomial in these variables is evaluated at the pinned partition vector of an instance I, the substitutions are

$$X_{\mathbf{a}} \longmapsto \mathbf{z}_I(\mathbf{a}), \qquad Y_{\mathbf{a}} \longmapsto \overline{\mathbf{z}_I(\mathbf{a})}. \tag{F.29}$$

Let $p_+, p_- \geq 0$. A monomial of X-degree p_+ and Y-degree p_- can be written as

$$M = \prod_{i\in[p_+]} X_{\mathbf{a}^{(i)}} \prod_{j\in[p_-]} Y_{\mathbf{b}^{(j)}}.$$

For $I \in \mathfrak{M}_k$, define

$$\chi_M(I) := M(\mathbf{z}_I, \overline{\mathbf{z}_I}),$$

using the simultaneous substitution in (F.29). The input tuples and their multiplicities are fixed by M. As I varies, these evaluations define the function χ_M on $\mathfrak{M}_k$. Thus $\chi_M(I)$ is a product of p_+ generated function values and p_- conjugated values, counted with multiplicity. The possibly repeated indices

$$\mathbf{a}^{(i)} = (a_1^{(i)}, \dots, a_k^{(i)}) \in D^k \quad (i \in [p_+]),$$
$$\mathbf{b}^{(j)} = (b_1^{(j)}, \dots, b_k^{(j)}) \in D^k \quad (j \in [p_-])$$

list the factors of M. The parenthesized superscript numbers a factor: $a_t^{(i)}$ is the entry in position t of the tuple indexing the ith X-factor.

To represent this product by one partition value, let each instance variable take a tuple in $D^{p_+ + p_-}$. The tuple records the values assigned to that variable for the ordinary factors, followed by those for the conjugated factors. For each constraint, multiply the evaluations of its original function on corresponding entries of these tuples, conjugating the last p_- evaluations. For each original constraint function $f : D^{r_f} \to K$, define $T_{p_+,p_-}(f) : (D^{p_+ + p_-})^{r_f} \to K$ by

$$T_{p_+,p_-}(f)(\mathbf{u}_1, \ldots, \mathbf{u}_{r_f}) = \prod_{\ell \in [p_+]} f(u_{1,\ell}, \ldots, u_{r_f,\ell}) \cdot \prod_{\ell = p_+ + 1}^{p_+ + p_-} \overline{f(u_{1,\ell}, \ldots, u_{r_f,\ell})},$$

where $\mathbf{u}_j \in D^{p_+ + p_-}$. When $p_+ = p_- = 0$, the domain is the singleton D^0 and $T_{0,0}(f)$ is the constant-one function of arity r_f on that domain. Write $T_{p_+,p_-}(\mathcal{F}) = \big(T_{p_+,p_-}(f) : f \in \mathcal{F}\big)$ for this f-indexed tensorized family.

The degrees determine the transformed constraint functions. The specific input tuples occurring in M determine the pins: at each label, we collect its prescribed values from all the factors into one tuple.

Definition F.5.3 (Tensor realization)**.** The pin $\beta_M : [k] \to D^{p_+ + p_-}$ is obtained by transposing these $p_+ + p_-$ assignments to the labels. For label $t \in [k]$, set

$$\beta_M(t) = \big(a_t^{(1)}, \ldots, a_t^{(p_+)}, b_t^{(1)}, \ldots, b_t^{(p_-)}\big) \in D^{p_+ + p_-}.$$

Equivalently, regarding the values $\beta_M(t)$ as columns,

$$\big(\beta_M(1) \ \cdots \ \beta_M(k)\big) = \begin{pmatrix} a_1^{(1)} & \cdots & a_k^{(1)} \\ \vdots & & \vdots \\ a_1^{(p_+)} & \cdots & a_k^{(p_+)} \\ b_1^{(1)} & \cdots & b_k^{(1)} \\ \vdots & & \vdots \\ b_1^{(p_-)} & \cdots & b_k^{(p_-)} \end{pmatrix}.$$

The block of $\mathbf{a}$-rows or $\mathbf{b}$-rows is omitted when $p_+ = 0$ or $p_- = 0$, respectively. For $I \in \mathfrak{M}_k$, let I^M be the k-labelled $T_{p_+,p_-}(\mathcal{F})$-instance obtained by replacing each occurrence of $f \in \mathcal{F}$ by $T_{p_+,p_-}(f)$ while retaining all variables, labels, and scopes.

The next lemma verifies that these transformed constraints and pins realize the product defining $\chi_M(I)$. Appendix F.A.7 illustrates the construction.

Lemma F.5.4. *For $I \in \mathfrak{M}_k$, the instance I^M from Definition F.5.3 satisfies*

$$Z_{\beta_M}^{T_{p_+,p_-}(\mathcal{F})}(I^M) = \chi_M(I) = \prod_{i \in [p_+]} \mathbf{z}_I(\mathbf{a}^{(i)}) \prod_{j \in [p_-]} \overline{\mathbf{z}_I(\mathbf{b}^{(j)})}.$$

Thus $\chi_M : \mathfrak{M}_k \to K$ is a unital multiplicative character.

Proof. An assignment $\eta : V(I) \to D^{p_+ + p_-}$ is uniquely a tuple $(\sigma_1, \ldots, \sigma_{p_+ + p_-})$ of assignments $\sigma_\ell : V(I) \to D$. By the definition of β_M in Definition F.5.3, the assignment σ_i pins the k labels to $\mathbf{a}^{(i)}$ for $i \in [p_+]$, while $\sigma_{p_+ + j}$ pins them to $\mathbf{b}^{(j)}$ for $j \in [p_-]$. For each component assignment σ_ℓ, the

product of its constraint weights is the instance value $F_I(\sigma_\ell)$ from (F.4). Expanding the tensorized constraint functions therefore gives

$$Z_{\beta_M}^{T_{p_+,p_-}(\mathcal{F})}(I^M) = \sum_{\sigma_1,\ldots,\sigma_{p_++p_-}} \prod_{i\in[p_+]} F_I(\sigma_i) \prod_{j\in[p_-]} \overline{F_I(\sigma_{p_++j})}$$
$$= \prod_{i\in[p_+]} \mathbf{z}_I(\mathbf{a}^{(i)}) \prod_{j\in[p_-]} \overline{\mathbf{z}_I(\mathbf{b}^{(j)})}.$$

The first sum ranges over the component assignments with the pins just specified. The second equality holds because these component assignments are independent, so the finite sum factors. Finite sums commute with complex conjugation. When $p_+ = p_- = 0$, the domain is D^0 and the value is the empty product 1.

Finally, Lemma F.5.1 gives $\chi_M(IJ) = \chi_M(I)\chi_M(J)$ and $\chi_M(1) = 1$, so χ_M is a unital multiplicative character. □

Now consider a second monomial N of (X, Y)-degrees (r_+, r_-). For each original function $f \in \mathcal{F}$, write

$$h_f := T_{p_+,p_-}(f), \qquad g_f := T_{r_+,r_-}(f).$$

These are the transformed versions of f for M and N, respectively. Both have arity r_f, on the respective domains $D_M = D^{p_++p_-}$ and $D_N = D^{r_++r_-}$. Use Definition F.5.3 to construct the pins β_M, β_N and the instances I^M, I^N by replacing each occurrence of f with h_f, g_f, respectively. By Lemma F.5.4, their pinned partition values are $\chi_M(I)$ and $\chi_N(I)$.

To apply Theorem F.5.2, the correspondence $h_f \leftrightarrow g_f$ must give a bijection between the two sets of functions. Tensorization can identify distinct original functions: for example, $T_{1,1}(\mathrm{i}f) = T_{1,1}(f)$, where $\mathrm{i}^2 = -1$. We therefore multiply the functions by positive rational scalars to make the functions within each family distinct at each arity. Choosing the same scalar λ_f for h_f and g_f multiplies both partition values by a common nonzero factor, preserving their equality. The next lemma makes this simultaneous choice for two indexed families.

Lemma F.5.5. *Let $(h_f : f \in \mathcal{F})$ and $(g_f : f \in \mathcal{F})$ be indexed families over finite domains D_M and D_N, respectively, where $h_f : D_M^{r_f} \to \mathbb{C}$ and $g_f : D_N^{r_f} \to \mathbb{C}$ are given by their exact values. Suppose that, within each family and each arity, at most one indexed function is identically zero. One can effectively choose positive rational numbers λ_f, $f \in \mathcal{F}$, such that the functions within each arity are pairwise distinct in both indexed families $(\lambda_f h_f : f \in \mathcal{F})$ and $(\lambda_f g_f : f \in \mathcal{F})$.*

Proof. Choose the tags sequentially, avoiding collisions with functions already treated. For a previously treated index e of the same arity, the equality $\lambda h_f = \lambda_e h_e$ excludes at most one positive rational value of λ, since the hypothesis ensures that at least one of h_f, h_e is nonzero. The same argument applies to the g-family. Thus only finitely many rational values are excluded at each stage. Enumerating the positive integers and testing the resulting function values exactly finds a valid tag. □

The tensorized families associated with monomial characters satisfy the hypothesis of Lemma F.5.5. At positive tensor degree, $T_{p_+,p_-}(f)$ is zero exactly when f is zero, and exact deduplication of the input leaves at most one zero function of each arity. At degree zero every function is the nonzero constant-one function.

For the two tensorized families, equality of their pinned partition values is exactly the desired equality $\chi_M = \chi_N$ by Lemma F.5.4. The next lemma decides this equality using the common scalar tags and Theorem F.5.2. We state the comparison for general indexed families so that it also applies to the degree-restricted construction in Section F.10.

Lemma F.5.6. *Let $\mathcal{H}_M = (h_f : f \in \mathcal{F})$ and $\mathcal{H}_N = (g_f : f \in \mathcal{F})$ be finite indexed families of exactly represented algebraic-valued functions on finite nonempty domains D_M and D_N. For each f, both h_f and g_f have arity r_f. Assume each family contains at most one zero function of each arity. Fix $k \geq 1$ and pins $\beta_M : [k] \to D_M$ and $\beta_N : [k] \to D_N$. For an original $\mathcal{F}$-instance I, form I^M and I^N by replacing each occurrence of f by h_f and g_f, respectively. It is decidable whether*

$$Z^{\mathcal{H}_M}_{\beta_M}(I^M) = Z^{\mathcal{H}_N}_{\beta_N}(I^N) \quad \text{for every } I \in \mathfrak{M}_k.$$

In particular, equality of two monomial characters on $\mathfrak{M}_k$ is decidable.

Proof. Abbreviate the two pinned partition values by $\chi_M(I)$ and $\chi_N(I)$:

$$Z^{\mathcal{H}_M}_{\beta_M}(I^M) = \chi_M(I), \qquad Z^{\mathcal{H}_N}_{\beta_N}(I^N) = \chi_N(I). \tag{F.30}$$

Thus χ_M and χ_N are functions on the same original labelled-instance monoid $\mathfrak{M}_k$.

Apply Lemma F.5.5 to choose common positive rational tags λ_f that make both indexed families duplicate-free, and put

$$\widehat{h}_f = \lambda_f h_f, \qquad \widehat{g}_f = \lambda_f g_f,$$
$$\widehat{\mathcal{H}}_M = (\widehat{h}_f : f \in \mathcal{F}), \quad \widehat{\mathcal{H}}_N = (\widehat{g}_f : f \in \mathcal{F}).$$

If $N_f(I)$ is the number of occurrences of f in the original instance I, interpreting that instance with the tagged functions multiplies both values in (F.30) by the same factor

$$\tau(I) = \prod_{f \in \mathcal{F}} \lambda_f^{N_f(I)}.$$

This factor is nonzero. Denoting the two tagged interpretations by $\widehat{I}^M$ and $\widehat{I}^N$, define

$$\widehat{Z}_M(I) := Z^{\widehat{\mathcal{H}}_M}_{\beta_M}(\widehat{I}^M) = \tau(I)\chi_M(I),$$
$$\widehat{Z}_N(I) := Z^{\widehat{\mathcal{H}}_N}_{\beta_N}(\widehat{I}^N) = \tau(I)\chi_N(I).$$

Since the same nonzero factor occurs on both sides, we have

$$\big(\forall I \in \mathfrak{M}_k,\ \chi_M(I) = \chi_N(I)\big) \iff \big(\forall I \in \mathfrak{M}_k,\ \widehat{Z}_M(I) = \widehat{Z}_N(I)\big). \tag{F.31}$$

The tagged families are sets of size $|\mathcal{F}|$, and $\widehat{h}_f \longleftrightarrow \widehat{g}_f \quad (f \in \mathcal{F})$ is an arity-preserving bijection. They are therefore similar in the sense of Section F.5.2. Because the functions are duplicate-free, every labelled $\widehat{\mathcal{H}}_M$-instance corresponds uniquely to an original $\mathcal{F}$-instance, and replacing $\widehat{h}_f$ by $\widehat{g}_f$ produces its $\widehat{\mathcal{H}}_N$ interpretation.

If the right-hand side of (F.31) holds, then by Theorem F.5.2, the domains have equal size and there is a bijection

$$\sigma : D_M \to D_N \tag{F.32}$$

that is an isomorphism of the two tagged families and satisfies

$$\sigma(\beta_M(t)) \text{ and } \beta_N(t) \text{ are twins in } \widehat{\mathcal{H}}_N \qquad (t \in [k]). \tag{F.33}$$

Conversely, such a bijection maps every assignment of $\widehat{I}^M$ to an assignment of $\widehat{I}^N$, preserves each constraint weight, and preserves the pins through the twin conditions. The converse direction of Theorem F.5.2 therefore gives equality of the two tagged pinned values for every I. Since $\tau(I) \neq 0$, division by the common factor gives $\chi_M(I) = \chi_N(I)$ for every $I \in \mathfrak{M}_k$.

If $|D_M| \neq |D_N|$, no bijection in (F.32) exists and the characters are different. Otherwise enumerate every bijection $\sigma : D_M \to D_N$. For each one, test exactly whether all corresponding tagged functions have equal values under σ and whether (F.33) holds. The characters agree if and only if some bijection passes. All domains and sets of functions are finite, so the enumeration terminates.

Finally, apply the criterion to the families $T_{p_+,p_-}(\mathcal{F})$ and $T_{r_+,r_-}(\mathcal{F})$ determined by M, N, with the pins from Definition F.5.3. Lemma F.5.4 identifies their pinned partition values with $\chi_M(I)$ and $\chi_N(I)$, respectively. These families satisfy the zero-function hypothesis of Lemma F.5.5, so the criterion decides equality of the two monomial characters. □

F.5.4 The universal identity oracle

Equality of the monomial characters is now decidable by Lemma F.5.6. We group monomials that define the same character and add their coefficients. To obtain an identity test, it remains to show that distinct characters cannot cancel in a nontrivial linear combination. The next lemma supplies this fact, which Theorem F.5.8 turns into finite coefficient tests.

Lemma F.5.7. *Let M be a monoid, let L be a field of characteristic zero, and let $\chi_1, \dots, \chi_s : M \to L$ be pairwise distinct unital multiplicative characters. If $c_1, \dots, c_s \in L$ satisfy*

$$\sum_{i \in [s]} c_i \chi_i(x) = 0 \qquad \text{for every } x \in M, \tag{F.34}$$

then $c_1 = \cdots = c_s = 0$. Equivalently, $\chi_1, \dots, \chi_s$ are linearly independent in the vector space L^M.

Proof. Suppose the conclusion is false, and choose a nontrivial relation of the form (F.34) involving the minimum possible number s of characters. After deleting zero terms and relabelling, we may assume

$$c_i \neq 0 \qquad (i \in [s]). \tag{F.35}$$

If $s = 1$, evaluation at the monoid identity gives $0 = c_1 \chi_1(1) = c_1$, contrary to (F.35). Hence $s \geq 2$.

Because χ_1 and χ_s are distinct functions on M, choose and fix $u \in M$ such that

$$\chi_1(u) \neq \chi_s(u). \tag{F.36}$$

For an arbitrary $v \in M$, evaluate the assumed relation at uv and use multiplicativity:

$$0 = \sum_{i \in [s]} c_i \chi_i(uv) = \sum_{i \in [s]} c_i \chi_i(u) \chi_i(v). \tag{F.37}$$

Evaluation at v, followed by multiplication by the fixed scalar $\chi_s(u)$, gives

$$0 = \chi_s(u) \sum_{i \in [s]} c_i \chi_i(v) = \sum_{i \in [s]} c_i \chi_s(u) \chi_i(v). \tag{F.38}$$

Subtracting (F.38) from (F.37) yields

$$0 = \sum_{i \in [s-1]} c_i (\chi_i(u) - \chi_s(u)) \chi_i(v) \qquad \text{for every } v \in M. \tag{F.39}$$

The term indexed by s has coefficient zero. With u fixed, the coefficients $b_i = c_i(\chi_i(u) - \chi_s(u)) \quad (i \in [s-1])$ lie in L and are independent of v. Moreover, $b_1 = c_1(\chi_1(u) - \chi_s(u)) \neq 0$ by Equations (F.35) and (F.36). Consequently, (F.39) is a nontrivial linear relation involving at most $s - 1$ of the characters, contradicting the minimality of s. □

Theorem F.5.8 (Universal identity oracle)**.** *Fix $k \geq 1$. The input is an arbitrary commutative polynomial $P \in \mathbb{C}[X_{\mathbf{a}}, Y_{\mathbf{a}} : \mathbf{a} \in D^k]$, given as a finite list of monomials with exactly encoded algebraic-complex coefficients. The $2q^k$ symbols $X_{\mathbf{a}}, Y_{\mathbf{a}}$ are independent formal variables. For $I \in \mathfrak{M}_k$, the notation $P(\mathbf{z}_I, \overline{\mathbf{z}_I})$ means the simultaneous substitution $X_{\mathbf{a}} = \mathbf{z}_I(\mathbf{a}), \quad Y_{\mathbf{a}} = \overline{\mathbf{z}_I(\mathbf{a})} \quad (\mathbf{a} \in D^k)$, as in* (F.29)*. There is a total exact algorithm deciding whether*

$$P(\mathbf{z}_I, \overline{\mathbf{z}_I}) = 0 \qquad \textit{for every } I \in \mathfrak{M}_k. \tag{F.40}$$

Proof. Let L be the number field generated over K by the finitely many coefficients of P. It is effectively constructible from their exact encodings. Every monomial character takes values in $K \subseteq L$, so we may regard all of them as L-valued characters.

Expand P into finitely many monomials and combine identical monomials. By Lemma F.5.6, we can partition these monomials into classes, placing M and N in the same class exactly when $\chi_M = \chi_N$. Let $\mathcal{E}_{\text{char}}$ be the finite set of these classes, and let χ_E denote the common character defined by the monomials in E. Then, for every $I \in \mathfrak{M}_k$,

$$P(\mathbf{z}_I, \overline{\mathbf{z}_I}) = \sum_{E \in \mathcal{E}_{\text{char}}} \left(\sum_{M \in E} c_M \right) \chi_E(I),$$

where $c_M \in L$ is the coefficient of M.

The characters χ_E are pairwise distinct, so Lemma F.5.7 makes them linearly independent over L. Therefore (F.40) holds if and only if

$$\sum_{M \in E} c_M = 0 \qquad \text{for every } E \in \mathcal{E}_{\text{char}}.$$

These are finitely many exact algebraic-number equality tests.

Constructing the tensorized functions, choosing scalar tags, enumerating domain bijections, and testing preservation of function values, twins, and coefficient sums all require only finitely many exact computations. If one class sum is nonzero, Lemma F.5.7 says that the displayed linear combination is not the zero function on $\mathfrak{M}_k$. Hence a finite labelled instance I satisfies $P(\mathbf{z}_I, \overline{\mathbf{z}_I}) \neq 0$, and the algorithm rejects. □

F.6 The Global Block Orthogonality Decision Procedure

To decide the joint condition $\mathsf{BO}(\mathcal{F})$ in Definition F.2.3, Corollary F.3.2 and Lemma F.3.4 reduce the problem to singleton purifications of generated functions of arity between two and $q^2 + 1$. For each such arity k, Corollary F.4.4 constructs a finite set $Q_k \subseteq K[X_{\mathbf{a}} : \mathbf{a} \in D^k]$. By (F.26), a k-ary generated function F has a block-orthogonal singleton purification exactly when $P(F) = 0$ for every $P \in Q_k$.

The remaining task is to test these equations on all generated functions of each tested arity. By Lemma F.5.1, these functions are precisely the $\mathbf{z}_I$ with $I \in \mathfrak{M}_k$. Each $P \in Q_k$ is a polynomial in the variables $X_{\mathbf{a}}$ alone, so Theorem F.5.8 decides whether $P(\mathbf{z}_I) = 0$ for every such instance. Thus finitely many identity tests decide the condition on the entire set $W_{\mathcal{F}}$.

Decision Procedure F.6.1 (Global Block Orthogonality)**.** On input $q \geq 1$ and a finite set $\mathcal{F}$ of algebraic complex-valued functions on D, perform the following steps.

1. Identify functions with the same arity and the same values. If $\mathcal{F} = \varnothing$, return YES. Otherwise construct the field K in (F.13) and compute $\mu(K)$.

2. For every integer $2 \le k \le q^2 + 1$, enumerate the Block Orthogonality certificates in Definition F.4.1 of arity k and construct Q_k as in Corollary F.4.4.

3. For every tested k and every $P \in Q_k$, use Theorem F.5.8 to decide whether

$$P(\mathbf{z}_I) = 0 \qquad \text{for every } I \in \mathfrak{M}_k.$$

   Return NO as soon as one identity fails.

4. If all tests succeed, return YES.

Theorem F.6.2 (Exact decision of global BO)**.** *Procedure F.6.1 is a total exact algorithm that returns* YES *on* $(q, \mathcal{F})$ *if and only if* $\mathsf{BO}(\mathcal{F})$ *holds.*

Proof. The case $\mathcal{F} = \varnothing$ was verified in Section F.2.5. Assume $\mathcal{F} \neq \varnothing$.

For termination, the arity loop is finite. At each tested arity, the row and column sets are finite, $\mu(K)$ is finite, and the certificate construction ranges over finitely many rectangle systems, row-pair modes, set partitions, anchors, and root-of-unity labels. Every root-sum test is an exact equality test in K. Each certificate has finitely many equations, and hence Q_k is finite and effectively computable. If a certificate's equations all have syntactically zero left-hand sides, this is detected when forming Q_k and gives $Q_k = \varnothing$.

Finally, Theorem F.5.8 is total and exact for each queried polynomial. Consequently every loop in Procedure F.6.1 is finite.

Suppose the procedure returns YES. Fix $2 \le k \le q^2 + 1$ and a k-ary $F \in W_{\mathcal{F}}$. By Lemma F.5.1, there is a finite $I \in \mathfrak{M}_k$ such that $F = \mathbf{z}_I$. Acceptance of all queries at arity k gives $P(F) = 0$ for every $P \in Q_k$. By (F.26), $F \in \mathcal{B}_k$. Thus every function in $W_{\mathcal{F}}$ of arity between two and $q^2 + 1$ has a block-orthogonal singleton purification.

If a function in $W_{\mathcal{F}}$ of larger arity had a non-block-orthogonal singleton purification, Lemma F.3.4 would produce such a member of $W_{\mathcal{F}}$ at a tested arity, a contradiction. Hence every member of $W_{\mathcal{F}}$ of arity at least two has a block-orthogonal singleton purification. By Corollary F.3.2, the joint condition $\mathsf{BO}(\mathcal{F})$ holds.

Conversely, suppose $\mathsf{BO}(\mathcal{F})$ holds. By Corollary F.3.2, every $F \in W_{\mathcal{F}}$ of arity at least two has a block-orthogonal singleton purification. At every tested arity k, Lemma F.5.1 therefore gives $\mathbf{z}_I \in \mathcal{B}_k$ for every $I \in \mathfrak{M}_k$. Equation (F.26) gives $P(\mathbf{z}_I) = 0$ for every $P \in Q_k$ and every $I \in \mathfrak{M}_k$. The universal identity oracle accepts every query, and the procedure returns YES.

If the procedure returns NO, a failed identity implies that there is a finite labelled instance I with $P(\mathbf{z}_I) \neq 0$ by Theorem F.5.8. Equation (F.26) then gives $\mathbf{z}_I \notin \mathcal{B}_k$, a witness to failure of the singleton condition. The equivalence for NO follows from termination and the two directions already proved for YES. □

F.7 Generated Supports and a Common Mal'tsev Polymorphism

Assume $\mathsf{BO}(\mathcal{F})$. We first construct a common Mal'tsev polymorphism for a finite family of generated supports, then use the finiteness of ternary operations on D to obtain one for all generated supports. For the finite family, we express preservation by constraints on the operation's values and impose the Mal'tsev identities one input triple at a time. Block Orthogonality will allow each new prescribed value to be added while retaining the earlier ones. To apply it, we first show that the solution relation of these constraints and its projections are generated supports.

F.7.1 Equality-free support realization

A product of function values is nonzero exactly when every factor is nonzero, so multiplication realizes conjunctions of support constraints. Projection requires summing over omitted variables, where nonzero contributions may cancel. We will raise each contribution to a common positive power before summing. The next lemma supplies an exponent for which every nonempty sum is nonzero, simultaneously for all retained assignments.

Lemma F.7.1. *Let $\mathcal{A}_1, \dots, \mathcal{A}_t$ be nonempty finite multisets of elements of $\mathbb{C}^\times$. There exist infinitely many integers $m \geq 1$ such that*

$$\sum_{a \in \mathcal{A}_i} a^m \neq 0 \qquad \text{for every } i \in [t]. \tag{F.41}$$

Moreover, such an m can be found by enumerating the positive integers and performing exact algebraic zero tests.

Proof. We will show that all sufficiently large multiples of one positive integer satisfy (F.41). We use the characteristic-zero form of the Skolem–Mahler–Lech theorem: the zero set of a sequence satisfying a homogeneous linear recurrence with constant coefficients is a finite set together with finitely many arithmetic progressions [Lec53]. For $i \in [t]$, put $s_i(n) = \sum_{a \in \mathcal{A}_i} a^n$. Let E be the shift operator, $(Es)(n) = s(n+1)$. The sequence a^n is annihilated by $E - a$, so

$$\prod_{a \in \operatorname{supp}(\mathcal{A}_i)} (E - a) s_i = 0.$$

Expanding this product gives the required constant-coefficient recurrence over the number field generated by $\mathcal{A}_i$. The theorem therefore applies to each s_i.

After absorbing finitely many initial terms into the finite exceptional set, write each progression in the zero set as $\{n \geq N : n \equiv r \pmod p\}$, $0 \leq r < p$. We claim that its residue r is nonzero. Otherwise, for some $p \geq 1$ and $N \geq 0$,

$$s_i(pn) = 0 \qquad (n \geq N). \tag{F.42}$$

Group the elements of the multiset $\mathcal{A}_i$ according to their pth powers. Let $\beta_1, \dots, \beta_h$ be the distinct resulting nonzero values, and let $\nu_j \geq 1$ be the multiplicity of β_j. Then

$$s_i(pn) = \sum_{a \in \mathcal{A}_i} (a^p)^n = \sum_{j \in [h]} \nu_j \beta_j^n.$$

Applying (F.42) at $n = N, N+1, \dots, N+h-1$ gives

$$\begin{pmatrix} 1 & \cdots & 1 \\ \beta_1 & \cdots & \beta_h \\ \vdots & & \vdots \\ \beta_1^{h-1} & \cdots & \beta_h^{h-1} \end{pmatrix} \begin{pmatrix} \nu_1 \beta_1^N \\ \vdots \\ \nu_h \beta_h^N \end{pmatrix} = \mathbf{0}. \tag{F.43}$$

The determinant of the first matrix is $\prod_{1 \leq j < k \leq h} (\beta_k - \beta_j) \neq 0$, so (F.43) forces $\nu_j \beta_j^N = 0$ for every j. Since every β_j is nonzero, this contradicts $\nu_j \geq 1$.

To choose a common exponent, let M be a common multiple of all progression periods occurring for all s_i, taking $M = 1$ if there are no such progressions. Every multiple of M has residue zero modulo each period, so it lies in none of these progressions. All sufficiently large multiples also avoid the finitely many exceptional zeros. Thus infinitely many exponents satisfy (F.41), and enumerating $m = 1, 2, \dots$ with exact zero tests finds one. □

We use this common exponent to realize the relation defined by an equality-free primitive-positive (pp) formula. Such a formula is formed from relation atoms by conjunction and existential quantification. The same variable may occur more than once, and equality atoms between distinct variables are excluded.

Lemma F.7.2. *Let $h \geq 1$ and $F_1, \ldots, F_h \in W_{\mathcal{F}}$. Let $\mathbf{x} = (x_1, \ldots, x_\ell)$, with $\ell \geq 1$, be the tuple of free variables, and let $\mathbf{y} = (y_1, \ldots, y_s)$, with $s \geq 0$, be the tuple of existential variables. Let $\mathcal{E}$ be a finite set indexing the atoms. For every $e \in \mathcal{E}$, choose $i(e) \in [h]$ and an ordered scope $\mathbf{z}_e$ of length $\mathrm{arity}(F_{i(e)})$ whose entries are variables from $\mathbf{x}$ and $\mathbf{y}$. The same logical variable may occur repeatedly within one scope or across several scopes. Define $R \subseteq D^\ell$ by*

$$R(\mathbf{x}) \quad \Longleftrightarrow \quad \exists \mathbf{y} \bigwedge_{e \in \mathcal{E}} \Phi_{F_{i(e)}}(\mathbf{z}_e).$$

Then there exists $H \in W_{\mathcal{F}}$ with $\Phi_H = R$. Given instances I_i with $F_i = F_{I_i}^{[r_i]}$, where $r_i = \mathrm{arity}(F_i)$, such an H can be obtained by a terminating exact search.

Proof. For $m \geq 1$, define

$$H_m(\mathbf{x}) = \sum_{\mathbf{y} \in D^s} \prod_{e \in \mathcal{E}} F_{i(e)}(\mathbf{z}_e)^m. \tag{F.44}$$

The empty product is 1. For fixed $\mathbf{a} \in D^\ell$, let $\mathcal{A}_{\mathbf{a}}$ be the multiset of all nonzero values $\prod_{e \in \mathcal{E}} F_{i(e)}(\mathbf{z}_e(\mathbf{a}, \mathbf{b}))$, $\mathbf{b} \in D^s$. Then $\mathcal{A}_{\mathbf{a}}$ is nonempty exactly when $\mathbf{a} \in R$, and

$$H_m(\mathbf{a}) = \sum_{w \in \mathcal{A}_{\mathbf{a}}} w^m, \tag{F.45}$$

where an empty sum is zero. Thus tuples outside R give zero term by term for every m. Apply Lemma F.7.1 to the finitely many nonempty multisets $\mathcal{A}_{\mathbf{a}}$, $\mathbf{a} \in R$. The resulting common exponent makes (F.45) nonzero for every tuple in R. If $R = \varnothing$, any m works. Hence $\Phi_{H_m} = R$.

We next realize H_m by an instance over $\mathcal{F}$. For each F_i, choose a $\#\mathsf{CSP}(\mathcal{F})$ instance I_i such that $F_i = F_{I_i}^{[r_i]}$, where $r_i = \mathrm{arity}(F_i)$. For every occurrence e and every copy index in $[m]$, take a fresh copy of $I_{i(e)}$, identify its retained variables with those in $\mathbf{z}_e$, and keep all of its hidden variables fresh. For fixed $(\mathbf{x}, \mathbf{y})$, summing over the copied hidden variables gives $\prod_{e \in \mathcal{E}} F_{i(e)}(\mathbf{z}_e)^m$. Summing over $\mathbf{y}$ then gives (F.44). Repeated variables in $\mathbf{z}_e$ are legal scope identifications, and the retained variables may be put in the displayed order. Therefore $H_m \in W_{\mathcal{F}}$. From the given instances I_i, all values of the F_i and the finite multisets $\mathcal{A}_{\mathbf{a}}$ can be computed exactly. The search for m terminates by Lemma F.7.1, so the construction is effective. Appendix F.A.8 illustrates it for two binary functions. □

By Lemma F.7.2, Block Orthogonality applies to relations defined by these formulas, including their projections. It forces any two nonempty row supports to be equal or disjoint. In the construction of a Mal'tsev operation, this property will let us add one prescribed value without changing those already fixed.

Corollary F.7.3 (Singleton rectangularity)**.** *Assume $\mathsf{BO}(\mathcal{F})$. Let $R \subseteq D^{r-1} \times D$, with $r \geq 2$, have an equality-free pp definition from finitely many relations Φ_F, $F \in W_{\mathcal{F}}$. For $\mathbf{u} \in D^{r-1}$, put $R_{\mathbf{u}} = \{a \in D : (\mathbf{u}, a) \in R\}$. Then any two nonempty sets $R_{\mathbf{u}}$ and $R_{\mathbf{v}}$ are equal or disjoint.*

Proof. By Lemma F.7.2, with the retained variables in the displayed order there is an $H \in W_{\mathcal{F}}$ with $\Phi_H = R$. Apply $\mathsf{BO}(\mathcal{F})$ to the singleton set $\{H\}$ and write $\mathrm{Pure}(H) = H'$. Purification preserves support, and $|H'|$ is block-rank one. Hence two nonzero rows that overlap have equal support. □

F.7.2 A common Mal'tsev polymorphism

For a finite family of generated supports, introduce variables for the outputs of a candidate ternary operation. The constraints require that applying it to the corresponding entries of any three tuples in each relation produces another tuple in that relation. The third projection preserves every relation and already satisfies one Mal'tsev identity. Starting from its values, we use Corollary F.7.3 to impose the other identity one input triple at a time. At each step, the prescribed values will extend to an operation preserving the entire family.

Lemma F.7.4. *Assume* $\mathsf{BO}(\mathcal{F})$. *Every finite family of generated support relations has a common Mal'tsev polymorphism.*

Proof. Let $\Delta \subseteq \{\Phi_F : F \in W_{\mathcal{F}}\}$ be finite. Empty relations may be removed because every operation preserves them. If nothing remains, fix $a_0 \in D$ and use

$$\varphi_0(a,b,c) = \begin{cases} c, & a = b, \\ a, & b = c, \\ a_0, & \text{otherwise.} \end{cases}$$

Thus assume that Δ contains a nonempty relation. For every $R \in \Delta$, fix $F_R \in W_{\mathcal{F}}$ such that $R = \Phi_{F_R}$, and write r_R for its arity.

The preservation constraints use only the input triples $(u_j^1, u_j^2, u_j^3) \in D^3$, where $R \in \Delta$, $\mathbf{u}^1, \mathbf{u}^2, \mathbf{u}^3 \in R$, and $j \in [r_R]$. Let $T \subseteq D^3$ be the set of these triples. For every $\tau \in T$, introduce a variable x_τ whose value in D is the candidate operation's output at τ. An assignment $\mathbf{p} = (p_\tau)_{\tau \in T} \in D^T$ therefore specifies a partial operation $p : T \to D$, with $x_\tau = p_\tau = p(\tau)$.

For every $R \in \Delta$ and $\mathbf{U} = (\mathbf{u}^1, \mathbf{u}^2, \mathbf{u}^3) \in R^3$, use the scope $\mathbf{x}[\mathbf{U}] = \big(x_{(u_j^1, u_j^2, u_j^3)}\big)_{j \in [r_R]}$. The constraints requiring preservation of every relation in Δ are

$$\Psi_\Delta((x_\tau)_{\tau \in T}) = \bigwedge_{R \in \Delta} \bigwedge_{\mathbf{U} \in R^3} \Phi_{F_R}(\mathbf{x}[\mathbf{U}]). \tag{F.46}$$

For a partial operation $p : T \to D$,

$$\Psi_\Delta(\mathbf{p}) \iff \forall R \in \Delta\ \forall \mathbf{u}^1, \mathbf{u}^2, \mathbf{u}^3 \in R, \quad \big(p(u_j^1, u_j^2, u_j^3)\big)_{j \in [r_R]} \in R.$$

These constraints require one common operation to preserve all relations in Δ. Every repeated input triple uses the same variable, so (F.46) is equality-free. An example appears in Appendix F.A.9.

Let $U_\Delta \subseteq D^T$ be the solution relation of Ψ_Δ. Every common ternary polymorphism of Δ restricts to an element of U_Δ. Conversely, any extension of a solution from T to D^3 preserves every relation in Δ: all input triples used in each preservation test belong to T, and the corresponding conjunct requires the resulting tuple to lie in the relation. Thus U_Δ consists exactly of these restrictions.

The Mal'tsev identities prescribe values on the following subsets of T:

$$A = \{(a,a,b) \in T : a, b \in D\},$$
$$C = \{(a,b,b) \in T : a, b \in D,\ a \neq b\},$$

Let $P = \mathrm{pr}_{A \cup C}(U_\Delta)$ be the relation obtained by restricting each solution to the inputs in $A \cup C$. The sets A and C are disjoint, and every constant triple in T lies in A. It suffices to find an element of P assigning b to each $(a,a,b) \in A$ and a to each $(a,b,b) \in C$, since membership in P guarantees

an extension to a common polymorphism. For each $k \in [3]$, let $\pi_k : D^3 \to D, \quad \pi_k(a_1, a_2, a_3) = a_k$, be the kth projection. It preserves every relation: for $R \in \Delta$ and $\mathbf{u}^1, \mathbf{u}^2, \mathbf{u}^3 \in R$,

$$\pi_k(\mathbf{u}^1, \mathbf{u}^2, \mathbf{u}^3) = \left(\pi_k(u_j^1, u_j^2, u_j^3)\right)_{j \in [r_R]} = \mathbf{u}^k \in R.$$

Hence

$$\pi_k|_T \in U_\Delta, \qquad \pi_k|_{A \cup C} \in P \qquad (k \in [3]). \tag{F.47}$$

Also, $A \neq \varnothing$: repeating any tuple of a nonempty positive-arity relation three times produces a constant triple in T.

Enumerate $C = \{c_1, \ldots, c_s\}$. For $0 \leq j \leq s$, we inductively construct a partial operation $p_j : A \cup C \to D$ satisfying the three invariants

$$\begin{aligned} p_j &\in P, \\ p_j(a, a, b) &= b && \text{for every } (a, a, b) \in A, \\ p_j(c_i) &= \pi_1(c_i) && \text{for every } i \in [j]. \end{aligned} \tag{F.48}$$

For $j = 0$, take $p_0 = \pi_3|_{A \cup C}$. Membership follows from (F.47), the first value condition follows from $\pi_3(a, a, b) = b$, and the second is vacuous.

Suppose $0 \leq j < s$ and the three invariants hold for p_j. To combine the values already prescribed by p_j with the required value at c_{j+1}, we first construct a common polymorphism of Δ that agrees with π_1 on A and at $c_1, \ldots, c_j$, and with p_j at c_{j+1}. Because $p_j \in P = \mathrm{pr}_{A \cup C}(U_\Delta)$, there exists a lift $\widehat{p}_j \in U_\Delta$ satisfying $\widehat{p}_j|_{A \cup C} = p_j$. This lift is effective: enumerate all maps $T \setminus (A \cup C) \to D$, combine each with the fixed values of p_j, and test (F.46). There are at most $q^{|T \setminus (A \cup C)|}$ candidates, and membership in P guarantees that one succeeds. Extend $\widehat{p}_j$ arbitrarily to a map $t_j : D^3 \to D$. The characterization of U_Δ shows that t_j is a common ternary polymorphism of Δ.

The minor $\widehat{t}_j(x, y, z) = t_j(x, y, y)$ is also a common ternary polymorphism: applying t_j to a first tuple and two copies of a second tuple in a relation still gives a tuple in that relation. Let $K_j = A \cup \{c_i : i \in [j]\}$. On K_j, $\widehat{t}_j$ agrees with π_1. For c_i with $i \in [j]$, the second and third entries coincide, so this follows from (F.48). If $(x, x, z) \in A$, then $(x, x, x) \in T$ by repeating the tuple that witnesses the occurrence of x. Since $(x, x, x) \in A$, $\widehat{t}_j(x, x, z) = t_j(x, x, x) = x = \pi_1(x, x, z)$.

We now impose the required value at c_{j+1} while keeping all values on K_j. Write $c_{j+1} = (a, b, b)$ and set

$$\begin{aligned} X = \pi_1|_{K_j} = \widehat{t}_j|_{K_j}, \qquad & Y = p_j|_{K_j}, \\ v = p_j(c_{j+1}) = \widehat{t}_j(c_{j+1}). & \end{aligned}$$

To apply Corollary F.7.3 with assignments on K_j as rows, let $Q_j = \mathrm{pr}_{K_j \cup \{c_{j+1}\}}(P)$, ordering the variable indexed by c_{j+1} last. This projection is defined by existentially quantifying the omitted variables in (F.46), so its pp definition is equality-free. Since $A \neq \varnothing$, its arity is at least two. The minor $\widehat{t}_j$ is a common polymorphism, so its restriction belongs to P. The same is true of p_j by the induction invariant and of π_1 by (F.47). Restricting $\widehat{t}_j$, p_j, and π_1 to $K_j \cup \{c_{j+1}\}$ therefore gives, respectively, the three corners

$$(X, v), \qquad (Y, v), \qquad (X, a) \in Q_j.$$

Figure F.1 summarizes both the finite lifting step and the missing-corner argument at this point of the induction.

By Corollary F.7.3, the two nonempty rows indexed by X and Y have equal support because they intersect at v. Consequently $(Y, a) \in Q_j$. Lifting this tuple to P gives $p_{j+1} \in P$ such that

$$\begin{aligned} p_{j+1}|_{K_j} &= Y = p_j|_{K_j}, \\ p_{j+1}(c_{j+1}) &= a = \pi_1(c_{j+1}). \end{aligned}$$

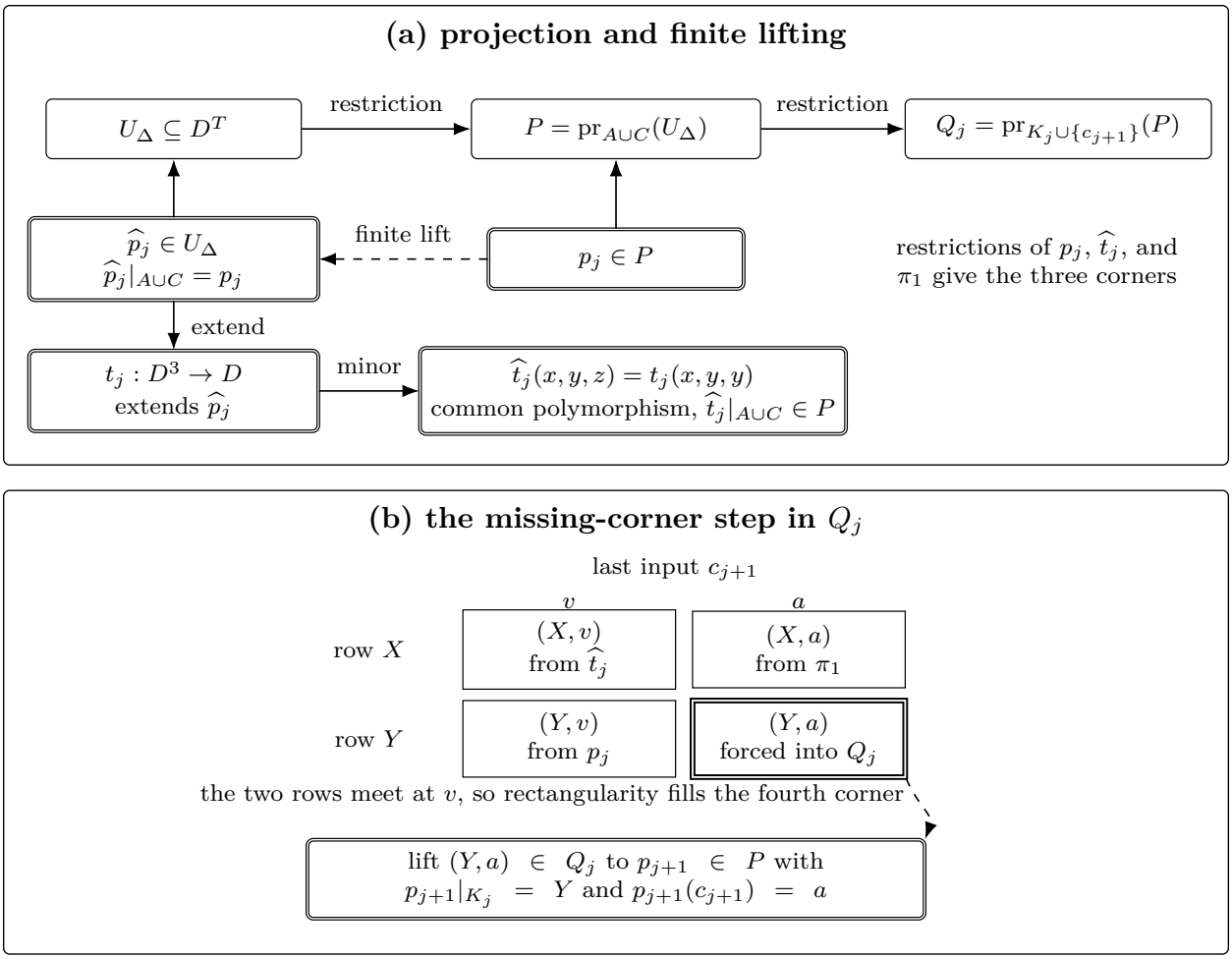


Figure F.1: One induction step in Lemma F.7.4. Top: membership $p_j \in P$ guarantees a lift $\widehat{p}_j \in U_\Delta$. Because all sets are finite, it is found by exhaustive search. Extending that lift and taking the displayed minor supplies the common polymorphism $\widehat{t}_j$. Solid arrows are restrictions, extensions, or minors, whereas dashed arrows are finite lifts. Bottom: in the row view of Q_j, the rows X and Y intersect at v. Singleton rectangularity makes their supports equal, so the double-bordered fourth corner (Y,a) must exist. Lifting that corner produces the next partial operation p_{j+1}.

Thus p_{j+1} retains the required values on $A, c_1, \ldots, c_j$ and has the required value at c_{j+1}, establishing all three invariants. This second lift is also a finite exact construction: enumerate $\widehat{p} \in D^T$ subject to

$$\widehat{p}|_{K_j} = Y, \qquad \widehat{p}(c_{j+1}) = a, \qquad \Psi_\Delta(\widehat{p}),$$

and take $p_{j+1} = \widehat{p}|_{A\cup C}$. Membership $(Y,a) \in Q_j$ guarantees that this search terminates.

At $j = s$, choose a lift $\widehat{p}_s \in U_\Delta$ with $\widehat{p}_s|_{A\cup C} = p_s$, using the same finite search. Its values on $A \cup C$ satisfy both Mal'tsev identities wherever their input triples lie in T. Extend it to D^3 by assigning the required values at triples of the forms (a,a,b) and (a,b,b) outside T, and arbitrary values elsewhere. The two prescriptions agree on their overlap, the constant triples. Denote the resulting operation by φ. Since every preservation constraint uses only inputs from T, φ preserves every relation in Δ. It satisfies

$$\varphi(a,b,b) = \varphi(b,b,a) = a \qquad (a,b \in D),$$

and is therefore a common Mal'tsev polymorphism of Δ. □

F.7.3 All generated supports

The operation in Lemma F.7.4 may depend on the finite family. To obtain one operation for all generated supports, we use the finiteness of the set of Mal'tsev operations on D: if no candidate preserved all supports, finitely many supports would exclude every candidate.

Theorem F.7.5 (Common Mal'tsev polymorphism of generated supports)**.** *If* $\mathsf{BO}(\mathcal{F})$ *holds, then*

$$\exists \varphi : D^3 \to D \quad \forall F \in W_{\mathcal{F}}, \quad \varphi \text{ is a Mal'tsev polymorphism of } \Phi_F. \tag{F.49}$$

Proof. Let $\mathcal{M}_q$ be the finite set of all Mal'tsev operations on D. By Lemma F.7.4, every finite family of generated supports is preserved by some member of $\mathcal{M}_q$.

Suppose, toward a contradiction, that no member of $\mathcal{M}_q$ preserves all generated supports. For every $\varphi \in \mathcal{M}_q$, choose $F_\varphi \in W_{\mathcal{F}}$ such that φ fails to preserve Φ_{F_φ}. The finite family $\Delta = \{\Phi_{F_\varphi} : \varphi \in \mathcal{M}_q\}$ has a common Mal'tsev polymorphism $\varphi^\star \in \mathcal{M}_q$ by Lemma F.7.4. But Δ contains $\Phi_{F_{\varphi^\star}}$, which was chosen not to be preserved by $\varphi^\star$, a contradiction. This proves (F.49). □

F.8 Realizing Row Equivalence as a Generated Support

Assume $\mathsf{BO}(\mathcal{F})$. Section F.7 gives one Mal'tsev operation preserving all generated supports. To show that the same operation preserves row equivalence, we construct, for each F, a generated function whose support is Ω_F. Its value at $(\mathbf{x}, \mathbf{y})$ will be a sum of products of positive powers of entries in the two rows indexed by $\mathbf{x}$ and $\mathbf{y}$. We first establish that powers remain generated and that the relative row phases are controlled by roots of unity. These facts, together with a power-sum argument, let us choose exponents for which the sum is nonzero exactly on pairs of nonzero, linearly dependent rows.

Let e_K be the exponent of the finite group $\mu(K)$, and define $L_q = \operatorname{lcm}(1, \dots, q)$. Since $-1 \in \mu(K)$ has order two, e_K is even and $e_K \geq 2$. Both integers are computable from the input, and

$$\xi^{e_K} = 1 \qquad \text{for every } \xi \in \mu(K). \tag{F.50}$$

For a function or vector H and $h \geq 1$, let $H^{\langle h \rangle}$ denote its entrywise hth power: each value $H(\mathbf{a})$ is replaced by $H(\mathbf{a})^h$. The angle brackets distinguish entrywise powers from matrix powers when a function is represented as a matrix.

Lemma F.8.1. *If* $F \in W_{\mathcal{F}}$ *and* $h \geq 1$*, then* $F^{\langle h \rangle} \in W_{\mathcal{F}}$*.*

Proof. Choose a finite $\mathcal{F}$-instance I with retained variables $\mathbf{a}$ and hidden variables $\mathbf{w}$ such that $F(\mathbf{a}) = \sum_{\mathbf{w} \in D^s} F_I(\mathbf{a}, \mathbf{w})$, where F_I is its instance function from (F.4). Take h copies of that instance, identify their retained variables $\mathbf{a}$, and make their internal tuples $\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(h)}$ pairwise fresh. Summing over them gives

$$\sum_{\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(h)}} \prod_{j \in [h]} F_I(\mathbf{a}, \mathbf{w}^{(j)}) = F(\mathbf{a})^h.$$

Thus $F^{\langle h \rangle} \in W_{\mathcal{F}}$. □

The closure under powers will keep the construction inside $W_{\mathcal{F}}$. We next compare the ratios between two overlapping rows. Dividing by the ratio at one nonzero entry separates a common scalar from the variation across entries. The next lemma shows that the remaining factors are roots of unity whose orders divide L_q. We can then choose powers that preserve these factors.

Lemma F.8.2. *Assume* $\mathsf{BO}(\mathcal{F})$. *Let* $F : D^r \to K$, $r \geq 2$, *belong to* $W_{\mathcal{F}}$, *and let* $\mathbf{u} = F(\mathbf{x}, *)$ *and* $\mathbf{v} = F(\mathbf{y}, *)$ *be nonzero rows with overlapping supports. Then their supports are equal. Fix any anchor* z_0 *in their common support* S, *and define*

$$\lambda := \frac{v_{z_0}}{u_{z_0}}, \qquad \zeta_z := \frac{v_z/u_z}{v_{z_0}/u_{z_0}} = \frac{v_z u_{z_0}}{u_z v_{z_0}} \qquad (z \in S). \tag{F.51}$$

Then $\lambda \in K^\times$ *and* $(\zeta_z)_{z\in S} \in \mu(K)^S$. *For every* $z \in S$,

$$v_z = \lambda \zeta_z u_z, \qquad \zeta_{z_0} = 1, \qquad \zeta_z^{L_q} = 1. \tag{F.52}$$

Proof. By Corollary F.3.2 and Corollary 1 of the Purification Lemma of Cai and Chen [CC17, Lemma 7 and Corollary 1], the original function F is block-orthogonal. Hence overlapping nonzero rows have equal supports and proportional magnitude vectors.

By Lemma F.8.1, $F^{\langle h\rangle} \in W_{\mathcal{F}}$ for every $h \geq 1$. Applying the same corollary again shows that each original function $F^{\langle h\rangle}$ is block-orthogonal. We show that some $1 \leq h \leq q$ makes the two powered rows linearly dependent, so their entry ratios have equal hth powers. Suppose, toward a contradiction, that $\mathbf{u}^{\langle h\rangle}$ and $\mathbf{v}^{\langle h\rangle}$ are linearly independent for every $1 \leq h \leq q$. Fix a nonempty magnitude block B of $\mathbf{u}$. Positive powering is injective on positive real numbers, so B remains the same magnitude block in every $\mathbf{u}^{\langle h\rangle}$. Block orthogonality of $F^{\langle h\rangle}$ gives the following power sums. Put $b = |B|$, $\quad x_z = u_z\overline{v_z} \quad (z \in B)$. Because B lies in the common support, every x_z is nonzero. Also $b \leq q$, and hence

$$p_h := \sum_{z\in B} x_z^h = \sum_{z\in B} u_z^h \overline{v_z^h} = 0, \qquad h = 1, \ldots, b.$$

Enumerate the x_z, $z \in B$, as $x_1, \ldots, x_b$, and let

$$e_k = \sum_{1\leq i_1<\cdots<i_k\leq b} x_{i_1} \cdots x_{i_k}, \qquad e_0 = 1,$$

be their elementary symmetric polynomials. Newton's identities state

$$k e_k = \sum_{j=1}^{k} (-1)^{j-1} e_{k-j} p_j \qquad (k = 1, \ldots, b). \tag{F.53}$$

Since the ambient field has characteristic zero, each integer $1 \leq k \leq b$ is invertible. As $p_1 = \cdots = p_b = 0$, induction on k in (F.53) gives $e_1 = \cdots = e_b = 0$. In particular,

$$0 = e_b = x_1 \cdots x_b = \prod_{z\in B} u_z\overline{v_z},$$

contradicting the fact that every factor is nonzero.

Consequently, $\mathbf{u}^{\langle h\rangle}$ and $\mathbf{v}^{\langle h\rangle}$ are dependent for some $1 \leq h \leq q$. Thus $(v_z/u_z)^h$ is independent of $z \in S$. By (F.51), for every $z \in S$,

$$\zeta_z^h = \frac{(v_z/u_z)^h}{(v_{z_0}/u_{z_0})^h} = 1.$$

Since h divides L_q, (F.52) follows. All ratios in (F.51) belong to $K^\times$, so each $\zeta_z \in \mu(K)$. The same display gives $\lambda \in K^\times$ and $\zeta_{z_0} = 1$. □

The remaining ingredient concerns cancellation on linearly dependent rows. After grouping equal exponential bases in the sum, we will use the following consequence of the Skolem–Mahler–Lech theorem to ensure that the resulting sum is nonzero for all sufficiently large values of its integer parameter.

Lemma F.8.3. *Let $s \geq 1$, let $a_1, \ldots, a_s, b_1, \ldots, b_s$ be nonzero algebraic numbers, and suppose b_i/b_j is not a root of unity whenever $i \neq j$. Then*

$$p(t) = \sum_{i \in [s]} a_i b_i^t$$

equals zero for only finitely many integers $t \geq 0$.

Proof. Let E be the shift operator from the proof of Lemma F.7.1. The operator $\prod_i (E - b_i)$ annihilates $p(t)$. The Skolem–Mahler–Lech theorem therefore says that its zero set is a finite set together with finitely many arithmetic progressions. Suppose one such progression is $\{r + mn : n \geq N\}$, where $m \geq 1$. Then

$$0 = p(r + mn) = \sum_{i \in [s]} (a_i b_i^r)(b_i^m)^n$$

for $n = N, N+1, \ldots, N+s-1$. The bases b_i^m are pairwise distinct because no b_i/b_j is a root of unity. The resulting Vandermonde system is invertible and forces $(a_i b_i^r)(b_i^m)^N = 0$ for every i, contradicting the nonzero hypotheses. Hence no infinite progression occurs. □

We now combine these facts in a function whose support is Ω_F. Taking $n = 1 + tL_q$, with $t \geq 0$, preserves the relative factors ζ_z in (F.52). The two row exponents below sum to $e_K n$, which removes root-of-unity ratios within each purified magnitude block. Block Orthogonality then makes the sum zero on independent rows. For nonzero dependent rows, Lemma F.8.3 will make it nonzero for all sufficiently large t. Copies of the original instance realize this function. We call the resulting function C_n the *row-equivalence detector*.

Theorem F.8.4 (Row-equivalence realization)**.** *Assume $\mathsf{BO}(\mathcal{F})$. Let $F : D^r \to K$, $r \geq 2$, be any member of $W_{\mathcal{F}}$. There is an integer $t \geq 0$ such that, for $n = 1 + tL_q$, the $2(r-1)$-ary function*

$$C_n(\mathbf{x}, \mathbf{y}) = \sum_{z \in D} F(\mathbf{x}, z)^{(e_K - 1)n} F(\mathbf{y}, z)^n \tag{F.54}$$

belongs to $W_{\mathcal{F}}$ and satisfies

$$\Phi_{C_n} = \Omega_F. \tag{F.55}$$

Moreover, there is an algorithm that, given the exact values of F and an instance I with $F = F_I^{[r]}$, computes such integers t, n and an instance J satisfying $C_n = F_J^{[2(r-1)]}$, with retained variables ordered as $(\mathbf{x}, \mathbf{y})$.

Proof. Fix an allowed exponent $n = 1 + tL_q$. Both exponents in (F.54) are positive. Let $\mathbf{u} = F(\mathbf{x}, *)$ and $\mathbf{v} = F(\mathbf{y}, *)$. If one row is zero, or if their supports are disjoint, every summand is zero. Suppose henceforth that the rows are nonzero and overlap. The global Block Orthogonality condition and support invariance under purification imply that their supports agree and that their purified magnitude rows are proportional. Write their common support as $S = \mathrm{supp}(\mathbf{u}) = \mathrm{supp}(\mathbf{v})$.

First suppose that $\mathbf{u}$ and $\mathbf{v}$ are linearly independent. Fix a legal singleton purification of F, and let B be one magnitude block of the purified row corresponding to $\mathbf{u}$. The two purified magnitude

rows are proportional, so B is simultaneously a magnitude block of the purified row corresponding to $\mathbf{v}$. Choose $z_B \in B$. By Lemma F.3.5, the anchor ratios

$$\eta_z = u_z/u_{z_B}, \qquad \theta_z = v_z/v_{z_B} \qquad (z \in B) \tag{F.56}$$

belong to $\mu(K)$. Purification preserves complex row dependence, so the two purified rows remain independent and are block-orthogonal. Root covariance then turns the Hermitian sum on B into

$$\sum_{z\in B} \eta_z \overline{\theta_z} = 0. \tag{F.57}$$

Apply Lemma F.8.2 with the same anchor z_B: $v_z = \lambda\zeta_z u_z$, where $\zeta_{z_B} = 1$ and $\zeta_z^{L_q} = 1$. Taking ratios gives $\theta_z = \zeta_z\eta_z$. Substitution in (F.57), followed by conjugation, yields

$$\sum_{z\in B} \zeta_z = 0.$$

Equation (F.56) and (F.50) give $u_z^{e_K} = u_{z_B}^{e_K}$. Since $n \equiv 1 \pmod{L_q}$, also $\zeta_z^n = \zeta_z$. The contribution of B to $C_n(\mathbf{x}, \mathbf{y})$ is therefore

$$\sum_{z\in B} u_z^{(e_K-1)n} v_z^n = \lambda^n \sum_{z\in B} \zeta_z^n u_z^{e_K n} = \lambda^n u_{z_B}^{e_K n} \sum_{z\in B} \zeta_z = 0. \tag{F.58}$$

The purified magnitude blocks partition the common support. Write $S = B_1 \dot\cup \cdots \dot\cup B_N$. Terms outside S are zero, and (F.58) applies to every B_i. Therefore

$$C_n(\mathbf{x}, \mathbf{y}) = \sum_{z\in S} u_z^{(e_K-1)n} v_z^n = \sum_{i\in[N]} \sum_{z\in B_i} u_z^{(e_K-1)n} v_z^n = 0.$$

Thus the detector equals zero on every overlapping independent ordered row pair for every allowed n.

It remains to treat a nonzero dependent pair, say $\mathbf{v} = \lambda\mathbf{u}$ with $\lambda \neq 0$. Using the common support S, we have

$$C_n(\mathbf{x}, \mathbf{y}) = \lambda^n \sum_{z\in S} u_z^{e_K n}.$$

Along $n = 1 + tL_q$, put $a_z = u_z^{e_K}$ and $b_z = u_z^{e_K L_q}$. The remaining sum is

$$p_{\mathbf{x},\mathbf{y}}(t) = \sum_{z\in S} a_z b_z^t. \tag{F.59}$$

To apply Lemma F.8.3, group the indices $z \in S$ having the same base b_z. We check that the grouped coefficients are nonzero and that the distinct bases have no root-of-unity quotients. If $b_z = b_w$, then $u_z/u_w \in \mu(K)$, so $a_z = a_w$ by (F.50). Thus each grouped coefficient is a positive integer multiple of one nonzero a_z. Moreover, the quotient of two distinct grouped bases is not a root of unity: otherwise a power of u_z/u_w would be one, so $u_z/u_w \in \mu(K)$ and the two bases would have been equal. After grouping, Lemma F.8.3 shows that (F.59) has only finitely many zeros.

There are finitely many ordered dependent nonzero row pairs in F. Choose t outside the union of their finite exceptional sets. Then the detector is nonzero on every member of Ω_F, whereas the zero-row, disjoint-support, and independent cases show that it equals zero on every nonmember. This proves (F.55).

The witness t is found by exact search. Compute Ω_F exactly: a row is nonzero if one entry is nonzero, and two nonzero rows are dependent exactly when

$$F(\mathbf{x}, z)F(\mathbf{y}, w) - F(\mathbf{x}, w)F(\mathbf{y}, z) = 0 \qquad (z, w \in D).$$

Enumerate $t = 0, 1, 2, \ldots$, evaluate (F.54) by exact algebraic arithmetic, and stop when its support equals Ω_F. The finite-exception argument proves termination.

Finally, we realize the two powers in (F.54) by copies of an instance representing F. Choose a finite $\mathcal{F}$-instance I with retained variables $(\mathbf{x}, z)$ and hidden variables $\mathbf{h}$ such that $F(\mathbf{x}, z) = \sum_{\mathbf{h} \in D^s} F_I(\mathbf{x}, z, \mathbf{h})$, where F_I is its instance function. Take $(e_K - 1)n$ copies sharing $(\mathbf{x}, z)$ and n copies sharing $(\mathbf{y}, z)$, with all copied hidden variables distinct. For fixed $(\mathbf{x}, \mathbf{y}, z)$, summing over these two collections of hidden variables gives

$$\left(\sum_{\mathbf{h}} F_I(\mathbf{x}, z, \mathbf{h})\right)^{(e_K-1)n} = F(\mathbf{x}, z)^{(e_K-1)n},$$

$$\left(\sum_{\mathbf{k}} F_I(\mathbf{y}, z, \mathbf{k})\right)^{n} = F(\mathbf{y}, z)^{n}.$$

Because the collections are disjoint, their contributions multiply. Summing over z now gives exactly $C_n(\mathbf{x}, \mathbf{y})$. Thus $C_n \in W_{\mathcal{F}}$. □

The realization theorem allows us to use the operation from Section F.7 for row equivalence as well. The following consequence applies to any operation preserving all generated supports.

Corollary F.8.5 (Support preservation extends to all row relations)**.** *Assume* $\mathsf{BO}(\mathcal{F})$*. If one operation* $\varphi : D^3 \to D$ *preserves* Φ_H *for every* $H \in W_{\mathcal{F}}$*, then the same operation preserves* Ω_F *for every* $F \in W_{\mathcal{F}}$ *of arity at least two.*

Proof. Fix the operation φ from the hypothesis, and let $F \in W_{\mathcal{F}}$ have arity at least two. By Theorem F.8.4, choose a row-equivalence detector $C_n \in W_{\mathcal{F}}$ with $\Phi_{C_n} = \Omega_F$. By hypothesis, φ preserves Φ_{C_n}, and hence Ω_F. Since F was arbitrary, this same φ preserves every generated row-equivalence relation. □

F.9 Type Partition and the Structural Collapse

Under $\mathsf{BO}(\mathcal{F})$, we have a common Mal'tsev operation preserving all generated supports, and Corollary F.8.5 shows that it also preserves all row-equivalence relations. It remains to derive the Type Partition condition. The next lemma does this for one function: preservation of row equivalence and the Mal'tsev identities transfer a row-equivalence class reached from one prefix to any same-length prefix whose type overlaps it.

For equal-length tuples $\mathbf{u}, \mathbf{v}, \mathbf{w}$, we write $\varphi(\mathbf{u}, \mathbf{v}, \mathbf{w})$ for coordinatewise application of φ.

Lemma F.9.1. *Let* $F : D^r \to \mathbb{C}$*,* $r \geq 2$*, and let* $\varphi : D^3 \to D$ *be a Mal'tsev polymorphism of* Ω_F*. Then the type map of* F *is a type-partition map.*

Proof. Write $\mathrm{Row}(F) = \{(S_1, \mathbf{v}_1), \ldots, (S_s, \mathbf{v}_s)\}$. Fix $0 \leq \ell \leq r - 1$ and $\boldsymbol{\alpha}, \boldsymbol{\beta} \in D^\ell$. Suppose their types intersect. Choose $j \in \mathrm{type}_F(\boldsymbol{\alpha}) \cap \mathrm{type}_F(\boldsymbol{\beta})$ and rows $\mathbf{a}, \mathbf{b} \in S_j$ extending $\boldsymbol{\alpha}, \boldsymbol{\beta}$, respectively.

Let $h \in \mathrm{type}_F(\boldsymbol{\alpha})$ be arbitrary. To show that $h \in \mathrm{type}_F(\boldsymbol{\beta})$, we construct a row index in S_h extending $\boldsymbol{\beta}$. Choose $\mathbf{c} \in S_h$ extending $\boldsymbol{\alpha}$. The three $2(r-1)$-tuples

$$(\mathbf{c}, \mathbf{c}), \qquad (\mathbf{a}, \mathbf{a}), \qquad (\mathbf{b}, \mathbf{a})$$

belong to Ω_F: the first two are diagonal pairs of nonzero rows, and the third belongs because $\mathbf{a}$ and $\mathbf{b}$ lie in the same row class S_j. Put $\mathbf{d} = \varphi(\mathbf{c}, \mathbf{a}, \mathbf{b})$. Applying φ coordinatewise to these three members of Ω_F gives

$$\varphi((\mathbf{c}, \mathbf{c}), (\mathbf{a}, \mathbf{a}), (\mathbf{b}, \mathbf{a})) = (\varphi(\mathbf{c}, \mathbf{a}, \mathbf{b}), \varphi(\mathbf{c}, \mathbf{a}, \mathbf{a})) = (\mathbf{d}, \mathbf{c}),$$

where the last equality applies the Mal'tsev identity $\varphi(x, y, y) = x$ at every tuple position. Preservation therefore gives $(\mathbf{d}, \mathbf{c}) \in \Omega_F$. For every $i \in [\ell]$, entry i of the first tuple is $\varphi(\alpha_i, \alpha_i, \beta_i) = \beta_i$. Thus the row indexed by $\mathbf{d}$ extends $\boldsymbol{\beta}$. Membership in Ω_F also says that this row is nonzero and belongs to the same row class S_h as $\mathbf{c}$. Therefore $h \in \mathrm{type}_F(\boldsymbol{\beta})$. Since h was arbitrary, $\mathrm{type}_F(\boldsymbol{\alpha}) \subseteq \mathrm{type}_F(\boldsymbol{\beta})$. Interchanging the prefixes proves the reverse inclusion. Thus any two types are equal or disjoint. Empty types and the zero function are covered as well. □

Applying Lemma F.9.1 with the common operation from Theorem F.7.5 and Corollary F.8.5 gives the Type Partition condition for every generated function. The same operation satisfies the Mal'tsev polymorphism condition, yielding the following implication.

Theorem F.9.2 (Structural collapse)**.** *For every $q \geq 1$ and every finite set $\mathcal{F}$ of algebraic complex-valued functions on D,*

$$\mathsf{BO}(\mathcal{F}) \quad \Longrightarrow \quad \mathsf{TP}(\mathcal{F}) \wedge \mathsf{Mal}(\mathcal{F}).$$

More explicitly, under $\mathsf{BO}(\mathcal{F})$ there is one Mal'tsev operation $\varphi : D^3 \to D$ such that, for every $F \in W_{\mathcal{F}}$,

$$\varphi \text{ preserves } \Phi_F, \qquad \mathrm{arity}(F) \geq 2 \implies \varphi \text{ preserves } \Omega_F.$$

Consequently,

$$\mathsf{CC}(\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{BO}(\mathcal{F}).$$

Proof. The case $\mathcal{F} = \varnothing$ was handled in Section F.2.5. Assume $\mathcal{F} \neq \varnothing$. Assume $\mathsf{BO}(\mathcal{F})$. By Theorem F.7.5, there is a common Mal'tsev polymorphism $\varphi : D^3 \to D$ of the relations Φ_H, $H \in W_{\mathcal{F}}$.

By Corollary F.8.5, the same φ preserves every Ω_F with $F \in W_{\mathcal{F}}$ of arity at least two. It therefore preserves all relations in $\Lambda_{\mathcal{F}}$, proving $\mathsf{Mal}(\mathcal{F})$.

Applying Lemma F.9.1 to every function in $W_{\mathcal{F}}$ of arity at least two proves $\mathsf{TP}(\mathcal{F})$. The argument applies in every ordering of variables. Relabelling the retained variables preserves membership in $W_{\mathcal{F}}$.

We have proved $\mathsf{BO}(\mathcal{F}) \Rightarrow \mathsf{TP}(\mathcal{F}) \wedge \mathsf{Mal}(\mathcal{F})$. The reverse implication in the final equivalence is immediate because $\mathsf{BO}(\mathcal{F})$ is a conjunct of $\mathsf{CC}(\mathcal{F})$. □

F.9.1 Proof of uniform decidability

The structural implication identifies the full tractability criterion with the Block Orthogonality condition. We can therefore apply the decision procedure from Section F.6 to decide that criterion, completing the proof of Theorem F.1.1.

Proof of Theorem F.1.1. On input $(q, \mathcal{F})$, run Procedure F.6.1. By Theorem F.6.2, this is a total exact procedure and returns YES if and only if $\mathsf{BO}(\mathcal{F})$ holds. By Theorem F.9.2, $\mathsf{BO}(\mathcal{F}) \iff \mathsf{CC}(\mathcal{F})$. Consequently, the same procedure halts on every input and returns YES exactly when $\mathsf{CC}(\mathcal{F})$ holds, proving both uniform decidability and (F.2). □

F.10 Extension to #CSP^d

We now prove Corollary F.1.2, using the definitions of $\#\mathsf{CSP}^d(\mathcal{F})$, $W^d_{\mathcal{F}}$, and the restricted tractability conditions from Sections F.2.2 and F.2.5. There are two tasks. First, we need an identity test over the degree-restricted instances. A root-of-unity filter will reduce it to the character comparisons from Section F.5. Second, we must verify that arity compression and the support and row-equivalence constructions preserve the degree restriction. These two steps extend the decision procedure and the structural implication to $W^d_{\mathcal{F}}$.

F.10.1 Universal identities on $W^d_{\mathcal{F}}$

Fix $k, d \geq 1$. Let $\mathfrak{M}_{k,d}$ be the submonoid of $\mathfrak{M}_k$ consisting of the labelled instances in which every variable has occurrence degree divisible by d. It is a submonoid: gluing labelled instances adds the degrees of identified variables, and the constraint-free labelled unit has degree zero at every variable. The same labelling and summation argument as in Lemma F.5.1 therefore gives

$$\{\mathbf{z}_I : I \in \mathfrak{M}_{k,d}\} = \{F \in W^d_{\mathcal{F}} : \mathrm{arity}(F) = k\}.$$

To enforce the restriction, we will use an auxiliary sum that is zero whenever some variable has degree not divisible by d. The following identity supplies this test for each variable. Write $C_d = \mathbb{Z}/d\mathbb{Z}$ and choose a primitive dth root of unity ω_d (with $\omega_1 = 1$). For fixed $n \in \mathbb{Z}$, the map $s \mapsto \omega_d^{sn}$ is a character of the additive group C_d. It is the constant-one character exactly when $d \mid n$. Otherwise its values form a nontrivial geometric progression whose sum is zero. Thus

$$\sum_{s \in C_d} \omega_d^{sn} = \begin{cases} d, & d \mid n, \\ 0, & d \nmid n. \end{cases} \tag{F.60}$$

To compare monomial characters after restriction to $\mathfrak{M}_{k,d}$, we use a root-of-unity filter that reduces the problem to finitely many ordinary character comparisons on $\mathfrak{M}_k$.

Lemma F.10.1. *Fix $k, d \geq 1$. Given two monomials M, N in the formal variables $\{X_{\mathbf{a}}, Y_{\mathbf{a}} : \mathbf{a} \in D^k\}$ and their associated characters $\chi_M, \chi_N : \mathfrak{M}_k \to K$, it is decidable whether*

$$\chi_M(I) = \chi_N(I) \qquad \text{for every } I \in \mathfrak{M}_{k,d}. \tag{F.61}$$

Proof. We construct two finite sums of characters on $\mathfrak{M}_k$. At each $I \in \mathfrak{M}_{k,d}$, their values will be $\chi_M(I)$ and $\chi_N(I)$ multiplied by the same nonzero factor. Both sums will be zero on the remaining instances. Their equality on the full monoid will then decide the desired equality on the restricted instances.

Use the tensorized families $\mathcal{H}_M = (h_f : f \in \mathcal{F})$ and $\mathcal{H}_N = (g_f : f \in \mathcal{F})$ constructed from M and N in Section F.5.3, on the domains D_M and D_N. Take the respective pins $\beta_M : [k] \to D_M$ and $\beta_N : [k] \to D_N$ from Definition F.5.3. For an original labelled $\mathcal{F}$-instance $I \in \mathfrak{M}_k$, let I^M and I^N denote its interpretations by these two families. By Lemma F.5.4,

$$Z^{\mathcal{H}_M}_{\beta_M}(I^M) = \chi_M(I), \qquad Z^{\mathcal{H}_N}_{\beta_N}(I^N) = \chi_N(I). \tag{F.62}$$

Attach an auxiliary value $s \in C_d$ to each variable. Each occurrence of that variable will contribute a factor ω_d^s, so multiplying over constraints will record its occurrence degree in the exponent. Let $K_d = K(\omega_d)$. For an input function $f \in \mathcal{F}$ of arity r_f, define the everywhere-nonzero function

$$u_f(s_1, \ldots, s_{r_f}) = \omega_d^{s_1 + \cdots + s_{r_f}}.$$

Augment h_f and g_f to functions over $D_M \times C_d$ and $D_N \times C_d$, respectively, by

$$h_f^{(d)}((a_1, s_1), \ldots, (a_{r_f}, s_{r_f})) = h_f(a_1, \ldots, a_{r_f}) u_f(s_1, \ldots, s_{r_f}).$$

Define $g_f^{(d)}$ by the same formula with g_f in place of h_f. The superscript (d) records the added cyclic factor. For $\mathbf{s} = (s_1, \ldots, s_k) \in C_d^k$, interpret an original instance I using the $h_f^{(d)}$-family and pin label t to $(\beta_M(t), s_t)$. Denote the resulting pinned value by $\rho_{\mathbf{s}}^M(I)$. Define $\rho_{\mathbf{s}}^N(I)$ analogously using the $g_f^{(d)}$-family and the pins $(\beta_N(t), s_t)$. Gluing labelled instances multiplies these pinned partition values, and the constraint-free labelled unit has value one. Thus every $\rho_{\mathbf{s}}^M, \rho_{\mathbf{s}}^N : \mathfrak{M}_k \to K_d$ is a unital multiplicative character.

We now compute the effect of summing over the auxiliary values. Fix $I \in \mathfrak{M}_k$. For an assignment $\theta : V(I) \to C_d$ to the cyclic factor, the product of all auxiliary factors is

$$\begin{aligned} &\prod_{(f; v_1, \ldots, v_{r_f}) \in \mathcal{C}_I} u_f(\theta(v_1), \ldots, \theta(v_{r_f})) \\ &\quad = \omega_d^{\sum_{(f; v_1, \ldots, v_{r_f}) \in \mathcal{C}_I} \sum_{j \in [r_f]} \theta(v_j)} \\ &\quad = \omega_d^{\sum_{v \in V(I)} \deg_I(v) \theta(v)}. \end{aligned} \tag{F.63}$$

Each occurrence of v contributes $\theta(v)$ to the exponent, including repeated occurrences within one scope. Its total contribution is therefore $\deg_I(v)\theta(v)$, where $\deg_I(v)$ is its occurrence degree.

Summing $\rho_{\mathbf{s}}^M(I)$ over all cyclic label pins $\mathbf{s} \in C_d^k$ is equivalent to summing θ over all cyclic assignments to $V(I)$. The assignments to the tensorized domain and to the cyclic group are independent, so Equations (F.62) and (F.63) give the following identity, where $\mathbf{1}_E$ is 1 when condition E holds and 0 otherwise:

$$\begin{aligned} \sum_{\mathbf{s} \in C_d^k} \rho_{\mathbf{s}}^M(I) &= \chi_M(I) \sum_{\theta : V(I) \to C_d} \omega_d^{\sum_v \deg_I(v)\theta(v)} \\ &= \chi_M(I) \prod_{v \in V(I)} \left(\sum_{t \in C_d} \omega_d^{t \deg_I(v)} \right) \\ &= d^{|V(I)|} \mathbf{1}_{\{I \in \mathfrak{M}_{k,d}\}} \chi_M(I). \end{aligned} \tag{F.64}$$

The last equality applies (F.60) separately at every variable. The identical calculation for N gives

$$\sum_{\mathbf{s} \in C_d^k} \rho_{\mathbf{s}}^N(I) = d^{|V(I)|} \mathbf{1}_{\{I \in \mathfrak{M}_{k,d}\}} \chi_N(I). \tag{F.65}$$

These two sums have the required behavior on and off $\mathfrak{M}_{k,d}$. Define them as functions on the full monoid $\mathfrak{M}_k$ by

$$R_M = \sum_{\mathbf{s} \in C_d^k} \rho_{\mathbf{s}}^M, \qquad R_N = \sum_{\mathbf{s} \in C_d^k} \rho_{\mathbf{s}}^N. \tag{F.66}$$

If $I \notin \mathfrak{M}_{k,d}$, both functions equal zero at I by Equations (F.64) and (F.65). If $I \in \mathfrak{M}_{k,d}$, the same equations give

$$R_M(I) = d^{|V(I)|} \chi_M(I), \qquad R_N(I) = d^{|V(I)|} \chi_N(I).$$

The common factor is nonzero because K_d has characteristic zero. Consequently,

$$\chi_M = \chi_N \text{ on } \mathfrak{M}_{k,d} \quad \Longleftrightarrow \quad R_M = R_N \text{ on } \mathfrak{M}_k. \tag{F.67}$$

It remains to compare the two sums on $\mathfrak{M}_k$. There are $2d^k$ character terms in (F.66). Apply Lemma F.5.6 to their augmented indexed families and fixed pins. At positive tensor degree, $T_{p_+,p_-}(f) \equiv 0$ if and only if $f \equiv 0$. At degree zero it is the constant-one function. Each tensorized family therefore has at most one zero function of each arity. Since u_f is everywhere nonzero, augmenting the functions preserves whether each is identically zero. The hypotheses of the comparison lemma hold, and it decides every pairwise equality.

Group these character terms by equality as functions on $\mathfrak{M}_k$, retaining their multiplicities in the two sums. Let $\mathcal{E}_{\mathrm{char}}$ be the resulting finite set of classes. For each class E, count the cyclic pins contributing a character in E on each side:

$$m_M(E) = |\{\mathbf{s} \in C_d^k : \rho_{\mathbf{s}}^M \in E\}|,$$
$$m_N(E) = |\{\mathbf{s} \in C_d^k : \rho_{\mathbf{s}}^N \in E\}|.$$

Choose one representative character ρ_E from each $E \in \mathcal{E}_{\mathrm{char}}$. Then, as functions on $\mathfrak{M}_k$,

$$R_M - R_N = \sum_{E \in \mathcal{E}_{\mathrm{char}}} (m_M(E) - m_N(E))\rho_E.$$

The representatives ρ_E, $E \in \mathcal{E}_{\mathrm{char}}$, are pairwise distinct. By Lemma F.5.7,

$$R_M = R_N \quad \Longleftrightarrow \quad m_M(E) = m_N(E) \text{ for every } E \in \mathcal{E}_{\mathrm{char}}. \tag{F.68}$$

Combining Equations (F.67) and (F.68) gives a finite exact decision procedure for (F.61). □

We can now group the monomials of a polynomial according to equality of their characters on $\mathfrak{M}_{k,d}$. As in Theorem F.5.8, linear independence reduces the identity test on $W_{\mathcal{F}}^d$ to testing the coefficient sum of each group.

Theorem F.10.2 (Universal identity oracle for $W_{\mathcal{F}}^d$)**.** *For every $k, d \geq 1$ and every polynomial $P \in \mathbb{C}[X_{\mathbf{a}}, Y_{\mathbf{a}} : \mathbf{a} \in D^k]$, given as a finite list of monomials with exactly encoded algebraic-complex coefficients, it is decidable whether*

$$P(\mathbf{z}_I, \overline{\mathbf{z}_I}) = 0 \qquad \textit{for every } I \in \mathfrak{M}_{k,d}.$$

Proof. Each monomial of P defines a unital multiplicative character on $\mathfrak{M}_{k,d}$. By Lemma F.10.1, equality of any two such characters is decidable. Group the finitely many monomials according to this equality and sum their coefficients in the number field generated by the input coefficients and K. Applying Lemma F.5.7 to the monoid $\mathfrak{M}_{k,d}$ shows that the identity holds exactly when every class sum is zero. All steps are finite and exact. □

F.10.2 Proof of uniform decidability

The restricted identity oracle supplies the new algorithmic ingredient. We now check that the remaining steps of the Block Orthogonality decision procedure and the constructions establishing the other two conditions stay within $W_{\mathcal{F}}^d$. In each instance construction, fresh internal variables retain their degrees and identified retained variables receive the sum of their degrees.

Proof of Corollary F.1.2. Fix $d \geq 1$. The proof of Corollary F.3.2 restricts verbatim to $W_{\mathcal{F}}^d$: $\mathsf{BO}^d(\mathcal{F})$ holds if and only if every member of $W_{\mathcal{F}}^d$ of arity at least two has a block-orthogonal singleton purification. At arity k, use the labelled submonoid $\mathfrak{M}_{k,d}$ and the oracle in Theorem F.10.2. The purification-invariance and finite-certificate arguments concern one function on a finite domain and

are unchanged. The arity compression also remains valid: identifying retained variables replaces their occurrence degrees by their sum, which is still divisible by d. Hence Procedure F.6.1, using the oracle for $W_{\mathcal{F}}^{d}$, is a total exact decision procedure for $\mathsf{BO}^{d}(\mathcal{F})$.

The structural proof likewise remains inside $W_{\mathcal{F}}^{d}$. More explicitly, every instance substitution used in Lemmas F.7.2 and F.8.1 and Theorem F.8.4 takes copies of $\#\mathsf{CSP}^{d}(\mathcal{F})$ instances, keeps their hidden variables fresh, and identifies some retained variables. A fresh variable retains a degree divisible by d, while an identified variable receives a sum of such degrees. Labelled products, diagonal minors, variable permutations, and repeated retained variables obey the same rule. Thus all functions realizing supports and row-equivalence detectors used in Sections F.7 to F.9 are members of $W_{\mathcal{F}}^{d}$. Applying the arguments of Theorems F.7.5 and F.9.2 to these restricted constructions gives

$$\mathsf{BO}^{d}(\mathcal{F}) \Longrightarrow \mathsf{TP}^{d}(\mathcal{F}) \wedge \mathsf{Mal}^{d}(\mathcal{F}),$$

with a common Mal'tsev polymorphism of all support and row-equivalence relations associated with $W_{\mathcal{F}}^{d}$. Since $\mathsf{BO}^{d}(\mathcal{F})$ is a conjunct of $\mathsf{CC}^{d}(\mathcal{F})$, this proves the claimed equivalence and uniform decidability of $\mathsf{CC}^{d}(\mathcal{F})$. Together with the dichotomy theorem for $\#\mathrm{CSP}^{d}$ proved by Lin [Lin21, Theorem 3], this decides the polynomial-time versus $\#\mathsf{P}$-hard alternative for $\#\mathsf{CSP}^{d}(\mathcal{F})$. □

F.11 Conclusion

We have given a total exact decision procedure for the three tractability conditions characterizing the polynomial-time side of the complex-weighted #CSP dichotomy theorem of Cai and Chen [CC17, Theorem 1 and Section 3.1]. The procedure decides these conditions on $W_{\mathcal{F}}$. Its finite reduction combines the arity bound $q^2 + 1$ with universal identity testing over all finite labelled instances.

The global Block Orthogonality condition also implies the Type Partition and Mal'tsev polymorphism conditions, with one common Mal'tsev operation preserving all generated support and row-equivalence relations. The support realization and row-equivalence constructions stay within $W_{\mathcal{F}}$ and control cancellation so that their supports are exactly the intended relations.

A cyclic root-of-unity degree filter and degree-preserving structural constructions extend uniform decidability and structural collapse to the degree-divisibility restriction $\#\mathrm{CSP}^{d}$ studied by Lin [Lin21, Definition 2 and Theorem 3]. Thus $\mathsf{CC}^{d}(\mathcal{F})$, the tractability criterion in that dichotomy theorem, is uniformly decidable for every positive degree modulus d.

F.A Examples

The following examples illustrate the tractability conditions in Section F.2, the variable-identification constructions in Lemmas F.3.3 and F.3.4, and the tensor realization used in Section F.5. The final two examples illustrate support realization and the constraints defining preserving operations in Section F.7.

F.A.1 Block-rank one

Take $q = 4$ and let the binary function A be

$x\backslash z$	0	1	2	3
0	1	2	0	0
1	3	6	0	0
2	0	0	1	4
3	0	0	2	8

.

For $\mathcal{F}_A = \{A\}$, the instance consisting of the single constraint $A(x, z)$ has $F_I^{[2]} = A$, so $A \in W_{\mathcal{F}_A}$. The first two magnitude rows are proportional, as are the last two, while rows from different pairs have disjoint supports. Equivalently, the support is the disjoint union of the rectangles $\{0,1\} \times \{0,1\}$ and $\{2,3\} \times \{2,3\}$. Thus A is block-rank one. In this example every pair of magnitude-proportional complex rows is already linearly dependent, so A is block-orthogonal.

F.A.2 Block Orthogonality and purification

On the same domain, let

$B =$

$x\backslash z$	0	1	2	3
0	$\sqrt{3}$	$\sqrt{3}$	$1/\sqrt{3}$	$1/\sqrt{3}$
1	$\sqrt{3}$	$-\sqrt{3}$	$1/\sqrt{3}$	$-1/\sqrt{3}$
2	0	0	0	0
3	0	0	0	0

.

Again $B \in W_{\{B\}}$ by the one-constraint instance. Its two nonzero rows $\mathbf{u}, \mathbf{v}$ are not linearly dependent, but their magnitude vectors are both $(\sqrt{3}, \sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$. Their equal-magnitude blocks are $T_1 = \{0,1\}$ and $T_2 = \{2,3\}$, and cancellation holds separately on both blocks:

$$\sum_{z \in T_1} u_z \overline{v_z} = 3 - 3 = 0, \qquad \sum_{z \in T_2} u_z \overline{v_z} = \tfrac{1}{3} - \tfrac{1}{3} = 0.$$

Thus $\mathbf{u} \perp_{\mathrm{B}} \mathbf{v}$ and B is block-orthogonal.

For singleton purification, choose $g_1 = \sqrt{3}$. Its nonzero integer powers are not roots of unity, and every nonzero value of B has the form εg_1^e with $\varepsilon \in \{1, -1\}$ and $e \in \{1, -1\}$. The construction replaces g_1 by the first prime $p_1 = 2$, while keeping the root-of-unity factor ε. Thus $\pm\sqrt{3} \mapsto \pm 2$ and $\pm 1/\sqrt{3} \mapsto \pm\frac{1}{2}$. The resulting function $\widetilde{B}$ has denominator-clearing factor $N = 2$, so its normalized purification is

$B' = 2\widetilde{B} =$

$x\backslash z$	0	1	2	3
0	4	4	1	1
1	4	-4	1	-1
2	0	0	0	0
3	0	0	0	0

.

The values have changed, while zeros, signs, and the two magnitude blocks are preserved. Within these blocks, the two nonzero rows of B' give $16 - 16 = 0$ and $1 - 1 = 0$, respectively. Hence B' is pure and block-orthogonal.

F.A.3 Type Partition

Take $q = 3$, with addition on D modulo three, and define $F : D^3 \to \mathbb{C}$ by

$(x_1, x_2)\backslash z$	0	1	2
$(0,0)$	1	0	0
$(0,1)$	0	1	0
$(0,2)$	0	0	1
$(1,0)$	0	2	0
$(1,1)$	0	0	2
$(1,2)$	2	0	0
$(2,0)$	0	0	4
$(2,1)$	4	0	0
$(2,2)$	0	4	0

.

Equivalently,

$$F(x_1, x_2, z) \neq 0 \quad \Longleftrightarrow \quad z \equiv x_1 + x_2 \pmod 3, \tag{F.69}$$

and the unique nonzero entry in a row has value 2^{x_1}. The one-constraint instance gives $F \in W_{\{F\}}$. Its three nonzero complex row classes are

$$\begin{aligned} S_1 &= \{(0,0),(1,2),(2,1)\}, \quad \mathbf{v}_1 = (1,0,0), \\ S_2 &= \{(0,1),(1,0),(2,2)\}, \quad \mathbf{v}_2 = (0,1,0), \\ S_3 &= \{(0,2),(1,1),(2,0)\}, \quad \mathbf{v}_3 = (0,0,1). \end{aligned}$$

For prefixes of length one, $\mathrm{type}_F(0) = \mathrm{type}_F(1) = \mathrm{type}_F(2) = [3]$. Thus one type contains three row-class labels and occurs at all three prefixes. For prefixes of length two, if $a + b \equiv j \pmod 3$ with $j \in \{0,1,2\}$, then $\mathrm{type}_F(a,b) = \{j+1\}$. Hence, at each fixed prefix length, any two types are equal or disjoint. The same function illustrates the Mal'tsev polymorphism condition in Appendix F.A.4.

F.A.4 A Mal'tsev polymorphism

Continue with the same domain D, with addition modulo three and the same function F from Appendix F.A.3. Consider the affine operation $\varphi(a,b,c) = a - b + c \pmod 3$. It is Mal'tsev because

$$\varphi(a,b,b) = a - b + b = a, \qquad \varphi(b,b,a) = b - b + a = a.$$

By (F.69), the support relation is the nontrivial affine relation $\Phi_F = \{(x_1, x_2, z) \in D^3 : z \equiv x_1 + x_2 \pmod 3\}$. Take any three tuples $\mathbf{u}^{(i)} = (x_1^{(i)}, x_2^{(i)}, z^{(i)}) \in \Phi_F$, $i \in [3]$, and let $\mathbf{u} = (u_1, u_2, u_3)$ be their coordinatewise image under φ. Then

$$\begin{aligned} u_3 &\equiv z^{(1)} - z^{(2)} + z^{(3)} \\ &\equiv (x_1^{(1)} - x_1^{(2)} + x_1^{(3)}) + (x_2^{(1)} - x_2^{(2)} + x_2^{(3)}) \\ &\equiv u_1 + u_2 \pmod 3. \end{aligned}$$

Thus $\mathbf{u} \in \Phi_F$, so φ preserves the support.

The row classes in Appendix F.A.3 also give

$$\Omega_F = \left\{ \begin{array}{l} (x_1, x_2, y_1, y_2) \in D^4 : \\ x_1 + x_2 \equiv y_1 + y_2 \pmod 3 \end{array} \right\}.$$

Indeed, two rows are proportional exactly when their unique nonzero entries occur in the same column. The same linearity calculation shows that φ preserves the equation $x_1 + x_2 - y_1 - y_2 = 0$ modulo three, and hence Ω_F. Thus this one operation preserves both relations.

F.A.5 Variable identification

Figure F.2 illustrates the construction in Lemma F.3.3. On a finite domain D, consider a $\#\mathsf{CSP}(\mathcal{F})$ instance I with retained variables $x_1, \ldots, x_5$ and hidden variables z_1, z_2. Set $F = F_I^{[5]}$ and take the constraint multiset

$$\mathcal{C}_I = \{f(x_1, z_1, x_3), g(z_1, x_2, x_5), \\ h(x_4, z_2, x_3), \ell(z_2, x_5)\}.$$

For the map $\pi : [5] \twoheadrightarrow [3]$ given by $\pi = (1, 2, 1, 3, 2)$, replace each x_i by $y_{\pi(i)}$. The resulting instance I_π has constraint multiset

$$\mathcal{C}_{I_\pi} = \{f(y_1, z_1, y_1), g(z_1, y_2, y_2), \\ h(y_3, z_2, y_1), \ell(z_2, y_2)\}.$$

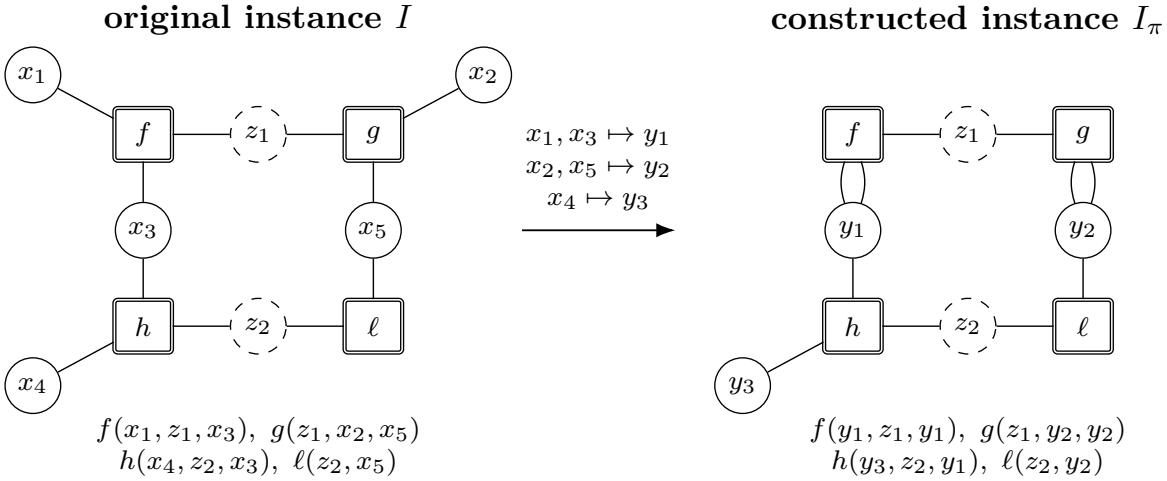


Figure F.2: Construction in Lemma F.3.3. Circles are variables, boxes are constraints, and dashed circles are hidden variables. Constraint nodes and hidden variables are unchanged. Only incidences of retained variables are relabelled. The parallel edges at f and g in I_π record repeated occurrences created by identification. Summing over the unchanged z_1, z_2 gives $F_{I_\pi}^{[3]}(y_1, y_2, y_3) = F(y_1, y_2, y_1, y_3, y_2) = H(y_1, y_2, y_3)$.

F.A.6 Arity compression

To illustrate the identification step in Lemma F.3.4, let $q = 2$ and let $F \in W_{\mathcal{F}}$ have arity nine. Choose the two row indices

$$\mathbf{x} = (0, 0, 1, 0, 1, 1, 0, 1), \\ \mathbf{y} = (0, 1, 0, 0, 1, 0, 1, 1).$$

Table F.1 lists the four classes of positions grouped by (x_i, y_i). Identify the variables within each class and keep the column variable z separate.

class	C_{00} (•)	C_{01} (∘)	C_{10} (▲)	C_{11} (⋄)
positions	$\{1,4\}$	$\{2,7\}$	$\{3,6\}$	$\{5,8\}$
new variable	u_{00}	u_{01}	u_{10}	u_{11}
x-value	0	0	1	1
y-value	0	1	0	1

Table F.1: The position classes for the displayed Boolean row-index tuples. Each black-and-white marker identifies one class of positions carrying the same ordered pair (x_i, y_i).

In this example the function obtained by identifying variables is

$$H(u_{00}, u_{01}, u_{10}, u_{11}, z) = F(u_{00}, u_{01}, u_{10}, u_{00}, \\ u_{11}, u_{10}, u_{01}, u_{11}, z).$$

Assigning $(u_{00}, u_{01}, u_{10}, u_{11})$ the values $(0,0,1,1)$ or $(0,1,0,1)$ recovers $\mathbf{x}$ or $\mathbf{y}$, respectively. There are four identified row variables and one separate column variable, so the resulting arity is $4 + 1 = q^2 + 1 = 5$.

F.A.7 Tensor realization

Take $q = 2$, $k = 2$, $p_+ = 2$, $p_- = 1$, and $M = X_{(0,1)}X_{(1,1)}Y_{(1,0)}$. The three factors fix the input tuples $(0,1), (1,1), (1,0)$. Transposing these tuples as in Definition F.5.3 gives $\beta_M(1) = (0,1,1), \quad \beta_M(2) = (1,1,0)$.

Figure F.3 visualizes this example and the construction in Lemma F.5.4.

F.A.8 Support realization

In Lemma F.7.2, the relation R consists of assignments to the free variables that extend to an assignment of the existential variables making every selected function nonzero.

For example, if $F_1, F_2 \in W_{\mathcal{F}}$ are binary, then

$$R(x_1, x_2) \quad \Longleftrightarrow \quad \exists y \; \Phi_{F_1}(x_1, y) \wedge \Phi_{F_2}(y, x_2).$$

Both atoms use the same existential variable y.

Figure F.4 shows how this binary example is implemented by one $\#\mathsf{CSP}(\mathcal{F})$ instance.

F.A.9 Preservation constraints

In the proof of Lemma F.7.4, the variable $x_{(a,b,c)}$ represents the output of the candidate operation at (a,b,c). A partial operation p assigns these outputs. For example, let $q = 2$ and suppose a binary relation R contains $\mathbf{u}^1 = (0,0)$, $\mathbf{u}^2 = (0,1)$, and $\mathbf{u}^3 = (1,1)$. The preservation constraint for these three tuples is $R(x_{(0,0,1)}, x_{(0,1,1)})$, which requires $(p(0,0,1), p(0,1,1)) \in R$ under the assignment p.

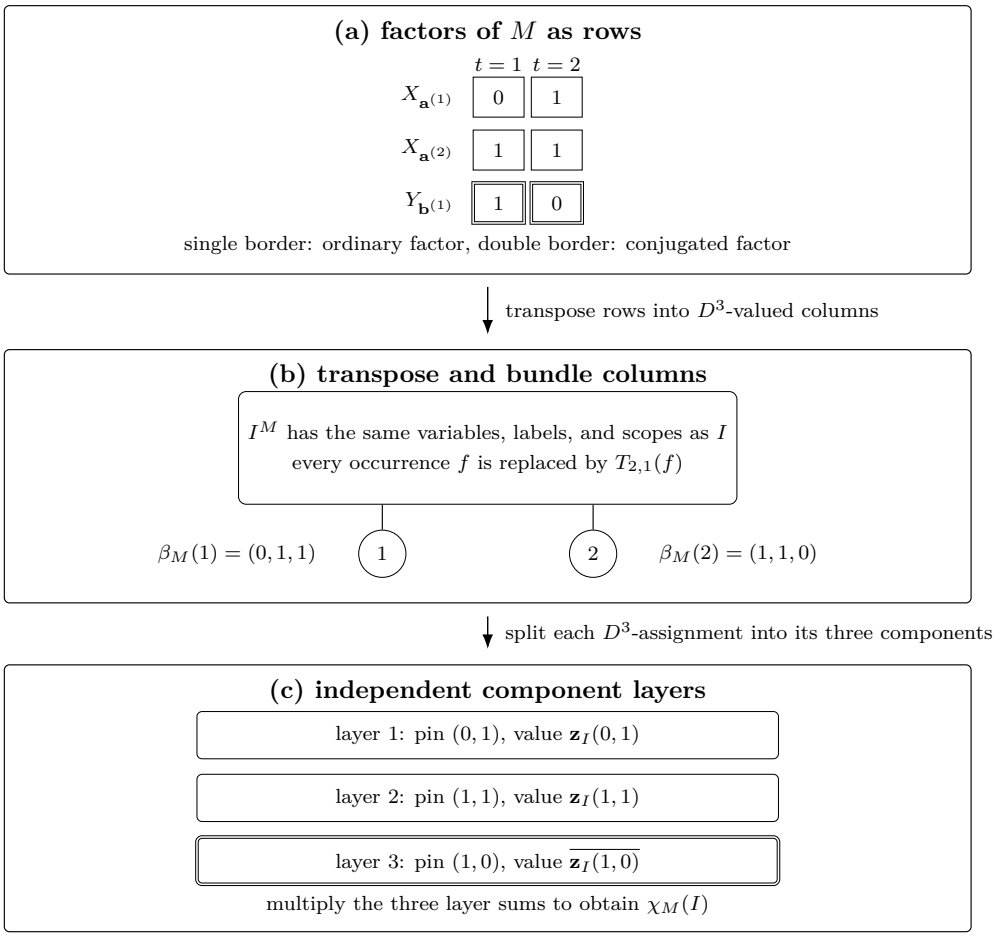


Figure F.3: Tensor realization in Definition F.5.3 for $M = X_{(0,1)}X_{(1,1)}Y_{(1,0)}$. The rows indexing the three monomial factors are transposed into one D^3-valued pin at each label. An assignment to I^M is then exactly three independent D-valued assignments to the same instance I. Double borders mark the factor and layer that use conjugated constraint weights. Hence its pinned partition value factors as $\mathbf{z}_I(0, 1)\mathbf{z}_I(1, 1)\overline{\mathbf{z}_I(1, 0)} = \chi_M(I)$.

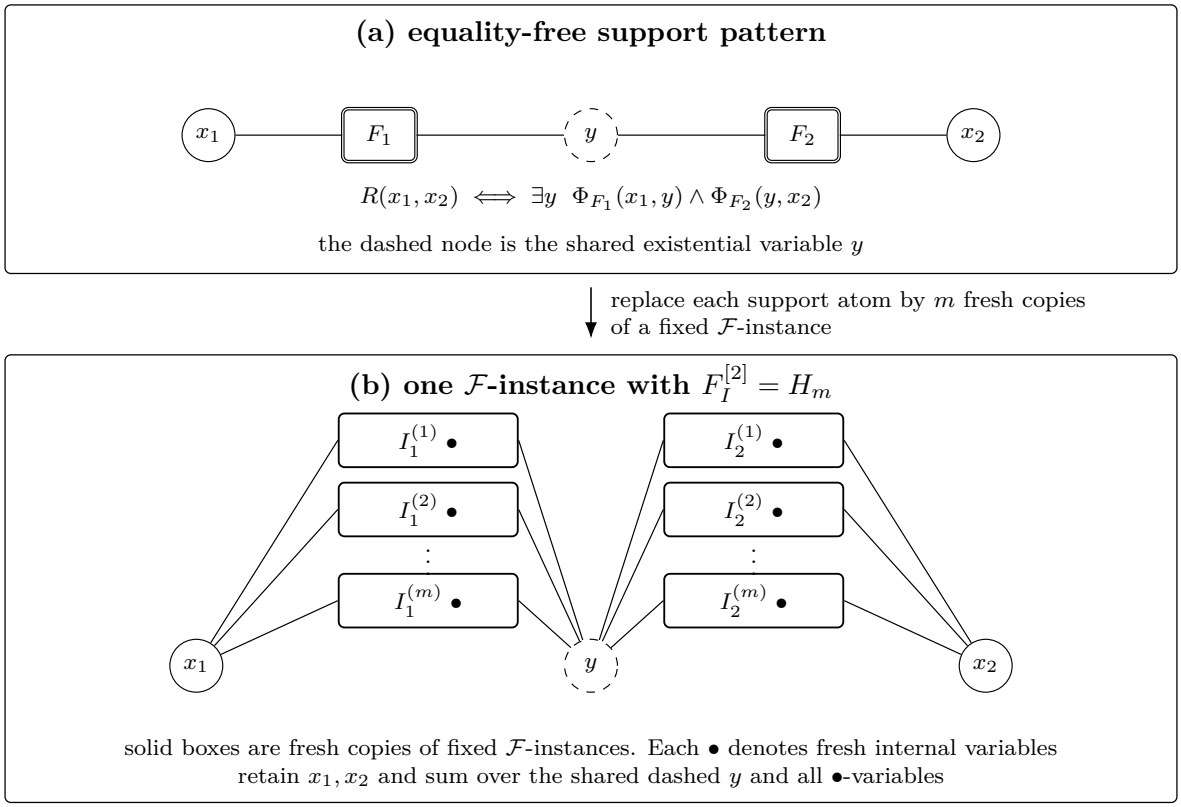


Figure F.4: The instance construction in Lemma F.7.2 for the binary example. For each atom, panel (b) schematically shows m fresh copies of a $\mathcal{F}$-instance representing the corresponding function, indexed by $[m]$. Identify their retained variables with the logical variables x_1, y, x_2. All internal variables, denoted by •, remain fresh. Summing out the internal variables first gives $F_1(x_1, y)^m F_2(y, x_2)^m$, and then summing over the one shared variable y gives $H_m(x_1, x_2)$.

F.B Why all labelled instances are needed

Young's main theorem [You25, Theorem 8] characterizes constraint-function isomorphism through partition values on all instances. In the finite-family setting used here, let $\mathcal{H}$ and $\mathcal{G}$ be finite families of constraint functions on finite domains $D_{\mathcal{H}}$ and $D_{\mathcal{G}}$, with values in a common field of characteristic zero and an arity-preserving bijection between the families. The theorem states that there is a single domain bijection preserving every pair of corresponding functions if and only if

$$Z^{\mathcal{H}}(I) = Z^{\mathcal{G}}(I_{\mathcal{H}\to\mathcal{G}})$$

for every finite unlabelled $\mathcal{H}$-instance I. Here $Z^{\mathcal{H}}(I)$ is the unpinned partition value, and $I_{\mathcal{H}\to\mathcal{G}}$ replaces each constraint function by its corresponding function while retaining the variables and scopes.

Young's Corollary 35 asserts a stronger statement involving pins and a restricted class of instances. Following [You25, Definition 4, p. 6], a labelled $\mathcal{H}$-instance is *simple* if the variables in each constraint scope are pairwise distinct and, for every fixed function $h \in \mathcal{H}$, no two occurrences of h have scope tuples that are permutations of one another.

Fix $k \geq 0$ and pins $\beta : [k] \to D_{\mathcal{H}}$ and $\gamma : [k] \to D_{\mathcal{G}}$. Corollary 35 [You25, p. 27] asserts that if

$$Z^{\mathcal{H}}_{\beta}(I) = Z^{\mathcal{G}}_{\gamma}(I_{\mathcal{H}\to\mathcal{G}})$$

for every finite simple k-labelled $\mathcal{H}$-instance I, then there is a domain isomorphism $\sigma : D_{\mathcal{H}} \to D_{\mathcal{G}}$ such that $\sigma(\beta(i))$ and $\gamma(i)$ are twins in $\mathcal{G}$ for every $i \in [k]$. Recall that twin domain elements can replace one another in any input position without changing any constraint-function value.

Theorem F.5.2 establishes this pinned conclusion under equality on all finite labelled instances. Its interpolation proof uses repeated constraints to form the powers in its indicator polynomials, and its single-constraint tests use repeated scope positions. The counterexample below shows that restricting the premise to simple instances is insufficient.

Remark F.B.1 (Simple-instance indistinguishability need not imply isomorphism). The counterexample uses singleton sets of strictly positive rational-valued unary functions with no distinct twin domain elements. It also refutes the simple-instance formulation of Theorem F.5.2 in the first version of this paper.

Take $q = 3$, $\mathcal{H} = \{f\}$, and $\mathcal{G} = \{g\}$, where

$$(f(0), f(1), f(2)) = (1, 2, 6), \qquad (g(0), g(1), g(2)) = (1, 3, 5).$$

The families are similar under $f \leftrightarrow g$. Take one label, with pins β, γ both assigning its variable u the value 0. Every simple $\mathcal{H}$-instance I places at most one copy of f on each variable. If r hidden variables carry a constraint and s hidden variables are isolated, independence of the hidden-variable sums gives

$$Z^{\mathcal{H}}_{\beta}(I) = f(0)^{\varepsilon}\Big(\sum_{a\in D} f(a)\Big)^{r} q^{s} = 9^{r}3^{s} = Z^{\mathcal{G}}_{\gamma}(I_{\mathcal{H}\to\mathcal{G}}),$$

where $\varepsilon \in \{0, 1\}$ records whether u carries a constraint. Since $f(0) = g(0) = 1$ and both unary sums equal 9, the equality holds for every simple instance, including empty constraint sets and either value of ε. No domain bijection σ is an isomorphism: it would require $2 = f(1) = g(\sigma(1))$, whereas $2 \notin \{1, 3, 5\}$. Both functions are injective, so allowing the prescribed pins to match only up to twins does not change the conclusion.

The obstruction occurs explicitly in the proof of [You25, Lemma 29, p. 20], which is used in the unary case of Theorem 34 and hence in Corollary 35. That proof places p_j copies of the same

unary constraint (F_j, v) on one hidden variable v and treats the resulting instance as simple. For $p_j \geq 2$ it violates Definition 4, so equality on simple instances does not supply the power-sum identities needed for the interpolation. Indeed, the unrestricted instance with two copies of the unary constraint on a single hidden variable, and an unused pinned label, has respective values

$$\sum_{a \in D} f(a)^2 = 41, \qquad \sum_{a \in D} g(a)^2 = 35.$$

The two-copy instance therefore distinguishes these families in the all-instance setting of [You25, Theorem 8] and Theorem F.5.2.